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Biomarkers for the diagnosis and treatment of fibrotic lung disease

Abstract

The present disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/624,500, filed Dec. 19, 2019, which application is a National Stage Application, filed under 35 U.S.C. § 371, of PCT/US2018/039573, filed Jun. 26, 2018, which claims the benefit of provisional application U.S. Ser. No. 62/525,087, filed Jun. 26, 2017 and U.S. Ser. No. 62/525,088, filed Jun. 26, 2017, the contents of each of which are herein incorporated by reference in their entirety.

INCORPORATION OF SEQUENCE LISTING

(1) The Sequence Listing XML associated with this application is provided electronically in XML format and is hereby incorporated by reference into the specification. The name of the XML file containing the Sequence Listing XML is “UNCO-018_C01US_SeqList_ST26.xml”. The XML file is 227,030 bytes, created on Sep. 7, 2022, and is being submitted electronically via USPTO Patent Center.

FIELD OF THE DISCLOSURE

(2) The disclosure is directed to molecular biology, genetics, and therapeutics for fibrotic lung disease.

BACKGROUND

(3) Fibrotic pulmonary diseases are progressive and irreversible. Standard therapies are mere palliative as they cannot address the underlying disease mechanism once the subject has progressed to a point at which symptoms are present. Thus, there is a long-felt but unmet need in the field for a method of treating asymptomatic subjects as well as those who are at risk of developing fibrotic pulmonary diseases to prevent onset of the disease, delay onset of the disease, or reduce the severity of disease symptoms. The methods of the disclosure provide a preventative or efficacious treatment, as opposed to a merely palliative treatment, for asymptomatic subjects as well as those subjects at risk of developing the disease.

SUMMARY

(4) The disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

(5) In some embodiments of the methods of the disclosure, the subject presents radiographic Usual Interstitial Pneumonia (UIP). In some embodiments, the subject has fibrotic interstitial lung disease (FILD). In some embodiments, the subject has a blood relative with familial interstitial pneumonia (FIP). In some embodiments, including those embodiments wherein the subject has a blood relative with familial interstitial pneumonia (FIP), the blood relative is a sibling. Alternatively, or in addition, in some embodiments, the subject has a mutation in a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase 1 (CAMKK1), zinc finger with KRAB and SCAN domains 1 (ZKSCAN1), isovaleryl-CoA dehydrogenase (IVD), ATPase phospholipid transporting 11A (AK025511) or Matrix Metalloprotease-7 (MMP-7).

(6) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

(7) In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is upregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the

disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitron (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(8) In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is downregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(9) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding MUC5B. In some embodiments, the mutation is a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950 comprising (SEQ ID NO: 7).

(10) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding TERC. In some embodiments, the mutation is a polymorphism in a sequence encoding TERC or a regulatory sequence thereof. In some embodiments the polymorphism is rs6793295 comprising (SEQ ID NO: 1).

(11) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic FAM13A. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic FAM13A or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2609260.

(12) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic TERT. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic TERT or a regulatory sequence thereof. In some embodiments, the

polymorphism is rs4449583.

(13) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic DSP. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic DSP or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2076295.

(14) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic ZKSCAN1. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic ZKSCAN1 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs6963345.

(15) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic OBFC1. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic OBFC1 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2488000.

(16) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding an AK025511 3' UTR. In some embodiments, the mutation is a polymorphism in a sequence encoding an AK025511 3' UTR or a regulatory sequence thereof. In some embodiments, the polymorphism is rs1278769.

(17) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding IVD. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic IVD or a regulatory sequence thereof. In some embodiments, the polymorphism is rs35700143.

(18) In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic DPP9. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic DPP9 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs12610495.

(19) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding FAM13A. In some embodiments, the mutation is a polymorphism in a sequence encoding FAM13A or a regulatory sequence thereof. In some embodiments the polymorphism is rs2609255 comprising (SEQ ID NO: 2).

(20) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding TERT. In some embodiments, the mutation is a polymorphism in a sequence encoding TERT or a regulatory sequence thereof. In some embodiments the polymorphism is rs2736100 comprising (SEQ ID NO: 3).

(21) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding DSP. In some embodiments, the mutation is a polymorphism in a sequence encoding DSP or a regulatory sequence thereof. In some embodiments the polymorphism is rs2076295 comprising (SEQ ID NO: 4).

(22) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding AZGP1. In some embodiments, the mutation is a polymorphism in a sequence encoding AZGP1 or a regulatory sequence thereof. In some embodiments the polymorphism is rs4727443 comprising (SEQ ID NO: 5).

(23) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding OBFC1. In some embodiments, the mutation is a polymorphism in a sequence encoding OBFC1 or a regulatory sequence thereof. In some embodiments the polymorphism is rs11191865 comprising (SEQ ID NO: 6).

(24) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding ATP11A. In some embodiments, the mutation is a polymorphism in a sequence encoding ATP11A or a regulatory sequence thereof. In some embodiments the polymorphism is rs12787690 comprising (SEQ ID NO: 8).

(25) In some embodiments of the methods of the disclosure, the subject has a mutation in a

sequence encoding IVD/DISP2. In some embodiments, the mutation is a polymorphism in a sequence encoding IVD/DISP2 or a regulatory sequence thereof. In some embodiments the polymorphism is rs2034650 comprising (SEQ ID NO: 9).

(26) In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding DPP9. In some embodiments, the mutation is a polymorphism in a sequence encoding DPP9 or a regulatory sequence thereof. In some embodiments the polymorphism is rs12610495 comprising (SEQ ID NO: 10).

(27) In some embodiments of the methods of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

(28) In some embodiments of the methods of the disclosure, the therapeutic agent comprises a N-acetylcysteine, pirfenidone, and nintedanib.

(29) In some embodiments of the methods of the disclosure, the therapeutic agent comprises pirfenidone. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the effective dosage is about 2400 mg/day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject about 800 mg of pirfenidone per day for a first week; (b) administering to the subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments of the escalating dosage regimen, the capsule or tablet comprises 267 mg of pirfenidone.

(30) In some embodiments of the methods of the disclosure, the therapeutic agent comprises nintedanib. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 300 mg/day. In some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is administered according to a modified or interrupted dosage regimen. In some embodiments, the modified or interrupted dosage regimen comprises (a) administering to the subject about 300 mg of nintedanib per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject about 200 mg of nintedanib per day until the subject presents the control level of liver enzymes; and (c) administering to the subject about 300 mg of nintedanib per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the modified or interrupted regimen comprises (a) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day until the subject

presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject two capsules or tablets comprising about 100 mg twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; and (c) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment.

(31) In some embodiments of the methods of the disclosure, the therapeutic agent prevents the onset or development of a sign or symptom of the fibrotic lung disease.

(32) In some embodiments of the methods of the disclosure, the therapeutic agent delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the therapeutic agent.

(33) In some embodiments of the methods of the disclosure, the therapeutic agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.

(34) In some embodiments of the methods of the disclosure, the therapeutic agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.

(35) In some embodiments of the methods of the disclosure, the at least one sign of the fibrotic lung disease is detectable before the subject presents a symptom of the fibrotic lung disease. In some embodiments, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough. In some embodiments, the symptom comprises shortness of breath during exercise, shortness of breath at rest, a dry and hacking cough, repeated bouts of coughing, and uncontrollable bouts of coughing.

(36) In some embodiments of the methods of the disclosure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, a secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

(37) The disclosure provides a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a dose of a composition that modifies transcription or translation of a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase Kinase 1 (CAMKK1) or Matrix Metalloprotease-7 (MMP-7), wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

(38) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a composition that modifies an activity of a product of a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

(39) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies

transcription or translation decreases or inhibits transcription or translation.

(40) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition decreases or inhibits transcription or translation of a sequence encoding a gene selected from the group consisting of Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), S100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitron (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(41) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies transcription or translation increases or activates transcription or translation.

(42) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition increases or activates transcription or translation of a sequence encoding a gene selected from the group consisting of Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(43) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies an activity decreases or inhibits the activity.

(44) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition decreases or inhibits the activity of a sequence encoding a gene selected from Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12

(S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitron (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(45) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies an activity increases or activates the activity.

(46) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition increases or activates the activity of a sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(47) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject is a mammal.

(48) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the mammal is genetically-modified.

(49) In some embodiments of the methods of the disclosure, the genetically-modified mammal is a model organism for the fibrotic lung disease.

(50) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA.

(51) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is pulmonary fibrosis or IPF.

(52) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is IPF.

(53) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject carries a mutation in a sequence encoding MUC5B.

- (54) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter.
- (55) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the polymorphism is rs35705950.
- (56) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject carries a mutation in a sequence encoding TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.
- (57) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition prevents the onset or development of a sign or symptom of the fibrotic lung disease.
- (58) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the composition.
- (59) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom when treated using a standard therapeutic intervention.
- (60) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the composition.
- (61) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom when treated using a standard therapeutic intervention.
- (62) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises a N-acetylcysteine, pirfenidone, and nintedanib.
- (63) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises pirfenidone.
- (64) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, an effective dosage of pirfenidone is about 2400 mg/day.
- (65) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered orally as a capsule or a tablet.
- (66) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered three times per day.
- (67) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered according to an escalating dosage regimen.
- (68) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the escalating dosage regimen comprises, administering to the non-human subject about 800 mg of pirfenidone per day for a first week; administering to the non-human subject about 1600 mg of pirfenidone per day for a second week;

and administering to the non-human subject about 2400 mg of pirfenidone per day for the remainder of the treatment.

(69) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the escalating dosage regimen comprises, administering to the non-human subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; administering to the non-human subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and administering to the non-human subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment.

(70) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the capsule or tablet comprises 267 mg of pirfenidone.

(71) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises nintedanib.

(72) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, an effective dosage of nintedanib is administered orally as a capsule or a tablet.

(73) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 300 mg/day.

(74) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

(75) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 200 mg/day.

(76) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

(77) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject presents at least one sign of the fibrotic lung disease.

(78) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough.

(79) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the compound prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease.

(80) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the compound prevents the onset for at 1 year, 2 years, 3 years, 4 years, 5 years or any whole or fractional number of years in between.

(81) In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

(82) The disclosure provides a composition for the treatment of a fibrotic lung disease identified by a method of the disclosure, including, a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure.

(83) The disclosure provides a method of treating fibrotic lung disease in a human subject of the disclosure comprising administering a therapeutically effective amount of a composition identified by a method of the disclosure, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease. In some embodiments, the subject is wild type (e.g. does not comprises a mutation or a sequence variation) with respect to a nucleic acid or amino acid sequence encoding one or more of TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

(84) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the human subject presents radiographic Usual Interstitial Pneumonia (UIP).

(85) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has fibrotic interstitial lung disease (FILD).

(86) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has a blood relative with familial interstitial pneumonia (FIP).

(87) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the blood relative is a sibling.

(88) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has a mutation or a sequence variation in a nucleic acid or an amino acid sequence encoding TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

(89) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter.

(90) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the polymorphism is rs35705950.

(91) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA.

(92) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is pulmonary fibrosis or IPF.

(93) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is IPF.

(94) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease.

(95) In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, a secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a map depicting an exemplary hierarchical clustering of differentially expressed genes for pre-pulmonary fibrosis subjects and normal subjects.
- (2) FIG. 2A-B is a pair of volcano plots showing serum sample quality control using Principal component analysis (PCA). FIG. 2A shows before outlier exclusion and FIG. 2B shows after outlier exclusion.
- (3) FIG. 3 is a volcano plot of 3315 plasma proteins, comparing results from 70 patients with established IPF and 70 controls. Solid red symbols represent 57 proteins that were significantly up-regulated and solid blue symbols 12 proteins that were significantly downregulated in patients with IPF after controlling for multiple comparisons and age/gender/smoking.
- (4) FIG. 4 is a survival plot showing receiver operator curves of predictive model for PrePF in asymptomatic relatives from FIP families. Area Under Curve (AUC) values for each model are as follows: Gene Expression alone (red)=0.83, Clinical Predictors (blue)=0.87, Clinical Predictors+MUC5B genotype (green)=0.87, Clinical Predictors+Gene Expression Score (yellow)=0.95, Clinical Predictors+MUC5B genotype+Gene Expression Score (black)=0.95, indicating that a peripheral blood biomarker panel may improve the diagnostic power of a predictive model for PrePF in an at-risk population.
- (5) FIG. 5 is a graph showing MUC5B expression in IPF (N=203) and unaffected subjects (N=139) stratified by MUC5B promoter variant (rs35705950) genotype.
- (6) FIG. 6A is a microscopic image demonstrating that MUC5B is produced in bronchoalveolar epithelia of patients with IPF (brown staining in photomicrographs). Staining is increased in the airways of patients positive for rs35705950 (TT) compared to WT (GG).
- (7) FIG. 6B is a graph showing the percentage of MUC5B positive area of bronchiolar epithelium. Unbiased stereological assessment of staining demonstrates that the volume fraction of stained airways (% positive area) is significantly greater in both the GT heterozygotes and the TT homozygotes.
- (8) FIG. 7A-B is series of bar graphs showing that Scgb1a1- and SFPTC promoter show significant worsening of fibrosis (hydroxyproline) after bleomycin while Muc5b^{-/-} mice are protected. FIG. 7A is a series of graphs and FIG. 7B is a series of confocal images showing that the concentration of Muc5b is directly related to the fibroproliferative response to bleomycin. Representative images from second harmonic generation (SHG) demonstrate increased lung collagen (red) in transgenic mice following bleomycin injury.
- (9) FIG. 8 is bar graph showing that the baseline expression of ER stress genes in lung tissue from WT and Scgb1a1 Muc5bTg mice. Muc5bTg mice have greater ER stress gene expression than their WT littermates (all genes in the ER stress pathway, with $p < 0.05$). Bleomycin also induces ER stress (data not shown).
- (10) FIG. 9 is a pair of microscopic images showing enhanced CHOP (Ddit3) protein in wild type (WT, top photograph) and Scgb1a1-Muc5bTg mice (bottom photograph) after repeat bleomycin.
- (11) FIG. 10 is a pair of microscopic images and corresponding graphs showing the expanded mucus layer and decreased mucociliary transport in SFTPC-Muc5bTg mice compared to littermate wild-type mice. Statistical differences were assessed by Mann-Whitney U Test.
- (12) FIG. 11 is a series of schematic diagram showing that the MUC5B variant and other biomarkers can identify an at-risk population or those with PrePF, establishing the opportunity for primary and secondary prevention of IPF. The 'at-risk' population and the population with PrePF is large (19% with the MUC5B promoter variant and 1.8% of individuals ≥ 50 years of age respectively), IPF is diagnosed in a small population with established, end-stage disease and PrePF can be identified using the MUC5B variant rs35705950. Results indicate that PrePF (detected via

chest CT scan) is associated with a poor prognosis suggesting that PrePF may be a harbinger of IPF.

(13) FIG. 12 is a schematic diagram showing a method of screening at-risk populations (family members of patients with IPF) to identify individuals with PrePF. Focus is placed on identifying the genetic variants and biomarkers that increase the yield of PrePF on HRCT scan, in addition to gender, age, and physiology scores.

(14) FIG. 13 is a table describing the baseline characteristics of patients with rheumatoid arthritis.

(15) FIG. 14 is a table describing the genotypic association of MUC5B rs35705950 single nucleotide polymorphism in patients with RA, with and without interstitial lung disease

(16) FIG. 15 is a table describing the dominant genotypic association of MUC5B rs35705950 single nucleotide polymorphism in patients with RA-ILD and a usual interstitial pneumonia or possible usual interstitial pneumonia pattern (RA-UIP) and in patients with RA-ILD and a pattern inconsistent with usual interstitial pneumonia (RA non-UIP).

(17) FIG. 16A is a forest plot of odds ratios {OR} and 95% confidence intervals {CI} depicting the lack of association of the MUC5B rs35705950 promoter variant with RA without 1LD {RA-nolLD}. The boxes indicate OR, and the horizontal lines indicate 95% CI for the best-fitting genetic model for each association test. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% CI, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

(18) FIG. 16B is a forest plot of odds ratios (OR) and 95% confidence intervals {CI} depicting the additive genotypic association of the MUC5B rs 35705950 promoter variant with RA-ILD. The red dotted line represent the mean value of overall OR value. The boxes indicate OR, and the horizontal lines indicate 95% CI for the best-fitting genetic model for each association test. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% CI, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

(19) FIG. 16C is a forest plot of odds ratios {OR} and 95% confidence intervals {CI} depicting dominant genotypic association of the MUC5B re35705950 promoter variant with ILD among patients with RA and those with the usual interstitial pneumonia or possible usual interstitial pneumonia (UIP) pattern. The boxes indicate OR, and the horizontal lines indicate 95% CI for the best-fitting genetic model for each association test. The red dotted line represent the mean value of overall OR value. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% CI, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

(20) FIG. 17 is a series of photographs depicting MUC5B expression in explanted lung tissue from rheumatoid arthritis associates interstitial lung disease. Representative lung tissue images from unaffected control (GG genotype, Panel A), RA-ILD case #1 (GG genotype, Panel B), and RA-ILD case #2 (GT genotype, Panel C). Low power views with high power view insets identified. Panel A—low power view of normal lung; top and middle insets with high power view of bronchiole with MUC5B staining; bottom inset with high power view of alveolar epithelia. Panel B and C—low power view of the usual interstitial pneumonia pattern in explanted lung tissue of RA-ILD; top inset with high power view of bronchiole with MUC5B staining; middle and bottom insets with high power view of MUC5B staining in metaplastic epithelia lining honeycomb cysts and MUC5B staining of mucous in honeycomb cysts.

(21) FIG. 18 is a flow chart depicting the screening and enrollment process for study subjects.

(22) FIG. 19A-D is a series of photographs depicting High-resolution CT (HRCT) images of: 19A) chest from a study subject whose scan was read as normal, without signs of interstitial lung disease or fibrosis. 19B) HRCT image from subject who was categorized as having “Probable Fibrotic ILD.” 19C) Representative HRCT image from subject who was characterized as having “Definite Fibrotic ILD.” 19D) HRCT image from a case of previously diagnosed, established Idiopathic Pulmonary Fibrosis (IPF) in one of the study families.

(23) FIG. 20 is a table depicting a summary of characteristics of study subjects used in quantitative CT Analyses.

(24) FIG. 21A-F is a series of photographs depicting representative axial HRCT images visually assessed as “No Fibrosis” (21A), “Probable Fibrotic ILD” (21C) and “Definite Fibrotic ILD” (E). Below each is the corresponding quantitative HRCT results for the above scan: (21B) “No Fibrosis” fibrosis extent 1.7% (fibrosis score=0.55), (21D) “Probable Fibrotic ILD” fibrosis extent 18.5% (fibrosis score 2.92), (F) “Definite Fibrotic ILD” fibrosis extent 35.5% (fibrosis score 3.60), Classification results color coded as follows: green=normal lung, blue=airway, yellow=reticular abnormality, magenta=ground glass opacity, red=honeycombing.

(25) FIG. 22 is a table depicting Screening Cohort Subject Characteristics. *DNA available on a total of 489 subjects (404 No Fibrosis and 75 PrePF subjects). **Odds ratios reported in this table were calculated from a mixed effects logistic regression model including age (as a continuous variable), male sex, ever smoker (yes/no), and MUC5B promoter variant (rs35705950) genotype. ***In the reported model, rs35705950 coded as a dominant allele; in log-additive genetic model, $p=0.05$, as well.

(26) FIG. 23 is a table depicting patterns of CT abnormalities in scans with probable or definite fibrotic ILD. *Because a confident single diagnosis was relatively uncommon, most cases included consideration of several patterns. For this reason, the percentages add up to more than 100%.

(27) FIG. 24 is a box plot depicting fibrosis score by visual diagnosis. Boxplots of fibrosis scores based on quantitative HRCT assessment for each visual diagnosis category. Fibrosis score means were significantly different (ANOVA, $p<0.0001$) across groups defined by visual diagnosis. Comparison of fibrosis score between groups showed significant differences for all comparisons ($p<0.01$ for all).

(28) FIG. 25A-C is a series of graphs depicting Receiver Operating Characteristic (ROC) curves for quantitative imaging measures of Fibrosis and PrePF. FIG. 5A depicts ROC curves for visual diagnosis compared to log HAA scores. FIG. 5B depicts ROC Curves for visual diagnosis compared to fibrosis scores. ROC analysis showed that fibrosis score discriminates subjects with visual diagnosis of PrePF. Average area under the curve (AUC) in fivefold cross validation was 0.85 (range 0.83-0.87) and average accuracy, sensitivity, and specificity in the test partitions were 0.83 (range 0.74-0.86), 0.74 (range 0.56-0.92) and 0.84 (range 0.76-0.89) respectively. Optimal threshold for fibrosis score ranged from 1.40-1.42. FIG. 5C depicts Density plots of fibrosis scores for visually diagnosed PrePF (pink) and No Fibrosis (blue) scans—the fibrosis score optimal threshold is indicated with the red line (1.40).

(29) FIG. 26 is a series of tables depicting Dyspnea questionnaire data. FIG. 26A depicts breathlessness responses for the cohort. FIG. 26B depicts breathlessness responses by Visual CT diagnosis.

(30) FIG. 27 is a graph that depicts the prevalence of PrePF in FIP Siblings Cohort by Age and MUC5B Genotype. PrePF prevalence in this FIP siblings cohort increases by age, as shown in this graph. By age >60 years, the prevalence of PrePF differed significantly based on MUC5B genotype (* $p=0.02$). Subjects with the variant are depicted by the red line, while those without it are depicted with the blue line.

(31) FIG. 28 is a table depicting subject characteristics based on Quantitative Fibrosis Score. Clinical characteristics and genotype breakdown of subjects with quantitative HRCT analyses. The cutoff of 1.4 for the logarithm of fibrosis score is based on analyses presented in the text. * p -value

compares characteristic between groups. Linear regression values regress fibrosis score on age, male sex, smoking history, and MUC5B promoter variant. **In the reported model, rs35705950 coded as a dominant allele given small number of TT subjects.

(32) FIG. 29 is a table depicting an exploratory genetic association study of 13 pulmonary fibrosis susceptibility variants in RA-ILD.

DETAILED DESCRIPTION OF THE DISCLOSURE

(33) The present disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

(34) Methods of Identifying a Therapeutic Agent of the Disclosure or Target Thereof

(35) The disclosure provides a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a dose of a composition that modifies transcription or translation of a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase Kinase 1 (CAMKK1) or Matrix Metalloprotease-7 (MMP-7), wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

(36) The disclosure provides method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a composition that modifies an activity of a product of a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

(37) In some embodiments of the methods of the disclosure, the composition that modifies transcription or translation decreases or inhibits transcription or translation. In some embodiments, the composition decreases or inhibits transcription or translation of a sequence encoding a gene selected from the group consisting of Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitronectin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7),

Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(38) In some embodiments of the methods of the disclosure, the composition that modifies transcription or translation increases or activates transcription or translation. In some embodiments, the composition increases or activates transcription or translation of a sequence encoding a gene selected from the group consisting of Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(39) In some embodiments of the methods of the disclosure, the composition that modifies an activity decreases or inhibits the activity. In some embodiments, the composition decreases or inhibits the activity of a sequence encoding a gene selected from Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), S100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitron (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITI5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(40) In some embodiments of the methods of the disclosure, the composition that modifies an activity increases or activates the activity. In some embodiments, the composition increases or activates the activity of a sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family

3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(41) In some embodiments of the methods of the disclosure, the non-human subject is a mammal. In some embodiments, mammal is genetically-modified. In some embodiments, the genetically-modified mammal is a model organism for the fibrotic lung disease.

(42) In some embodiments of the methods of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

(43) In some embodiments of the methods of the disclosure, the non-human subject carries a mutation in a sequence encoding MUC5B. In some embodiments, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950. Alternatively, or in addition, in some embodiments, the non-human subject carries a mutation in a sequence encoding TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

(44) In some embodiments of the methods of the disclosure, the composition prevents the onset or development of a sign or symptom of the fibrotic lung disease.

(45) In some embodiments of the methods of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the composition. In some embodiments, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom when treated using a standard therapeutic intervention.

(46) In some embodiments of the methods of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the composition. In some embodiments, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom when treated using a standard therapeutic intervention.

(47) In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises a N-acetylcysteine, pirfenidone, and nintedanib.

(48) In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises pirfenidone. In some embodiments, an effective dosage of pirfenidone is about 2400 mg/day. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, the effective dosage is administered three times per day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, the escalating dosage regimen comprises (a) administering to the non-human subject about 800 mg of pirfenidone per day for a first week; (b) administering to the non-human subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the non-human subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, the escalating dosage regimen comprises (a) administering to the non-human subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the non-human subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the non-human subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments, the capsule or tablet comprises 267 mg of pirfenidone.

(49) In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises nintedanib. In some embodiments, an effective dosage of nintedanib is administered orally as a capsule or a tablet. In some embodiments, the effective dosage is about 300 mg/day. In

some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

(50) In some embodiments of the methods of the disclosure, the non-human subject presents at least one sign of the fibrotic lung disease. In some embodiments, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough.

(51) In some embodiments of the methods of the disclosure, the compound prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, the compound prevents the onset for at 1 year, 2 years, 3 years, 4 years, 5 years or any whole or fractional number of years in between. In some embodiments, the secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

(52) The disclosure provides a composition for the treatment of a fibrotic lung disease identified by a method of the disclosure for identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease.

(53) Subjects of the Disclosure

(54) The disclosure provides a method of treating a fibrotic lung disease in a human subject comprising administering to the subject the composition for the treatment of a fibrotic lung disease identified by a method of the disclosure for identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

(55) In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the human subject presents radiographic Usual Interstitial Pneumonia (UIP). In some embodiments, the human subject has fibrotic interstitial lung disease (FILD). In some embodiments, the human subject has a blood relative with familial interstitial pneumonia (FIP). In some embodiments, the blood relative is a sibling. Alternatively, or in addition, in some embodiments, the human subject has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7. In some embodiments, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950.

(56) In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

(57) In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, the secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

(58) Idiopathic Pulmonary Fibrosis (IPF)

(59) IPF is localized to the lung and is characterized by a pattern of heterogeneous, subpleural patches of fibrotic, remodeled lung, and often results in death within 3-5 years of diagnosis. IPF affects 5 million people worldwide, disproportionately affects men, is associated with cigarette smoking, increases with age, is inexplicably increasing in prevalence, and is likely underdiagnosed. Most patients with IPF are discovered in the advanced stage when little can be done to influence survival. There is a critical unmet need in idiopathic pulmonary fibrosis (IPF) for an early detection and prevention of IPF. Earlier diagnosis of IPF detects subjects with a lower burden of fibrotic lung

disease providing an opportunity for secondary prevention of this progressive disease and changes the clinical approach to patients with IPF from palliative to preventive.

(60) Early detection and prevention of idiopathic pulmonary fibrosis (IPF) is critical. As demonstrated herein, treatment of subjects at risk for developing PrePF is based on two central concepts of first, understanding that PrePF is essential for primary and secondary prevention of IPF and second, that similar to asymptomatic family members of familial IPF (FIP; ≥ 2 family members with IPF), asymptomatic family members of sporadic IPF represent an at-risk population for PrePF. These central concepts are supported by the observation that 1) IPF has a pre-symptomatic phase and PrePF appears to be a harbinger of IPF, 2) familial and sporadic IPF are similar etiologically, 3) MUC5B promoter variant is critical to early disease recognition and 4) identification of PrePF represents an opportunity to prevent extensive lung fibrosis. As shown herein, a common gain-of-function MUC5B promoter variant rs35705950 is a strong risk factor (genetic and otherwise), accounting for at least 30% of the total risk of developing IPF. The MUC5B promoter variant rs35705950 may be used to identify individuals with PrePF. MUC5B promoter variant rs35705950 is also predictive of radiographic progression of PrePF and is present in over 50% of non-Hispanic white patients with IPF and is also associated with unique clinical and biological IPF phenotypes. PrePF can be predicted using a combination of clinical risk factors, the MUC5B promoter variant rs35705950, and a panel of biomarkers. This disclosure provides methods of treating subjects with Preclinical Pulmonary Fibrosis (PrePF) and who may also be at risk for developing IPF. The methods of the disclosure fundamentally change the clinical approach to treating subjects with IPF, shifting the focus from a merely palliative to a proactive and preventive therapy.

(61) Rheumatoid Arthritis-Associated Interstitial Lung Disease (RA-ILD)

(62) Rheumatoid arthritis (RA) is a common inflammatory and autoimmune disease that is associated with progressive impairment, systemic complications and increased mortality. Interstitial lung disease (RA-ILD) is detected in up to 60% of patients with RA on high-resolution computed-tomography (HRCT), is clinically significant in 10%, and is a leading cause of morbidity and mortality in patients with RA.

(63) RA-ILD shares several characteristics with idiopathic pulmonary fibrosis (IPF), including common environmental risk factors, the high prevalence of the usual interstitial pneumonia (UIP) pattern, the progressive nature of the disease, and poor survival. The hypothesis of a shared genetic background between IPF and RA-ILD was recently suggested by a whole-exome sequencing (WES) genetic association study in patients with RA-ILD, revealing an excess of mutations in genes in RA-ILD previously associated with familial interstitial pneumonia (FIP) including TERT, RTEL1, PARN and SFTPC.

(64) The common gain-of-function promoter variant rs3570595013 of the gene encoding mucin5B (MUC5B) is the strongest genetic risk factor for IPF, observed in at least 50% of the cases of IPF and accounting for 30% of the risk of developing this disease. The MUC5B promoter variant is associated with increased expression of MUC5B in lung parenchyma of unaffected controls and cases of IPF. Consequently, it is hypothesized that the MUC5B promoter variant rs35705950 would also contribute to the occurrence of RA-ILD. To test this hypothesis, a multi-ethnic association study of the MUC5B promoter variant and RA-ILD in seven distinct case series was performed.

(65) The MUC5B promoter variant rs35705950, the strongest genetic risk factor for IPF, is also a strong risk factor for RA-ILD, especially among those with radiographic evidence of UIP. Of note, the effect of the MUC5B promoter variant on the development of ILD associated with RA was similar in magnitude and direction to that observed in IPF.

(66) The relationship between the MUC5B promoter variant and RA-ILD may be specific to UIP and may not be generalizable to other autoimmune conditions of the lung. The MUC5B promoter variant has not been found to be associated with risk of ILDs linked to systemic sclerosis or autoimmune myositis. Unlike these other types of ILD, RA-ILD shares more characteristics with IPF, notably the increased frequency of the UIP pattern (both radiologic and histologic), an

increased prevalence of male sex and older age, and genetic susceptibility as assessed by an excess of mutations in genes linked to FIP in a cohort of RA-ILD, and now the MUC5B promoter variant rs35705950.

(67) The disclosure demonstrates that the MUC5B promoter variant is a risk factor for UIP, and not simply limited to IPF and RA-ILD. In fact, emerging studies have identified the MUC5B promoter variant as a risk factor for chronic hypersensitivity pneumonitis, another condition known to have a sub-phenotype of UIP. Further, since HRCT underestimates the presence of ILD and the UIP pattern of fibrosis, our point estimates for association with the MUC5B variant are likely conservative. Similar to IPF, early forms of RA-ILD can be identified using the MUC5B promoter variant as biomarker.

(68) The disclosure demonstrates that Muc5b is overexpressed by the bronchoalveolar epithelia and MUC5B mRNA is co-expressed by cells expressing surfactant protein C, as has been shown in IPF. These findings suggest either type 2 alveolar epithelial cells can express MUC5B or that in patients with RA-ILD, the cells in the distal airspace de-differentiate. Importantly, the disclosure demonstrates for the first time that cells that overexpress MUC5B are undergoing ER stress, a recognized mechanism of cell injury and repair. In aggregate, these findings indicate that the gain-of-function MUC5B promoter variant rs35705950 injures alveolar epithelia by inducing ER stress.

(69) RA-ILD is a complex genetic phenotype with the minor allele of the MUC5B promoter variant rs35705950 identified as a risk factor for the disease. The odds ratios for the association of MUC5B promoter variant with RA-ILD is equivalent to that observed with IPF and substantively higher than those for the most other common risk variants for RA-ILD, including cigarette smoking and the human leukocyte antigen locus for RA.

(70) The MUC5B promoter variant is a risk factor for UIP in general and may prove relevant beyond RA-ILD and IPF.

(71) Expression of MUC5B in the bronchoalveolar epithelia co-incident with markers of ER stress suggest that the MUC5B promoter variant may be causing pulmonary fibrosis by initiating microscopic foci of injury and repair.

(72) The MUC5B promoter variant appears to predict ILD in the RA population, identifying potential opportunities for early ILD detection in patients with RA.

(73) Preclinical Idiopathic Pulmonary Fibrosis

(74) Better understanding and recognition of early pulmonary fibrosis is critical because medical therapies have been shown to slow progression, not to reverse or even stabilize established fibrosis—therefore, intervention before irreversible fibrosis has become extensive has the potential to improve quality of life and decrease morbidity. While IPF affects approximately 5 million people worldwide, between 1.8 and 14% of the general population ≥ 50 years of age have radiologic findings of undiagnosed pulmonary fibrosis. Large cohort studies indicate that interstitial lung abnormalities, postulated to represent early pulmonary fibrosis, are associated with increased mortality, and that most of these abnormalities progress over time. Members of families with 2 or more cases of pulmonary fibrosis (FIP, Familial Interstitial Pneumonia) have been identified as an “at-risk” population. In a previous study of FIP relatives, 14% had interstitial lung abnormalities on high resolution computed tomography (HRCT), and 35% had an abnormal transbronchial biopsy indicating interstitial lung disease.

(75) HRCT provides visualization of the lung parenchyma and plays a key role in the diagnosis of the Idiopathic Interstitial Pneumonias (IIPs), including IPF. Currently, visual diagnosis by thoracic radiologists, in conjunction with multidisciplinary clinical conference, is the gold standard for diagnosing TIPS. However, visual assessment is imprecise and hampered by inter-observer variation. Quantitative HRCT (qHRCT) evaluation provides measures of fibrosis extent that, in subjects diagnosed with IPF, correlate with degree of physiologic impairment at baseline, and may be more sensitive to subtle changes in disease status than routinely used physiological metrics. The design and utility of quantitative methods in the context of early forms of fibrotic ILD requires

further study. Deep learning methods have been increasingly used in imaging to identify and classify CT patterns, and may be particularly valuable in detection of early lung fibrosis.

(76) PrePF is prevalent among FIP relatives, and a texture-based quantitative method of HRCT analyses is useful in identifying these abnormalities in this population, and key risk factors, including the MUC5B promoter variant, predict those at risk of this disease. PrePF subjects are older, more likely to be male, and more likely to have smoked than the unaffected subjects; additionally, the gain-of-function MUC5B promoter variant rs35705950, which has been shown in prior studies to be associated with pulmonary fibrosis, is more common in PrePF subjects when compared to their unaffected family members. Given the subtlety of the fibrotic change in many of these cases of PrePF, the high prevalence of potential UIP pattern on HRCT scan suggests that PrePF subjects may progress to IPF over time.

(77) Methods for Detecting a Genetic Variant

(78) The present disclosure also provides methods of detecting the biomarkers of the present disclosure. Methods of detecting a genetic variant are further described in US Application US 2016-0060701A1 (the contents of which are incorporated herein by reference in their entirety). The practice of the present disclosure employs, unless otherwise indicated, conventional methods of analytical biochemistry, microbiology, molecular biology and recombinant DNA techniques within the skill of the art. Such techniques are explained fully in the literature. (See, e.g., Sambrook, J. et al. *Molecular Cloning: A Laboratory Manual*. 3rd, ed., Cold Spring Harbor Laboratory, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N. Y., 2000; *DNA Cloning: A Practical Approach*, Vol. I & II (D. Glover, ed.); *Oligonucleotide Synthesis* (N. Gait, ed., Current Edition); *Nucleic Acid Hybridization* (B. Hames & S. Higgins, eds., Current Edition); *Transcription and Translation* (B. Hames & S. Higgins, eds., Current Edition); *CRC Handbook of Parvoviruses*, Vol. I & II (P. Tijessen, ed.); *Fundamental Virology*, 2nd Edition, Vol. I & II (B. N. Fields and D. M. Knipe, eds.)).

(79) The methods of the invention are not limited to any particular way of detecting the presence or absence of a genetic variant (e.g. SNP) and can employ any suitable method to detect the presence or absence of a variant(s), of which numerous detection methods are known in the art. Dynamic allele-specific hybridization (DASH) can be used to detect a genetic variant. DASH genotyping takes advantage of the differences in the melting temperature in DNA that results from the instability of mismatched base pairs. The process can be vastly automated and encompasses a few simple principles. Thus, the aspects and embodiments described herein provide methods for assessing the presence or absence of SNPs in a sample (e.g. biological sample) from a subject suspected of having or developing an interstitial lung disease (e.g., because of family history). In certain embodiments, one or more SNPs are screened in one or more samples from a subject. The SNPs can be associated with one or more genes, e.g., one or more genes or other genes associated with mucous secretions as disclosed herein.

(80) Typically, the target genomic segment is amplified and separated from non-target sequence, e.g., through use of a biotinylated primer and chromatography. A probe that is specific for the particular allele is added to the amplification product. The probe can be designed to hybridize specifically to a variant sequence or to the dominant allelic sequence. The probe can be either labeled with or added in the presence of a molecule that fluoresces when bound to double-stranded DNA. The signal intensity is then measured as temperature is increased until the T_m can be determined. A non-matching sequence (either genetic variant or dominant allelic sequence, depending on probe design), will result in a lower than expected T_m .

(81) DASH genotyping relies on a quantifiable change in T_m , and is thus capable of measuring many types of mutations, not just SNPs. Other benefits of DASH include its ability to work with label free probes and its simple design and performance conditions.

(82) Molecular beacons can also be used to detect a genetic variant. This method makes use of a specifically engineered single-stranded oligonucleotide probe. The oligonucleotide is designed such

that there are complementary regions at each end and a probe sequence located in between. This design allows the probe to take on a hairpin, or stem-loop, structure in its natural, isolated state. Attached to one end of the probe is a fluorophore and to the other end a fluorescence quencher. Because of the stem-loop structure of the probe, the fluorophore is in close proximity to the quencher, thus preventing the molecule from emitting any fluorescence. The molecule is also engineered such that only the probe sequence is complementary to the targeted genomic DNA sequence.

(83) If the probe sequence of the molecular beacon encounters its target genomic DNA sequence during the assay, it will anneal and hybridize. Because of the length of the probe sequence, the hairpin segment of the probe will be denatured in favor of forming a longer, more stable probe-target hybrid. This conformational change permits the fluorophore and quencher to be free of their tight proximity due to the hairpin association, allowing the molecule to fluoresce.

(84) If on the other hand, the probe sequence encounters a target sequence with as little as one non-complementary nucleotide, the molecular beacon will preferentially stay in its natural hairpin state and no fluorescence will be observed, as the fluorophore remains quenched. The unique design of these molecular beacons allows for a simple diagnostic assay to identify SNPs at a given location. If a molecular beacon is designed to match a wild-type allele and another to match a mutant of the allele, the two can be used to identify the genotype of an individual. If only the first probe's fluorophore wavelength is detected during the assay then the individual is homozygous to the wild type. If only the second probe's wavelength is detected then the individual is homozygous to the mutant allele. Finally, if both wavelengths are detected, then both molecular beacons must be hybridizing to their complements and thus the individual must contain both alleles and be heterozygous.

(85) A microarray can also be used to detect genetic variants. Hundreds of thousands of probes can be arrayed on a small chip, allowing for many genetic variants or SNPs to be interrogated simultaneously. Because SNP alleles only differ in one nucleotide and because it is difficult to achieve optimal hybridization conditions for all probes on the array, the target DNA has the potential to hybridize to mismatched probes. This can be addressed by using several redundant probes to interrogate each SNP. Probes can be designed to have the SNP site in several different locations as well as containing mismatches to the SNP allele. By comparing the differential amount of hybridization of the target DNA to each of these redundant probes, it is possible to determine specific homozygous and heterozygous alleles.

(86) Restriction fragment length polymorphism (RFLP) can be used to detect genetic variants and SNPs. RFLP makes use of the many different restriction endonucleases and their high affinity to unique and specific restriction sites. By performing a digestion on a genomic sample and determining fragment lengths through a gel assay it is possible to ascertain whether or not the enzymes cut the expected restriction sites. A failure to cut the genomic sample results in an identifiably larger than expected fragment implying that there is a mutation at the point of the restriction site which is rendering it protected from nuclease activity.

(87) PCR- and amplification-based methods can be used to detect genetic variants. For example, tetra-primer PCR employs two pairs of primers to amplify two alleles in one PCR reaction. The primers are designed such that the two primer pairs overlap at a SNP location but each matches perfectly to only one of the possible alleles. As a result, if a given allele is present in the PCR reaction, the primer pair specific to that allele will produce product but not the alternative allele with a different allelic sequence. The two primer pairs can be designed such that their PCR products are of a significantly different length allowing for easily distinguishable bands by gel electrophoresis, or such that they are differently labeled.

(88) Primer extension can also be used to detect genetic variants. Primer extension first involves the hybridization of a probe to the bases immediately upstream of the SNP nucleotide followed by a 'mini-sequencing' reaction, in which DNA polymerase extends the hybridized primer by adding a

base that is complementary to the SNP nucleotide. The incorporated base that is detected determines the presence or absence of the SNP allele. Because primer extension is based on the highly accurate DNA polymerase enzyme, the method is generally very reliable. Primer extension is able to genotype most SNPs under very similar reaction conditions making it also highly flexible. The primer extension method is used in a number of assay formats, and can be detected using e.g., fluorescent labels or mass spectrometry.

(89) Primer extension can involve incorporation of either fluorescently labeled ddNTP or fluorescently labeled deoxynucleotides (dNTP). With ddNTPs, probes hybridize to the target DNA immediately upstream of SNP nucleotide, and a single, ddNTP complementary to the SNP allele is added to the 3' end of the probe (the missing 3'-hydroxyl in didioxynucleotide prevents further nucleotides from being added). Each ddNTP is labeled with a different fluorescent signal allowing for the detection of all four alleles in the same reaction. With dNTPs, allele-specific probes have 3' bases which are complementary to each of the SNP alleles being interrogated. If the target DNA contains an allele complementary to the 3' base of the probe, the target DNA will completely hybridize to the probe, allowing DNA polymerase to extend from the 3' end of the probe. This is detected by the incorporation of the fluorescently labeled dNTPs onto the end of the probe. If the target DNA does not contain an allele complementary to the probe's 3' base, the target DNA will produce a mismatch at the 3' end of the probe and DNA polymerase will not be able to extend from the 3' end of the probe.

(90) The iPLEX® SNP genotyping method takes a slightly different approach, and relies on detection by mass spectrometer. Extension probes are designed in such a way that many different SNP assays can be amplified and analyzed in a PCR cocktail. The extension reaction uses ddNTPs as above, but the detection of the SNP allele is dependent on the actual mass of the extension product and not on a fluorescent molecule. This method is for low to medium high throughput, and is not intended for whole genome scanning.

(91) Primer extension methods are, however, amenable to high throughput analysis. Primer extension probes can be arrayed on slides allowing for many SNPs to be genotyped at once. Broadly referred to as arrayed primer extension (APEX), this technology has several benefits over methods based on differential hybridization of probes. Comparatively, APEX methods have greater discriminating power than methods using differential hybridization, as it is often impossible to obtain the optimal hybridization conditions for the thousands of probes on DNA microarrays (usually this is addressed by having highly redundant probes).

(92) Oligonucleotide ligation assays can also be used to detect genetic variants. DNA ligase catalyzes the ligation of the 3' end of a DNA fragment to the 5' end of a directly adjacent DNA fragment. This mechanism can be used to interrogate a SNP by hybridizing two probes directly over the SNP polymorphic site, whereby ligation can occur if the probes are identical to the target DNA. For example, two probes can be designed; an allele-specific probe which hybridizes to the target DNA so that its 3' base is situated directly over the SNP nucleotide and a second probe that hybridizes the template upstream (downstream in the complementary strand) of the SNP polymorphic site providing a 5' end for the ligation reaction. If the allele-specific probe matches the target DNA, it will fully hybridize to the target DNA and ligation can occur. Ligation does not generally occur in the presence of a mismatched 3' base. Ligated or unligated products can be detected by gel electrophoresis, MALDI-TOF mass spectrometry or by capillary electrophoresis.

(93) The 5'-nuclease activity of Taq DNA polymerase can be used for detecting genetic variants. The assay is performed concurrently with a PCR reaction and the results can be read in real-time. The assay requires forward and reverse PCR primers that will amplify a region that includes the SNP polymorphic site. Allele discrimination is achieved using FRET, and one or two allele-specific probes that hybridize to the SNP polymorphic site. The probes have a fluorophore linked to their 5' end and a quencher molecule linked to their 3' end. While the probe is intact, the quencher will remain in close proximity to the fluorophore, eliminating the fluorophore's signal. During the PCR

amplification step, if the allele-specific probe is perfectly complementary to the SNP allele, it will bind to the target DNA strand and then get degraded by 5'-nuclease activity of the Taq polymerase as it extends the DNA from the PCR primers. The degradation of the probe results in the separation of the fluorophore from the quencher molecule, generating a detectable signal. If the allele-specific probe is not perfectly complementary, it will have lower melting temperature and not bind as efficiently. This prevents the nuclease from acting on the probe.

(94) Förster resonance energy transfer (FRET) detection can be used for detection in primer extension and ligation reactions where the two labels are brought into close proximity to each other. It can also be used in the 5'-nuclease reaction, the molecular beacon reaction, and the invasive cleavage reactions where the neighboring donor/acceptor pair is separated by cleavage or disruption of the stem-loop structure that holds them together. FRET occurs when two conditions are met. First, the emission spectrum of the fluorescent donor dye must overlap with the excitation wavelength of the acceptor dye. Second, the two dyes must be in close proximity to each other because energy transfer drops off quickly with distance. The proximity requirement is what makes FRET a good detection method for a number of allelic discrimination mechanisms.

(95) A variety of dyes can be used for FRET, and are known in the art. The most common ones are fluorescein, cyanine dyes (Cy3 to Cy7), rhodamine dyes (e.g. rhodamine 6G), the Alexa series of dyes (Alexa 405 to Alexa 730). Some of these dyes have been used in FRET networks (with multiple donors and acceptors). Optics for imaging all of these require detection from UV to near IR (e.g. Alex 405 to Cy7), and the Atto series of dyes (Atto-Tec GmbH). The Alexa series of dyes from Invitrogen cover the whole spectral range. They are very bright and photostable.

(96) Example dye pairs for FRET labeling include Alexa-405/Alexa-488, Alexa-488/Alexa-546, Alexa-532/Alexa-594, Alexa-594/Alexa-680, Alexa-594/Alexa-700, Alexa-700/Alexa-790, Cy3/Cy5, Cy3.5/Cy5.5, and Rhodamine-Green/Rhodamine-Red, etc. Fluorescent metal nanoparticles such as silver and gold nanoclusters can also be used (Richards et al. (2008) J Am Chem Soc 130:5038-39; Vosch et al. (2007) Proc Natl Acad Sci USA 104:12616-21; Petty and Dickson (2003) J Am Chem Soc 125:7780-81 Available filters, dichroics, multichroic mirrors and lasers can affect the choice of dye.

(97) In Vitro Complexes

(98) Provided herein are nucleic acid complexes, e.g., formed in in vitro assays to indicate the presence of a genetic variant sequence. One of skill will understand that a nucleic acid complex can also be formed to detect the presence of a dominant allelic sequence, depending on the design of the probe or primer, e.g., in assays to distinguish homozygous and heterozygous subjects.

(99) In some embodiments, the complex comprises a first nucleic acid hybridized to a genetic variant nucleic acid, wherein the genetic variant nucleic acid is a genetic variant in a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7. In some embodiments, the genetic variant nucleic acid is an amplification product. In some embodiments, the genetic variant nucleic acid is on genomic DNA, e.g., from a subject that has or is suspected of having an interstitial lung disease. In some embodiments, the first nucleic acid is an amplification product or a primer extension product. In some embodiments, the first nucleic acid is labeled. In some embodiments, the nucleic acid complex further comprises a second nucleic acid hybridized to the genetic variant nucleic acid. In some embodiments, the second nucleic acid is labeled e.g., with a FRET or other fluorescent label. In some embodiments, the first and second nucleic acids form a FRET pair when hybridized to a genetic variant sequence.

(100) In some embodiments, the nucleic acid complex further comprises an enzyme, such as a DNA polymerase (e.g., standard DNA polymerase or thermostable polymerase such as Taq) or ligase.

(101) The present disclosure includes but is not limited to the following embodiments:

(102) A method for determining if an individual is predicted to develop and/or progress rapidly

with an interstitial pneumonia comprising: detecting in a biological sample from the individual, at least one of: a) the presence of a marker polymorphism selected from the group consisting of: rs35705950; and/or, b) a level of gene expression of a marker gene or plurality of marker genes selected from the group consisting of: a marker gene having at least 95% sequence identity with at least one sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DSP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; c) polypeptides encoded by the marker genes of b) d) fragments of polypeptides of c); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b); wherein the presence of the plurality of markers is indicative of whether an individual will develop a disease. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in b). In some embodiments, the presence or level of at least one of the plurality of markers is determined and compared to a standard level or reference set. In some embodiments, the standard level or reference set is determined according to a statistical procedure for risk prediction. In some embodiments, the statistical procedure for risk prediction comprises using the sum of the gene expression of the marker or markers or the presence or absence of a set of markers, weighted by a Proportional Hazards coefficient. In some embodiments, the presence of the at least one marker is determined by detecting the presence or absence or expression level of a polypeptide. In some embodiments, the method further comprises detecting the presence of the polypeptide using a reagent that specifically binds to the polypeptide or a fragment thereof. In some embodiments, the reagent is selected from the group consisting of an antibody, an antibody derivative, and an antibody fragment. In some embodiments, the presence of the marker is determined by obtaining the sequence of genomic DNA at the locus of the polymorphism. In some embodiments, the presence of the marker is determined by obtaining RNA from the biological sample; generating cDNA from the RNA; amplifying the cDNA with probes or primers for marker genes; obtaining from the amplified cDNA the expression levels of the genes or gene expression products in the sample. In some embodiments, the individual is a human.

(103) In some embodiments, the method further comprises: a) comparing the expression level of the marker gene or plurality of marker genes in the biological sample to a control level of the marker gene(s) selected from the group consisting of: a control level of the marker gene that has been correlated with interstitial lung disease, the risk of developing interstitial lung disease, or having a interstitial lung disease; and a control level of the marker that has been correlated with slow or no progression of interstitial lung disease, or low risk of developing an interstitial lung disease; and b) selecting the individual as being predicted to progress rapidly in the development of interstitial pneumonia, if the expression level of the marker gene in the individual's biological sample is statistically similar to, or greater than, the control level of expression of the marker gene that has been correlated with interstitial lung disease, or c) selecting the individual as being predicted to not develop interstitial lung disease, or to progress slowly, if the level of the marker gene in the individual's biological sample is statistically less than the control level of the marker gene that has been correlated with interstitial lung disease.

(104) In some embodiments, the method further comprising the presence of a polymorphism, in the biological sample to a set of genetic variants or polymorphic markers from an individual or control group having developed interstitial lung disease, and, selecting the individual as being predicted to develop or to progress with interstitial pneumonia if the polymorphic markers present in the biological sample are identical to or statistically similar to a set of polymorphic markers from the individual or control group or, selecting the individual as being predicted to develop or rapidly progress with interstitial pneumonia, if the polymorphic markers present in the biological sample are not identical to or statistically similar to the set of genetic variants or polymorphic markers from the individual or control group.

(105) A method for monitoring the progression of interstitial lung disease in a subject, comprising: i) measuring expression levels of a plurality of gene markers in a first biological sample obtained

from the subject, wherein the plurality of markers comprise a plurality of markers selected from the group consisting of: a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of a) MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; b) polypeptides encoded by the marker genes of a); c) fragments of polypeptides of d); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b); ii) measuring expression levels of the plurality of markers in a second biological sample obtained from the subject; and iii) comparing the expression level of the marker measured in the first sample with the level of the marker measured in the second sample. In some embodiments, the marker genes detected share 100% sequence identity with the corresponding marker gene in a). In some embodiments, the method further comprises performing a follow-up step selected from the group consisting of CT scan of the chest and pathological examination of lung tissues from the subject. In some embodiments, the first biological sample from the subject is obtained at a time t_0 , and the second biological sample from the subject is obtained at a later time $t_{sub.1}$. In some embodiments, the first biological sample and the second biological sample are obtained from the subject are obtained more than once over a range of times.

(106) A method of assessing the efficacy of a treatment for interstitial lung disease or interstitial pneumonia in a subject, the method comprising comparing: i) the expression level of a marker measured in a first sample obtained from the subject at a time t_0 , wherein the marker is selected from the group consisting of a) a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; b) polypeptides encoded by the marker genes of a) c) fragments of polypeptides of b); and d) a polynucleotide which is fully complementary to at least a portion of a marker gene of a); ii) the level of the marker in a second sample obtained from the subject at time $t_{sub.1}$; and, iii) performing a follow-up step selected from CT scan of the chest and pathological examination of lung tissues from the subject; wherein a decrease in the level of the marker in the second sample relative to the first sample is an indication that the treatment is efficacious for treating interstitial pneumonia in the subject. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in a). In some embodiments, the time t_0 is before the treatment has been administered to the subject, and the time t_1 is after the treatment has been administered to the subject. In some embodiments, the comparing is repeated over a range of times.

(107) An assay system for predicting individual prognosis therapy for interstitial pneumonia comprising a means to detect at least one of: a) the presence of a marker polymorphism selected from the group consisting of: rs35705950; and/or, b) a level of gene expression of a marker gene or plurality of marker genes selected from the group consisting of: a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; c) polypeptides encoded by the marker genes of b) d) fragments of polypeptides of c); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b). In some embodiments, the means to detect comprises nucleic acid probes comprising at least 10 to 50 contiguous nucleic acids of the marker polymorphisms or gene(s), or complementary nucleic acid sequences thereof. In some embodiments, the means to detect comprises binding ligands that specifically detect polypeptides encoded by the marker genes. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in b). In some embodiments, the means to detect comprises at least one of nucleic acid probe and binding ligands disposed on an assay surface. In some embodiments, the assay surface comprises a chip, array, or fluidity card. In some embodiments, the probes comprise complementary nucleic acid sequences to at least 10 to 50

nucleic acid sequences of the marker genes. In some embodiments, the binding ligands comprise antibodies or binding fragments thereof. In some embodiments, the assay system further comprises: a control selected from information containing a predetermined control level or set of genetic variants or polymorphic markers that has been correlated with diagnosis, development, progression, or life expectancy in interstitial lung disease patients.

(108) A method of detecting a level of gene expression of one or more marker genes in a human subject with interstitial pneumonia, comprising, optionally, obtaining a biological sample from a human individual with interstitial pneumonia; detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof in one or more cells from the biological sample from the individual.

(109) A method of treating an interstitial lung disease in a subject in need of such treatment, comprising: detecting a level of one or more marker genes selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof in a biological sample obtained from the human subject; and, administering an effective amount of an effective treatment. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual.

(110) Detection of Genetic Variants

(111) Methods of detecting a genetic variant are further described, for example, in U.S. Pat. No. 8,673,565 (the contents of which are herein incorporated by reference in their entirety). Genetic variations in the mucin genes are associated with pulmonary diseases. These genetic variations can be found in any part of the gene, e.g., in the regulatory regions, introns, or exons. Relevant genetic variations may also be found the intergene regions, e.g., in sequences between mucin genes. Insertions, substitutions, and deletions are included in genetic variants. Single nucleotide polymorphisms (SNPs) are exemplary genetic variants.

(112) In particular, 14 independent SNPs are associated with pulmonary disorders (e.g. FIP or IPF). The studies disclosed herein demonstrate that presence of one or more of these SNPs associated with MUC5B can lead to predisposition to a pulmonary disorder. In addition, in some embodiments, if present, some of these SNPs are related to a transcription factor binding site. The transcription factor binding site can effect modulation of MUC5B expression, for example E2F3 loss, and HOXA9 and PAX-2 generation.

(113) The disclosure thus provides methods for assessing the presence or absence of SNPs in a sample from a subject suspected of having or developing a pulmonary disorder (e.g., because of family history). In certain embodiments, one or more SNPs are screened in one or more samples from a subject. The SNPs can be associated with one or more genes, e.g., one or more MUC genes

or other genes associated with mucous secretion. In some embodiments, a MUC gene associated SNP is associated with MUC5B and/or another MUC gene, such as MUC5AC or MUC1. SNPs contemplated for diagnostic, treatment, or prognosis can include SNPs found within a MUC gene and/or within a regulatory or promoter region associated with a MUC gene. For example, one or more SNPs can include, but are not limited to, detection of the SNPs of MUC5B alone or in combination with other genetic variations or SNPs and/or other diagnostic or prognostic methods.

(114) Methods for detecting genetic variants such as a SNP are known in the art, e.g., Southern or Northern blot, nucleotide array, amplification methods, etc. Primers or probes are designed to hybridize to a target sequence. For example, genomic DNA can be screened for the presence of an identified genetic element of using a probe based upon one or more sequences, e.g., using a probe with substantial identity to a subsequence of the MUC5B gene. Expressed RNA can also be screened, but may not include all relevant genetic variations. Various degrees of stringency of hybridization may be employed in the assay. As the conditions for hybridization become more stringent, there must be a greater degree of complementarity between the probe and the target for duplex formation to occur. Thus, high stringency conditions are typically used for detecting a SNP.

(115) Thus, in some embodiments, a genetic variant MUC5B gene in a subject is detected by contacting a nucleic acid in a sample from the subject with a probe having substantial identity to a subsequence of the MUC5B gene, and determining whether the nucleic acid indicates that the subject has a genetic variant MUC5B gene. In some cases, the sample can be processed prior to amplification, e.g., to separate genomic DNA from other sample components. In some cases, the probe has at least 90, 92, 94, 95, 96, 98, 99, or 100% identity to the MUC5B gene subsequence. Typically, the probe is between 10-500 nucleotides in length, e.g., 10-100, 10-40, 10-20, 20-100, 100-400, etc. In the case of detecting a SNP, the probe can be even shorter, e.g., 8-20 nucleotides in length. In some cases, the MUC5B gene sequence to be detected includes at least 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some embodiments, the sequence to be detected includes 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides.

(116) The degree of stringency can be controlled by temperature, ionic strength, pH and/or the presence of a partially denaturing solvent such as formamide. For example, the stringency of hybridization is conveniently varied by changing the concentration of formamide within the range up to and about 50%. The degree of complementarity (sequence identity) required for detectable binding will vary in accordance with the stringency of the hybridization medium and/or wash medium. In certain embodiments, in particular for detection of a particular SNP, the degree of complementarity is about 100 percent. In other embodiments, sequence variations can result in <100% complementarity, <90% complementarity probes, <80% complementarity probes, etc., in particular, in a sequence that does not involve a SNP. In some examples, e.g., detection of species homologs, primers may be compensated for by reducing the stringency of the hybridization and/or wash medium.

(117) High stringency conditions for nucleic acid hybridization are well known in the art. For example, conditions may comprise low salt and/or high temperature conditions, such as provided by about 0.02 M to about 0.15 M NaCl at temperatures of about 50° C. to about 70° C. Other exemplary conditions are disclosed in the following Examples. It is understood that the temperature and ionic strength of a desired stringency are determined in part by the length of the particular nucleic acid(s), the length and nucleotide content of the target sequence(s), the charge composition of the nucleic acid(s), and by the presence or concentration of formamide, tetramethylammonium chloride or other solvent(s) in a hybridization mixture. Nucleic acids can be completely complementary to a target sequence or exhibit one or more mismatches.

(118) Nucleic acids of interest can also be amplified using a variety of known amplification techniques. For instance, polymerase chain reaction (PCR) technology may be used to amplify target sequences (e.g., genetic variants) directly from DNA, RNA, or cDNA. In some

embodiments, a stretch of nucleic acids is amplified using primers on either side of a targeted genetic variation, and the amplification product is then sequenced to detect the targeted genetic variation (using, e.g., Sanger sequencing, Pyrosequencing, Nextgen® sequencing technologies). For example, the primers can be designed to hybridize to either side of the upstream regulatory region of the MUC5B gene, and the intervening sequence determined to detect a SNP in the promoter region. In some embodiments, one of the primers can be designed to hybridize to the targeted genetic variant. In some cases, a genetic variant nucleotide can be identified using RT-PCR, e.g., using labeled nucleotide monomers. In this way, the identity of the nucleotide at a given position can be detected as it is added to the polymerizing nucleic acid. The Scorpion™ system is a commercially available example of this technology.

(119) Thus, in some embodiments, a genetic variant MUC5B gene in a subject is detected by amplifying a nucleic acid in a sample from the subject to form an amplification product, and determining whether the amplification product indicates a genetic variant MUC5B gene. In some cases, the sample can be processed prior to amplification, e.g., to separate genomic DNA from other sample components. In some cases, amplifying comprises contacting the sample with amplification primers having substantial identity to MUC5B genomic subsequences, e.g., at least 90, 92, 94, 95, 96, 98, 99, or 100% identity. Typically, the sequence to be amplified is between 30-1000 nucleotides in length, e.g., 50-500, 50-400, 100-400, 50-200, 100-300, etc. In some cases, the sequence to be amplified or detected includes at least 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some embodiments, the sequence to be amplified or detected includes 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some aspects, the contiguous nucleotides include nucleotide 28.

(120) Amplification techniques can also be useful for cloning nucleic acid sequences, to make nucleic acids to use as probes for detecting the presence of a target nucleic acid in samples, for nucleic acid sequencing, for control samples, or for other purposes. Probes and primers are also readily available from commercial sources, e.g., from Invitrogen, Clontech, etc.

(121) Detection of Expression Levels

(122) Expression of a given gene, e.g., MUC5B or another mucin, pulmonary disease marker, or standard (control), is typically detected by detecting the amount of RNA (e.g., mRNA) or protein. Sample levels can be compared to a control level.

(123) Methods for detecting RNA are largely cumulative with the nucleic acid detection assays described above. RNA to be detected can include mRNA. In some embodiments, a reverse transcriptase reaction is carried out and the targeted sequence is then amplified using standard PCR. Quantitative PCR (qPCR) or real time PCR (RT-PCR) is useful for determining relative expression levels, when compared to a control. Quantitative PCR techniques and platforms are known in the art, and commercially available (see, e.g., the qPCR Symposium website, available at qpersymposium.com). Nucleic acid arrays are also useful for detecting nucleic acid expression. Customizable arrays are available from, e.g., Affimatrix. An exemplary human MUC5B mRNA sequence, e.g., for probe and primer design, can be found at GenBank Accession No. AF086604.1.

(124) Protein levels can be detected using antibodies or antibody fragments specific for that protein, natural ligands, small molecules, aptamers, etc. An exemplary human MUC5B sequence, e.g., for screening a targeting agent, can be found at UniProt Accession No. 000446.

(125) Antibody based techniques are known in the art, and described, e.g., in Harlow & Lane (1988) *Antibodies: A Laboratory Manual* and Harlow (1998) *Using Antibodies: A Laboratory Manual*; Wild, *The Immunoassay Handbook*, 3d edition (2005) and Law, *Immunoassay: A Practical Guide* (1996). The assay can be directed to detection of a molecular target (e.g., protein or antigen), or a cell, tissue, biological sample, liquid sample or surface suspected of carrying an antibody or antibody target.

(126) A non-exhaustive list of immunoassays includes: competitive and non-competitive formats, enzyme linked immunosorption assays (ELISA), microspot assays, Western blots, gel filtration and

chromatography, immunochromatography, immunohistochemistry, flow cytometry or fluorescence activated cell sorting (FACS), microarrays, and more. Such techniques can also be used in situ, ex vivo, or in vivo, e.g., for diagnostic imaging.

(127) Aptamers are nucleic acids that are designed to bind to a wide variety of targets in a non-Watson Crick manner. An aptamer can thus be used to detect or otherwise target nearly any molecule of interest, including a pulmonary disease associated protein. Methods of constructing and determining the binding characteristics of aptamers are well known in the art. For example, such techniques are described in U.S. Pat. Nos. 5,582,981, 5,595,877 and 5,637,459. Aptamers are typically at least 5 nucleotides, 10, 20, 30 or 40 nucleotides in length, and can be composed of modified nucleic acids to improve stability. Flanking sequences can be added for structural stability, e.g., to form 3-dimensional structures in the aptamer.

(128) Protein detection agents described herein can also be used as a treatment and/or diagnosis of pulmonary disease or predictor of disease progression, e.g., propensity for survival, in a subject having or suspected of developing a pulmonary disorder. In certain embodiments, MUC5B antibodies can be used to assess MUC5B protein levels in a subject having or suspected of developing a pulmonary disorder. It is contemplated herein that antibodies or antibody fragments may be used to modulate MUC5B production in a subject having or suspected of developing a pulmonary disease. In certain embodiments, one or more agents capable of modulating MUC5B may be used to treat a subject having or suspected of developing a pulmonary disorder. One or more antibodies or antibody fragments may be generated to detect one or more of the SNPs disclosed herein by any method known in the art.

(129) In certain embodiments, MUC5B diagnostic tests may include, but are not limited to, alone or in combination, analysis of rs35705950 SNP in MUC5B gene, MUC5B mRNA levels, and/or MUC5B protein levels.

(130) Additional Pulmonary Disease Markers

(131) The above methods of detection can be applied to additional pulmonary disease markers. That is, the expression level or presence of genetic variants of at least one additional pulmonary disease marker gene can be determined, or the activity of the marker protein can be determined, and compared to a standard control for the pulmonary disease marker. The examination of additional pulmonary disease markers can be used to confirm a diagnosis of pulmonary disease, monitor disease progression, or determine the efficacy of a course of treatment in a subject.

(132) In some cases, pulmonary disease is indicated by an increased number of lymphocytes, e.g., CD4+CD28⁻ cells.

(133) Genetic variations in the following genes are associated with pulmonary disease: Surfactant Protein A2, Surfactant Protein B, Surfactant Protein C, TERC, TERT, IL-1RN, IL-1 α , IL-1 β , TNF, Lymphotoxin α , TNF-RII, IL-10, IL-6, IL-12, IFN γ , TGF β , CR1, ACE, IL-8, CXCR1, CXCR2, MUC1 (KL6), or MUC5AC. Thus, the invention further includes methods of determining whether the genome of a subject comprises a genetic variant of at least one gene selected from these genes. The presence of a genetic variant indicates that the subject has or is at risk of developing pulmonary disease. Said determining can optionally be combined with determining whether the genome of the subject comprises a genetic variant MUC5B gene, or determining whether the subject has an elevated level of MUC5B RNA or protein to confirm or strengthen the diagnosis or prognosis.

(134) Abnormal expression in the following genes can also be indicative of pulmonary disease: Surfactant Protein A, Surfactant Protein D, KL-6/MUC1, CC16, CK-19, Ca 19-9, SLX, MCP-1, MIP-1a, ITAC, glutathione, type III procollagen peptide, sIL-2R, ACE, neopterin, beta-glucuronidase, LDH, CCL-18, CCL-2, CXCL12, MMP7, and osteopontin. Thus, the expression of one of these genes can be detected and compared to a control, wherein an abnormal expression level indicates that the subject has or is at risk of developing pulmonary disease. Said determining can optionally be combined with determining whether the genome of the subject comprises a

genetic variant MUC5B gene, or determining whether the subject has an elevated level of MUC5B RNA or protein to confirm or strengthen the diagnosis or prognosis.

(135) Biomarkers

(136) The present disclosure provides a peripheral blood biomarker profile for IPF to demonstrate the use of a predictive biomarker profile in cases of preclinical pulmonary fibrosis (PrePF) derived from families with familial IPF. The present disclosure also provides biomarker identification for association between each genetic, epigenetic or protein (gene product) biomarker with PrePF and the predictive value of the combination of biomarkers associated with PrePF.

(137) A large cohort of families with familial IPF for genetic research was established, including 937 families with ≥ 2 cases of IPF, and 2375 family members that have been previously phenotyped as unaffected. This study focuses on subjects with PrePF to elucidate the processes active in early disease pathogenesis and to predict or prevent the irreversible fibroproliferative process. Genetic risk factors, especially the MUC5B promoter variant, identifies individuals with preclinical interstitial changes on chest CT scan that progress and are associated with reduced survival. Biomarkers may be used to identify those subjects with PrePF among those at-risk for IPF. Given the irreversible nature of IPF, even approved treatments (pirfenidone and nintedanib) only modestly slow progression and have not been shown to alter the 3-5 year survival. Pirfenidone and nintedanib are effective in patients with mild disease, suggesting that patients with PrePF may be targeted for early intervention, before most of the lung has been irreversibly remodeled.

(138) Table 1 below shows additional gene expression changes present in subjects with IPF compared to controls. Specifically, the expression of the genes listed in Table 1 are upregulated in IPF compared to the expression of these same genes in control subjects. Accordingly, the discovery of elevated expression levels of one or more genes listed in Table 1 compared to a control in an asymptomatic subject may indicate that the subject has PrePF and/or that the subject is at risk for developing IPF.

(139) In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is upregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitronectin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metalloproteinase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metalloproteinase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif

Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

(140) TABLE-US-00001 TABLE 1 TARGET_GENE_SYM- ORGAN- B-H Fold BOL ISM p-value q-value Change LTA4H Human 8.70E-43 3.13E-39 3.912 SFTP8 Human 1.17E-37 2.10E-34 3.399 BCAR3 Human 4.28E-25 3.85E-22 2.906 CXCL13 Human 1.30E-29 1.56E-26 2.904 EPHA2 Human 9.62E-23 6.93E-20 2.651 SAA1 Human 6.01E-07 7.84E-06 2.631 PLA2G2A Human 8.19E-21 2.95E-18 2.171 Igfbp3 Mouse 1.18E-18 2.66E-16 2.149 CCL28 Human 1.22E-22 7.30E-20 2.135 S100A12 Human 1.06E-20 3.45E-18 2.125 TBXAS1 Human 1.60E-21 7.20E-19 2.11 LECT1 Human 4.17E-19 1.00E-16 2.082 C3 Human 7.08E-07 8.95E-06 2.062 GRP Human 8.35E-09 1.66E-07 1.988 CSP Human 1.36E-08 2.61E-07 1.957 VIT Human 2.47E-17 4.45E-15 1.929 IGFBP1 Human 4.32E-11 1.56E-09 1.914 FAM173A Human 2.19E-13 1.84E-11 1.904 NPPA Human 5.02E-12 2.58E-10 1.877 SFRP1 Human 1.74E-20 5.23E-18 1.866 EZR Human 6.41E-10 1.72E-08 1.809 ITIH5 Human 5.11E-21 2.04E-18 1.705 PSD2 Human 5.38E-18 1.08E-15 1.689 LGALS38P Human 8.06E-22 4.15E-19 1.678 1.18E-05 0.000102 1.668 CTNNB1 Human 5.66E-12 2.87E-10 1.625 CDYL2 Human 4.11E-07 5.59E-06 1.622 MMP7 Human 1.56E-19 4.02E-17 1.621 APOB Human 8.73E-13 6.42E-11 1.597 PRELP Human 1.13E-10 3.53E-09 1.595 EIF1AX Human 2.13E-06 2.31E-05 1.59 MANF Human 0.00458 0.015006 1.585 TNFRSF13C Human 1.77E-11 7.31E-10 1.573 C3 Human 2.40E-16 3.93E-14 1.566 DEAF1 Human 0.000221 0.001192 1.565 TPT1 Human 1.22E-12 7.82E-11 1.548 UNC5B Human 2.06E-34 2.18E-12 1.547 PEBP1 Human 4.92E-11 1.72E-09 1.544 STX8 Human 8.82E-12 4.13E-10 1.537 PIGR Human 1.29E-09 3.19E-08 1.532 APRT Human 1.51E-07 2.26E-06 1.525 MMP3 Human 9.50E-07 1.15E-05 1.524 LGALS7 Human 7.51E-05 0.000474 1.514 BTK Human 1.47E-09 3.52E-08 1.511 NSFL1C Human 7.33E-11 2.40E-09 1.506 FER Human 2.24E-07 3.24E-06 1.503 REG1B Human 6.68E-11 2.25E-09 1.502 SMAD2 Human 4.39E-10 1.25E-08 1.493 IL1RL1 Human 9.55E-07 1.15E-05 1.492 CCL18 Human 1.25E-13 1.07E-11 1.491 ACP2 Human 3.73E-08 6.33E-07 1.488 EIF4E2 Human 1.67E-12 1.02E-10 1.483 NRXN3 Human 2.33E-17 4.42E-15 1.48 IGFL1 Human 5.07E-10 1.40E-08 1.474 NME1 Human 1.43E-10 4.39E-09 1.463 KCNE1L Human 3.93E-20 1.09E-17 1.462 NXPH2 Human 9.66E-30 2.47E-08 1.451

(141) Table 2 below shows additional gene expression changes present in subjects with IPF compared to controls. Specifically, the expression of the genes listed in Table 2 are downregulated in IPF compared to the expression of these same genes in control subjects. Accordingly, the discovery of decreased expression levels of one or more genes listed in Table 2 compared to a control in an asymptomatic subject may indicate that the subject has PrePF and/or that the subject is at risk for developing IPF.

(142) In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is downregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

(143) TABLE-US-00002 TABLE 2 TARGET_GENE_SYM- ORGAN- B-H Fold BOL ISM p-value q-value Change SFTPD Human 8.19E-15 9.83E-13 -2.262 GAPDH Human 1.46E-09 3.52E-08 -2.096 HIST1H1C Human 3.68E-18 7.80E-16 -2.011 3.63E-16 5.69E-14 -1.964 YTHDC1 Human 1.19E-11 5.38E-10 -1.699 PLXNA1 Human 1.64E-12 1.02E-10 -1.64

SPINK6 Human 3.68E-07 5.04E-06 -1.635 LRPA1 Human 2.65E-15 3.53E-13 -1.521
SCGB3A1 Human 3.35E-07 4.61E-06 -1.518 H2AFZ Human 3.91E-14 3.91E-12 -1.501
2.95E-11 1.16E-09 -1.493 C1orf162 Human 1.29E-84 7.52E-04 -1.458

(144) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

(145) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Telomerase RNA Component (TERC). In some embodiments the polymorphism is rs6793295 comprising (SEQ ID NO: 1).

(146) TABLE-US-00003 (SEQ ID NO: 1) AGAAAGAAGT CATGAAAGTA
GGAACCACAT TTTTACTCAT CTTTCTGTCT CCAGCAAGCA GCTTACTGCT
TTTCATACAC ATTTTGCTTT TATTACTCAT GATTTCAAAG GTGTAATGGT
TCAGCCACAT CAATGTAACA AACAGTTCAC ACTGGGCTCT TATAGTCTGG
CCTTTAAAC CTTCACTATT TATGCTTTCA TCTTAAC TTTGACCCTC
ACAGGTTTAC TACTAAGAA CTTGAGTTTC AAGAGAAAAG ATGACATGTT
TGCTGCTTAA ACAAGCAATA TCTAAAAGCA TATTTAGTTA TAAACGTCTT
ACCAAGAATT GATATAATTT TCATTTAAAC ATTTTATAA ATAGTAGTTT
ACAAGATATA GTAAGTACAT CTCTAAAAT ACAGTGTATT CATGTACCTT
GACATAAACT TG TAGTAGTA CCTTAGTTTT ATTCATGTTG TTATATTAAC
TACCATCACT TTGAATACAT ACCTGTTTAC B GTACAGTATA GGTCGGTTTA
GGTTTATTGC CTTAATTGCT TGGTTTTGAG TTAGTACTGT AGCAAATGCT
ATCACACTTT GCATTCCTA AAAACAGGTA AATTCATTAA GGAAACAGAC
AAAGTATATA ATAATCTCGC TACATAAATA TTTCAAGATC AGCTATCTGC
ATTCTGATAAAATTGTTTT AAAATTAAAG CATTCTTGG ACTTTGAATT
GTAAGTTGAT CAAATTCAAA AATGAATTGT TACTGTATTC TTCTCTCCTG
GCCCTAAAAT CTATCTAAA CATGGCATGG GGAGTTTCTT AATGTTTCAG
TGTCATTTC CTGGGTGTTT CCCTCTAGGT TTTTTTCT CACCCCTCAA
GCTTCTATGT GGATCCCAGC TAGAGCTCAT ACTACTTATC CAACACACAT
CATTGTGCAA GCACTCTTTT ATATTCATAC TAGTACTTTT AAGTGTGTGT
GCGGTGGGAA AAGGTTACCA ATCACATTTT

(147) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Family with sequence similarity 13 member A (FAM13A). In some embodiments the polymorphism is rs2609255 comprising (SEQ ID NO: 2).

(148) TABLE-US-00004 (SEQ ID NO: 2) GTATTCATCA ACTCCTATTT
CATTCCCTCT TCCTGTGCTC ACTGGAAGAT GACATTTCCC AGACTTCCAA
GAATGTTACT GAGTTCTGGA ATGTAAGTAG AAGGGATAAG TATCACTTCT
GTGCTGTGGC GGTTATGGAC CTGTGAACTT TGCACACGCC TTCTATCTTC
TTTTTCAGTG TCCATTTTCA AGGGCATGTT TTCAGATGAA ACCAGTAGAA
GATGGAAGCA GCCTGTGACT AGAATCACTG CTTAGGGTCT TGCTGCCTAG
GAATCCCACT CTACCTGCAA CAGACTGTGA AAGAACCGAG AAATACACTG
ATTTTGAACA TAGCCCATAC TATAATGGGG ATGTTTGTTA CAGCAGTTAG
CATTAAAAAC CTTGGCTAGG CATTGGTCAT AATTGTAGAA CACAGCAAAT
GAAGGGAAAC TGGAACATAG AGGCCAGTGA GAACTTTAGG GTTAATGAAA
AATGAGGGCA ACCAGGATAA TTTGGTTCTT K GCCAAATAGG AAGGTGAAAC
CAAAGGTAGA CTGGAGGTCA GAAAATCAGT CCAGCACATG TGATGTTTTT
ATTTAGTTGC CTGTATGTCT GTCTGGTCTC CAGCTCAGCC TGGCTCCTTG
AGGTAAGAGG CAGTGGCTGT TCACCTTTGC ATCCAGCAC CTGGCATACA
ATAGATGGGA TGAAATGTTT AAAGTGAAGC TAAGCTTCAG GGTGCTTATC
AAAGCAGGGA AGATACACAA GAGGAGATGA TTCAGGTCCA

GGGACAGTCA GGTATCTAAA CCGAAGCTCT TAGGAAGCTG GATCTCCGA
ACCAGGGAGA ACAAGCTGGA TATGCACTGG ATTTCCCAGC AGTACTGATC
TAGAGACTCT CATAGAGTCC CTTTTATTCC TTGGCCTAGG GTTACAACCTG
CTTATAGCAT CTGGAAAGAC TCAACACCTC AAAAGAGACT TTCAGTAGAT
ACAGCAAATA CACTCATGGA ATTGATAATT AAGCTTCAAT

(149) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Telomerase Reverse Transcriptase (TERT).

In some embodiments the polymorphism is rs2736100 comprising (SEQ ID NO: 3).

(150) TABLE-US-00005 (SEQ ID NO: 3) ATTGTCGTTG TTGCTTTTG
TTTATTGAGA CAGTCTCACT CTGTCACCCA GGCTGGAGTG TAATGGCACA
ATCTCGGCTC ACTGCAACCT CTGCCTCCTC GGTTC AAGCA GTTCTCATTC
CTCAACCTCA TGAGTAGCTG GGATTACAGG CGCCCACCAC CACGCCTGGC
TAATTTTTTGT ATTTT TAGTA GAGATAGGCT TTCACCATGT TGGCCAGGCT
GGTCTCAAAC TCCTGACCTC AAGTGATCTG CCCGCCTTGG CCTCCCACAG
TGCTGGGATT ACAGGTGCAA GCCACCGTGC CCGGCATACC TTGATCTTTT
AAAATGAAGT CTGAAACATT GCTACCCTTG TCCTGAGCAA TAAGACCCTT
AGTGTATTTT AGCTCTGGCC ACCCCCCAGC CTGTGTGCTG TTTTCCCTGC
TGACTTAGTT CTATCTCAGG CATCTTGACA CCCCCACAAG CTAAGCATTA
TTAATATTGT TTTCCGTGTT GAGTGTTTCT K TAGCTTTGCC CCCGCCCTGC
TTTTCCTCCT TTGTTCCCCG TCTGTCTTCT GTCTCAGGCC CGCCGTCTGG
GGTCCCCTTC CTTGTCCTTT GCGTGGTTCT TCTGTCTTGT TATTGCTGGT
AAACCCCAGC TTTACCTGTG CTGGCCTCCA TGGCATCTAG CGACGTCCGG
GGACCTCTGC TTATGATGCA CAGATGAAGA TGTGGAGACT CACGAGGAGG
GCGGTCATCT TGGCCCGTGA GTGTCTGGAG CACCACGTGG CCAGCGTTCC
TTAGCCAGTG AGTGACAGCA ACGTCCGCTC GGCCTGGGT CAGCCTGGAA
AACCCCAGGC ATGTCGGGGT CTGGTGGCTC CGCGGTGTCG AGTTTGAAAT
CGCGCAAACC TGCGGTGTGG CGCCAGCTCT GACGGTGCTG CCTGGCGGGG
GAGTGTCTGC TTCCTCCCTT CTGCTTGGGA ACCAGGACAA AGGATGAGGC
TCCGAGCCGT TGTCGCCCAA CAGGAGCATG

(151) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Desmoplakin (DSP). In some embodiments the polymorphism is rs2076295 comprising (SEQ ID NO: 4).

(152) TABLE-US-00006 (SEQ ID NO: 4) ATTTGGGAAC CTTTAAAAA
TATTCTGGCT TCAAAAATAC TCCATATTTA CATCTTTGGT TCTATCTGAA
GTAAAGCCGT GATGGTGTGC GTAAGTGAAA CAGGTGCAAA GGGGCAACAA
CAAAGGGCGC CTCTCTTTGT CTTTGTGTCG CAGGCGGAGA TGGACATGGT
GGCCTGGGGT GTGGACCTGG CCTCAGTGGA GCAGCACATT AACAGCCACC
GGGGCATCCA CAACTCCATC GGCGACTATC GCTGGCAGCT GGACAAAATC
AAAGCCGACC TGGTACTTGT CTGTGTTTCA TTTTAGAGTC TTCAAATAT
CTACCGAAGG ATCGTGTAAT TACTCAATCC CAGGGAGTTT CTTCTGAAAC
ATTGCTATTA TTTCTTTCCC AGAAGACTGG AAATGTTTAG AAATCCCCT
TCTTAAATGG GGAAGTGGAATCAGTAGCCC TATTAGAGAT TATGTTAACA
CTTGAAGAGG AGTTAAACCA GAGGCTGAGG K TGTGCAAACA CTCATTTGCA
GTTTGTGAAT AAGTCTCTTT AGGGGTGGCA GTTTGTTTCT GCGGTAAGCA
GAACATCTTT TTGAATAGGG GAAATGCAAC AGTCTTATAC AGTAGTTTGT
GTCATTGGTG AATCCTTTCC TAGGTGGTAA TTAAAACATT ATTTCTACTG
AGCAAAGCCA TATGTCATCC CGACACCCGC TCCCATGCTG AAAAAAGTCA
GACTTGAAAC TGGGTTGAGA ATTACAGCAT AAAATCATAA CTGATCTTAA
GTGCTTAGTT TCCCGCAGGT CTCTACACTT GTAAATCACT AAACTTTTTT
TTTTTTTTTT TACCTGAGAC CATAGCTTCT CATCCTCATT TCTTCTTCTG

GCTTTTGGG GCTTACTTTT GTCCTTCTGACC GCCCTTCTCTCC
TTCATTTCTC TAAGACCTAG GGAATCCTAA ATGATGTCTT TAAACTTTAA
GACAATTTTC TAACACGTGA GTCCTTAAAGT

(153) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Zinc-alpha 2-Glycoprotein 1 (AZGP1). In some embodiments the polymorphism is rs4727443 comprising (SEQ ID NO: 5).

(154) TABLE-US-00007 (SEQ ID NO: 5) CCCAACCCAA ATAAGCACTA
TAACCTCTTG TTATTCACCT CTCATGCAAC CAGTCTTCTG TTCTCTGTGA
GTCTTTAGGA AATGAGGAGC ATGATCTTCT AGCAGTAAAA CACCTGTAGA
GAATTGCCTT ATGTTTTTTT TTTGTTTATT TGTGTGTGTG CTTTGGTTTG
GTTTGCTTTT TTTTTTTTTT TTTTTTTTTT TTTGAGATGG AGTCTCGCCC
TGTTGCCAG GCTGGAGTGT AGTGGCGAAA TCTCGGCTCA CTGCAACCTC
CACCTCCCTG GTTCAAGCAA TTCCCCTGTC TCAGCCTCCC GAGTAGCTGA
GATTACAGGT GCACACCACC ACGCCCGGCT AATTTTTTTG TATTTTATAGT
AGAGATGGGG TTTCACCATG TTGGCCAGAC TGGTCTCGAA CTTCTGACCT
CAGGCAATCC GCCTGCCTCA GCCTCCCAA GCGCTGGGAT TACAGGCATG
AGCCACTGCG CCCC GCCTCC ATGTTAATCAM TCTTTCTGAT TTCAAATAAC
TCATTATCCC CATGACCTTA TGGATTGTGTT TTTCTCTTC ATCCACAAAA
TTCTCCAGAG AAGTCTCCCT TGTTATCTCT TGGCTGTGCT TTCTATCTCA
CCAGTTATCT TTCTCCAAAG AGCTTCCTCT GCAAAGAAGC TTTGTATATG
AAGACCATGT GGGGGCTGAA TCAAGACCAA GTTTCACAAC CTAAAAGTAG
TTCACAAAGC TTCCTTGCCT CTATTCTCTG CAAATCTGTA AACTCTTCAG
CTGACCCAAT TTCTCTCTTT AGCCTTCAGA GATTATTTTA TTTTATTTTA
TTTCATTTCA TTTCATTTCA TTTTGACAGA ATCTAGCTCT GTCGCCCAGG
CTGGAGTGCA GTGGCACCAT CTTTGCTCAC TGCAACCTCC CCCTCACAGG
TTCAAGCAAC TGTCTGCCT CAGCCTCCCG AGTAGCTGGG ATTACAGGCG
TGAGCCACCA CGCCCAGCTG ATTTTTTTTT

(155) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1). In some embodiments the polymorphism is rs11191865 comprising (SEQ ID NO: 6).

(156) TABLE-US-00008 (SEQ ID NO: 6) CCTCTACTGC CGTACACCCC
ACCACTCAGC CTTGGAGTGC CTGTGTGCAG AGCAGGGCTG AGGCATGGTG
CTGCTTTGGT GGTCTAGGTT TGCTGCAGGG CCAGGTGGCC TGAGCTCCAG
GCAGGATCTC TGGCTGCACT CAGCCCTTTC TGCCTCCCCA AATGCTCTAT
ATCACTATTT GTACACTGAG CAGAGTAAAG TTAGAGAGAA CTGTTTTATA
GAATAGGGCT GGCCCCCGCT CCCCTGGCCT ACGTGATGGT CCTTCCTGGC
TGCCAGGTAC TTGTTTGTAT TAGAGACAGA CACTCCACAG GGTCTGTTGT
GGCCACAGC ACATAGGCAA TCAGAGGCAG AAAGCAGAGC TGTTTGACC
CACAGAGGGC CGGCTGTCTG CCACTGAAAT GTCTTTCCAG TTGGTTGAGA
AGCAGCAGGA TGCTCTGCTG GTGATGTCTG AAAGTCCCAG GATTCTTTGG
GTCTCCAAGG AGATCCTAGC ATATACTACT R TCGTGGTTTT AATAAAGAGC
AAAAACACTT TCAGATGGGG AGAAGAGTGG AACAAAAGGT ATTCTTCCTG
GGTTGAAGTC TGGGGGAAAG GCATTGAGAA GACTGGGCTA ATGGCACAAA
CCAATGAAGT ACTCAAGTCA CCTGTGATGG AGGCCAGTCA TCCAATGGTA
TCAACTTTGT ATGTGGCAAC ACTTAATAAA AATCTGAACA GGTCTTCACT
TGTGGACACA GTAGACTTTC TTGAAAAAGG ACAGAAAAGT GAGCCCTGTG
AATTTTCATC TCACGGACTG ACAACAATGA CTTGCCTTTA AGGACAGTCA
CTCAAGATGA AGATGCAACA AAACCCTTCC AGTTCCAAGT GGCTGATGAA
AAAAAAAAA TCTTAAAAGC ATCACAGAAC AACGGAGAAA

GAGATCAGAA GACTATAACA GATATTGGA ATTTTAAAC TCAGAGAAAA
GCAACTGAGG AGGAAATACA CTGCTTAGAA AGAAGAAACT

(157) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Mucin 5B (MUC5B). In some embodiments the polymorphism is rs35705950 comprising (SEQ ID NO: 7).

(158) TABLE-US-00009 SEQ ID NO: 7) TGGACGGCCT CTGAAGGGGT
CTGTGGGGTC CTGGACGGGT CCCCATTCAT GGCAGGATTA ACCCCCCTCG
GGTTCTGTGT GGTCCAGGCC GCCCCTTTGT CTCCACTGCC CCCTGGCCAG
AATGAGGGAC AGTGACCCAC CCAGGGCTGG GCCTGGCTCA GACTCCGTCA
GAGCCGCAGG GCAAGTTCCT GGCACGTCCG AGGTGGGAGG CTCCTCTGCG
CTCCAGGAGG CTGTGCCTGG CCCCCCTTCC CGGCAGGAAC CGGCTGTGTC
CCTTTCCTTC CTTTATCTTC TGTTTTCAGC D CCTTCAACTG TGAAGAGGTG
AACTCTTCAA ACACGCTGAG CAAACAGGCC CGACTCCCAG GGCCGCATCC
GGGATGTCTC AATAGCTGTG GCCTTGACGT CCACCTCGGA CCCCTGCCCC
GGACCCAGCC CAGTTCCCAA TGGGCCCTCT GCCCGGGGAG GTGCCTAGTG
GGAGGGACGA GGGCAAAGTC GGGGCCCCCA CTTGTTTGGT GTCACTGTGT
GCCAGCGGCC ACTGGCGGGC GAGGCTGTTC CAGGGTGGAG
GCGGGGAGGG TTGGACCACA GGC ACTGAGC GGGGACAGAG

(159) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding ATPase Phospholipid Transporting 11A (ATP11A). In some embodiments the polymorphism is rs12787690 comprising (SEQ ID NO: 8).

(160) TABLE-US-00010 (SEQ ID NO: 8) GTCATTGGTC AAATGTGGCC
TGTATCTAAA TTCCA ACTGT TAGAATCATA GACATCTAGA GCTTACGTCA
GTTTTAGATA TTTCTTATGA ATTCTCAGAA TTCATAGATT CTCATTTTAA
TTCTTAGACT TCTCAGATAT TCCGTTTTTG ATAGTATACC CTTCTGAGTC
TAATATGTCC TAAAGTGCGA ACTTGATACAA TTTtttttt tttttttt tttttttt t K
tgataaggag tttactctg tcacccaggc tggagtgcag tgaccgatc tcggctcact gcaacctctg cctcccgggt
tcaagtgatt gtgatgtctc agtctcccaa gtagctggga ttacaggctc ctgccaccac atgcctagct aattgttata
cttagtaga aatggggctt cgccgtgta gtcaggctgg tctgtactc ctgacctcag ttgatctgcc taccttggcc
cccaagggtgc tgggattaca ggcattagacc accgcgcctg accCAGCTTC TTAAATTATT
CTGGGGCCACC AGTAATGTGA ATCATGtaaa ttaaaatata taattaaaCA AAATCATATA
GCGATTAGAG ATAATAGTTG TGAAATGCTT GAAAAATCAT AGGCATTTAA
TAAATAGAAG CCATTCCAAT TAGGATTCTT CTTGATTTTT TTTCAAGACC
AAAAAAATAC TCttttaaat atttattata ataCTCCATG

(161) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2). In some embodiments the polymorphism is rs2034650 comprising (SEQ ID NO: 9).

(162) TABLE-US-00011 (SEQ ID NO: 9) aggctgcagt tagtcatgac tgcgcgctgc
actccagcct gggtgacaaa gtgaggccct gtctcaaaaa caataaaaa TTTAAAAGAG
CTGAGCATGG AGGCcacttt gggaggctga ggcaggcaga tctcttaagc ccaggagtct
gagaccagcc tgggcgacat gatgaagccc catctctaca aaaaatacaa aaaaattagc tgagctttat
ggcaaatccc tgtaatccca gttacctagg aggcccaggc aggaagatgg cttgagccca aaagggttag
gctgtagtga gctgtgatca tgaacagagt gagaccctgt ttcaaaacaa aatgaaaaac aaacaaacaa
aaaaaCCAAG AAAACAAGAA AACAAAAACT ATACAATGAT GAGCCAAAAA
GCAAGATATG GAAGAatata tatatatata tatatatata tataGTATGA GTCCAGCTAT
AGAAAGTTTG AAATCAGGCA ACCTAAACAA TATTGTTTCA GATCTATAC
AGAGGCAGGA AGCCATTGAG AAAGGTAAGG GGAGGATTAT CACCAAATTC
AGGATGGTGG CTCCCCTGGG GAGAATATGT CAAGGAGGGG CACATGGGCT
TGAATACTG TCTTCATTGA CCTGCGTGTT GGGTACACAG GAGTTTGTGA

TTTTCACACA TGCATATGTG CATCTCCATA TATACCATGC
 ATTTACACACA AGAACACAAA GGCTGTGTGG CTCTGCTCTG CCCCTTTCCC
 CTTCCAGCTC CCATTCTCGT C Y TCAGCTAGCA
 GAGGAGGGTC AGGGTCTTTT AGCACAGCTT CTTTCTGTCT CTGAGTGGGT
 CAGAGGAGTA CGGGGATGAG GGCCTCCCTT CTGCGGCTGG GCTCTGGCCA
 CTCCAGGGTG GGAAGGCCTG GAGAAAACAG GGCCAGGCAA
 AGCCGGCTGG CCCTGCTGTT TCTGCCAATG CTGGGATTAG GCCAGGGCTC
 TGGCCCACCT GTCATTTTAC TCATTACAGCA TGAACATAGC CACTGAGCAC
 TTA CTGTGAG CCCC GGGTGC TATTGGGAGA GTTCAGATAA GTGAGAGAGG
 GTCTTTGACC TCAAAGATCT TACAGAGAGG ACCGTATACA CAAATAACAG
 TATACCAGCA AAATGTGAGC TAAGTGTCAT GTGACTACTC atctactctt
 tcaataaata tttgtgtgc acctattaca tgccaggaac tgtgtctggat ggtgatcatg taaagacagt
 caaatcacag tcctagctct cagattcaca gcctgcctaa tgctggggaa acTGGAAT

(163) In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Dipeptidyl Peptidase 9 (DPP9). In some embodiments the polymorphism is rs12610495 comprising (SEQ ID NO: 10).

(164) TABLE-US-00012 (SEQ ID NO: 10) CCAGCCAGAA GGGGCGCAGT
 TTGTTAGTTC AGCTCCTCCT GAGACAGAAA TAAAGACACG AACCAAAGGA
 CATCAGCACT TACAGGGCTC TCAGGTCACA CACAGGATGT CCGCGCCCAC
 TGCAGAGCTG CAGGTCCCCT CCAGGGCAGT GGGGAGCCAC AAGCAGCGTT
 AGGCAGCGGC TGGGACCAGG ACCGCCTGAG CACTCAAGAA CCCCCACTGC
 CCAAGCACT GCTGGCAGCA AGCCCAGAAA ACTGAGCCCG GGGAGCTCCT
 CTGAGCGGCC TAAGCACCCC TCTAAGCTGT GCTGCCCCAA TTCAAGCCTG
 GCTCACGGCA GCAAAGAAAA AATGTGACCT TCGGAGCTCC CAAAGGGGCC
 ACCCATAAGC TGAGAGCCTG CCCGGAAGCA CTTATAGACC CGCGTGCTT
 GTTTTCATTG CAAAGAACAA TAAAAATTAT CTTGCCTCTG ATCACCCTG
 ATAGCCCAAG AAGCAAAAAT TCGATCCCGG D GATGAGAAAT
 GAAATGAAAC ATCGCGAGAA ACTTCCAGGA ATCTTCTGGA TGTGGCTAGA
 CTCTTTAGCT TGAGCTTCCA GACAGGCCGA GGCTTGGTGC TGGAGCCTGG
 CCCTCCGCTG ACCTCTCTTC TACCCGGGGG CACAGCCCGG ATTGCAGAGA
 GGCTGGCGCA AGAGTGAGGG AGCGAGGGCT AGCCTGTGAT GGGCTTTCTC
 CACCTAGCAC CACCCTATGC TGTGGCTCAG GGGAGTCAAG AGTTTACACA
 GCTGCAGAGA TGGATTCCAG GCCACTTACT CAAGTCTACC TACTCCTTCC
 TTCGGCCAAT CAGCTGGGTG CCTCTGCGGC CTGTGACACC ACCAGCAAAC
 AGCTCCAGAC CTCCTAGCAT GGTCTCTGTC AAGGCTGGGT GGCAGATCTG
 TGATCTCCTT TTAAATTTT TCATTTTTT TAAGAGATGG GTTCTTGCTA
 TATTGCCAG GCTGGTCTCA AACTCCTGGG CTCCAGCGAT

(165) In some embodiments of the methods of the disclosure, the wild type human MUC5B gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_002458.2):

(166) TABLE-US-00013 (SEQ ID NO: 11)

1	cacccggccc	ggctccctcc
ctgcccgtcc	ccgtccccc	acccgtgcca
61	tggttgcccc	gagcgcgtgc
cggacgctgg	tggttgctct	ggcggccatg
121	cgcaggcaga	gacccagggc
cctgtggagc	cgagctggga	gaatgcaggg
181	atggcggtgc	cccagactcc
tcgccaccc	ggcgcgtgag	ctttgtcca
241	tcttccccag	cctgagcccc
ctgaacccgg	cgcacatgg	gcgggtgtgc
301	gtgacttcca	ctacaagacc
ttcgacggcg	acgtctccg	cttcctggc
361	acgtgttctc	tgagcactgc
acgaggactt	caacgtccag	ctacgccag
421	gcctagtggg	ctccaggcct
gtgtgtcat	caaggcccag	gggctggtgc
481	tggaggcgtc	caacggctcc
gtcctcatca		
541	gcactggcct	cctggtggag
cagagcgggg		

actacatcaa	ggtcagcatc	cggctggtgc	601	tgacattcct	gtggaacgga	gaggacagtg
ccctgctgga	gctggatccc	aaatacgcca	661	accagacctg	tggcctgtgt	ggggacttca
acggcctccc	ggccttcaac	gagttctatg	721	cccacaacgc	caggctgacc	ccgctccagt
ttgggaacct	gcagaagttg	gatgggcccc	781	cggagcagtg	cccggacccg	ctgcccttgc
cggccggcaa	ctgcacggac	gaggagggca	841	tctgccaccg	caccctgctg	gggccggcct
ttgcggagtg	ccacgcactg	gtggacagca	901	ctgcgtacct	ggccgcctgc	gcccaggacc
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gcgcgcagct	ctgcgaggac	cactgtgtgg	1141	acggctgctt	ctgccccca	ggcacgggtgc
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gccgcaccta	cagcccgggc	acctccttca	1261	acaccacctg	cagctcctgc	acctgctccg
ggggggctatg	gcagtgccag	gacctgccgt	1321	gccctggcac	ctgctctgtg	cagggcgggg
cccacatctc	cacctatgat	gagaaactct	1381	acgacctgca	tggtgactgc	agctacgttc
atgtcccgcac	agcagcttca	1441	ccgtgctggc	tgagctgcgg	aagtgcggcc	tgacggacaa
cgagaactgc	ctgaaagcgg	1501	tgacgctcag	cctggacggc	ggggacacgg	ccatccgggt
ccaagcggac	ggcggcgtgt	1561	tctcaactc	catctacacg	cagctgcccc	tgtcggcagc
caacatcacc	ctgttcacac	1621	cctcgagctt	cttcatcgtg	gtgcagacag	gcctggggct
gtgcagctgg	1681	tgccactcat	gcaggtgttt	gtcaggctgg	accccgcca	ccagggccag
1741	tgtgtgggaa	cttcaaccag	aaccaggctg	acgacttcac	ggccctcagc	ggggtggtgg
1801	aggccacggg	cgcagccttc	gccaacacct	ggaaggccca	ggctgcctgt	gccaatgcc
1861	ggaacagctt	tgaggacccc	tgctccctca	gtgtggagaa	tgagaactac	gcccggcact
1921	gggtgctcgcg	cctgaccgat	cccaacagtg	ccttctcgcg	ctgccactcc	atcatcaacc
1981	ccactcgaac	tgcatgtttg	acacctgcaa	ctgtgagcgg	agcgaggact	2041
2041	cgcgctgtcc	tcctatgtgc	acgcctgtgc	cgccaagggc	gtacagctca	2101
2101	ggacggcgtc	tgaccaagt	acatgcagaa	ctgccccaa	tcccagcgt	2161
2161	gggtgatgcc	tgccagccca	cttgccgcgg	cctgagttag	gccgacgtca	2221
2221	ttccttcgtg	cctgtggacg	gctgcacctg	ccccgcgggc	accttcttca	2281
2281	cccgccagg	agtgcctctg	ctacgctcac	ggcaccgtgc	2341	tggctcctgg
2341	cacgacgagg	gcgccgtgtg	ttcatgtacg	gggtgggaagc	2401	taagctgcct
2401	ctgcagaaaa	gcacaggggtg	tgacgcccc	atggtgtacc	2461	tggactgcag
2461	gcggggcaccc	ctggggccga	gtgcctccgg	agctgccaca	2521	cgctggacgt
2521	agcacacact	gcgtgtccgg	ctgtgtctgt	ccccggggc	2581	tgggtgtcgga
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2881	gctccaaggc	catcaagctc	ttcgtggaga	2941	gctacgagct	gatcctccaa
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3001	tcttctggt	catcgagacc	cacgggatgg	3061	ccgtgtcctg	ggaccggaag
3061	tcatccgact	gcaccaggac	tacaagggca	3121	gggtctgcgg	cctgtgcggg
3121	acaatgccat	caatgacttt	gccacgcgta	3181	gccgggtccgt	ggtgggggac
3181	ttgggaacag	ctggaagctc	tccccctct	3241	gcccggacgc	cctggcacc
3241	gcacggccaa	ccccctccgc	aagtctctggg	3301	cccagaagca	gtgcagcatc
3301	ccacctcgc	cgctgcccgc	tcccaggttg	3361	actccaccaa	gtactacgag
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tcaactccag	ggacaacacc	tatccccca	gtgctgacca	8221			ccaccgccac
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tccacgggga	cttcccacac	cccagcagca	accaccggta	8461			ccaccagca
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cacgacaagg	gccaccagtt	9421					ccatgtccac
cctcacagag	ctgaccacag	9481					cagccactac
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acactacaa agtgccgact accacaacca cgggcttcac agtcacccc tcctccagc 11701
cagggacggc acgcacgcct ccagtgtgga tcagcacaac caccacacc acaaccagt 11761
gtccacgggt gacccctcc tcgtcccg ggaccacca cccccaca gtgctgacca 11821
ccaccaccac aactgtggcc actggttcta tggcaacacc ctctctagc acacagacca 11881 gtggtactcc
cccatcactg atcaccaggg ccactacgat cacggccacc ggctccacca 11941 ccaaccctc ctcaactcca
gggacaacac ctatcccc agtgctgacc accaccgcca 12001 ccacacctgc agccaccagc
agcacagtga ctccctctc tgccctagg accaccaca 12061 cacccccagt gccgaacacc
acggccacca cacacgggcg atccctgtc cccagcagtc 12121 cccacacggt gcgcacagcc
tggtctcgg ccacctagg caccttgggc accaccaca 12181 tcacagagcc ttccacggg acttccaca
ccccagcagc aaccaccggt accaccagc 12241 actcgactcc agccctgtc agccctcacc
ctagcagcag gaccaccgag taccctct 12301 cccagggac gaccacccg ggccacacca
cggccacctc caggaccag gccacggcca 12361 caccagcaa gaccgcacc tcgacctgc
tgccagcag cccacatcg gccccataa 12421 ccacggtggt gaccacgggc tgtgagcccc
agtgtgcctg gtcagagtgg ctggactaca 12481 gctacccat gccggggccc tctggcggg actttgacac
ctactccaac atccgtgcgg 12541 ccggaggggc cgtctgtgag cagccctgg gcctcgagt
ccgtgcccag gccagcctg 12601 gtgtccctc gggggagttg ggccaggtcg tggaatgcag
cctggactt ggctgtgt 12661 gcaggaaccg tgagcaggtg gggaagtca agatgtgtt caactatgaa
atccgtgtgt 12721 tctgtgcaa ctacggccac tgccccagca cccggccac cagctctac gccatgcct
12781 cctccactcc ggggacgacc tggatctca cagagctgac cacaacagc actacgactg 12841

catccactgg	atccacggcc	acccccgtct	ccaccccggg	aacagctccc	cctcccaaag 12901	
tgctgaccag	cccggccacc	acaccacag	ccaccagttc	caaagccact	tctctctcca 12961	
gtccaaggac	tgcaaccacc	cttccagtgc	tgacaagcac	agccacaaa	tccacagcta 13021	ccagcggtac
accatcccc	tctccaccc	ttgggaccac	cgggaccctc	ccagaacaga 13081	ccaccacacc	
cgtggccacc	atgtccaaa	tccacccctc	ctccactccg	gagaccacc 13141	acacctccac	agtgctgacc
acgaaggcca	ccacgacaag	ggccaccagt	tccacgtcca 13201	ccccctctc	cactccgggg	
acgacctgga	tctcacaga	gctgaccaca	gcagccacta 13261	caactgcagc	cactggcccc	
acggccaccc	cgtcctccac	cccagggacc	acctggatcc 13321	tcacagagct	gaccacaaca	
gccactacga	ctgcgtccac	tggatccacg	gccaccccg 13381	cctccacccc	agggaccacc	
tggatcctca	cagagccgag	cactacagcc	accgtgacgg 13441	tgcccaccgg	atccacggcc	
accgcctct	ccaccaggg	aactgctggc	acccacatg 13501	tgagcaccac	ggccacgaca	
cccacagtca	ccagctccaa	agccactccc	tctccagtc 13561	cagggactgc	aactgccctt	ccagcactga
gaagcacagc	caccacaccc	acagctacca 13621	gctttacagc	catcccctcc	tctctcctgg	gcaccacctg
gacccgccta	tcacagacca 13681	ccacacccac	ggccaccatg	tccacagcca	caccctctc	
cactccagag	actgtccaca 13741	cctccacagt	gcttaccgcc	acggccacca	caaccggggc	
caccggctct	gtggccaccc 13801	cctcctccac	cccaggaaca	gtcacacta	ccaaagtgcc	
gactaccaca	accacgggct 13861	tcacagccac	cccctctctc	agcccaggga	cggcactcac	
gcctccagt	tggatcagca 13921	caaccaccac	accacaacc	accacacca	caaccagtgg	
ctccacggtg	acccctctct 13981	ccatcccggg	gaccacccac	accgccagag	tgctgaccac	
caccaccaca	actgtggcca 14041	ctggttctat	ggcaacaccc	tctctagca	cacagaccag	tggtactccc
ccatcactga 14101	ccaccacggc	cactacgatc	acggccaccg	gctccaccac	caaccctcc	tcaactccag
14161	ggacaacacc	catcaccca	gtgctgacca	gcacggccac	cacacccgca	gccaccagct 14221
ccaaagccac	ttctctctcc	agtccaagga	ctgcaaccac	ccttcagtg	ctgacaagca 14281	cagccacaaa
atccacagct	accagcttta	cacccatccc	ctcctccacc	ctgtggacca 14341	cgtggaccgt	cccagcacag
accaccacac	ccatgtccac	catgtccaca	atccacacct 14401	cctctactcc	agagaccacc	cacacctcca
cagtgtctgac	caccacagcc	accatgacaa 14461	gggccaccaa	ttccacggcc	acacctctct	
ccactctggg	gacgaccgg	atcctcactg 14521	agctgaccac	aacagccact	acaactgcag	
ccactggatc	cacggccacc	ctgtcctcca 14581	ccccagggac	cacctggatc	ctcacagagc	
cgagcactat	agccaccgtg	atggtgcca 14641	ccggttcac	ggccaccgcc	tctccactc	
tgggaacagc	tcacacccc	aaagtgggta 14701	ccaccatggc	cactatgccc	acagccactg	
cctccacggt	tcccagctcg	tccaccgtgg 14761	ggaccacccg	caccctgca	gtgtcccca	
gcagcctgcc	aaccttcagc	gtgtccactg 14821	tgctctctc	agtctcacc	acctgagac	ccactggctt
ccccagctcc	cacttctcta 14881	ctccctgctt	ctgcagggca	tttgacagt	tttctcgcc	cggggaagtc
atctacaata 14941	agaccgaccg	agccggctgc	catttctacg	cagtgtgcaa	tcagcactgt	gacattgacc
15001	gcttcagggg	cgctgtccc	acctccccac	cgccagtgtc	ctccgccccg	ctgtcctcgc 15061
cctcccctgc	ccctggctgt	gacaatgcca	tcctctccg	gcaggtgaat	gagacctgga 15121	ccctggagaa
ctgcacggtg	gccaggtgcg	tgggtgacaa	ccgtgtcgtc	ctgtggacc 15181	caaagcctgt	
ggccaacgtc	acctgcgtga	acaagcacct	gccccatcaa	gtgtcggacc 15241	cgagccagcc	
ctgtgacttc	cactatgagt	gcgagtgcac	ctgcagcatg	tggggcggtc 15301	cccactattc	cacctttgac
ggcacctctt	acacctccg	gggcaactgc	acctatgtcc 15361	tcatgagaga	gatccatgca	cgctttggga
atctcagct	ctacctggac	aaccactact 15421	gcacggcctc	tgccactgcc	gctgccgccc	gctgcccccg
cgcctcagc	atccactaca 15481	agtccatgga	tatcgtctc	actgtacca	tgggtcatgg	gaaggaggag
ggcctgatcc 15541	tgtttgacca	aattccggtg	agcagcggtt	tcagcaagaa	cggcgtgctt	gtgtctgtgc
15601	tggggaccac	caccatgcgt	gtggacattc	ctgccctggg	cgtgagcgtc	accttcaatg 15661
gccaaagtctt	ccaggccccg	ctgccctaca	gcctcttcca	caacaacacc	gagggccagt 15721	
gcggcacctg	caccaacaac	cagagggacg	actgtctcca	gcgggacgga	accactgccg 15781	
ccagttgcaa	ggacatggcc	aagacgtggc	tgggtccccga	cagcagaaag	gatggctgct 15841	
gggccccgac	tggcacaccc	cccactgcca	gccccgcagc	cccgggtgtc	agcacacca 15901	
ccccacccc	atgccacca	cagccgctct	gtgatctgat	gctgagccag	gtctttgctg 15961	agtgccaaa
ccttggtccc	ccgggcccac	tcttcaacgc	ctgcatcagc	gaccactgca 16021	ggggccgcct	tgaggtgccc

tgccagagccc tggaggctta cgcagagctc tgccgcgccc 16081 ggggagtggt cagtgaactgg
 cgaggtgcaa ccggtggcct gtgcgacctc acctgcccac 16141 ccaccaaagt gtacaagcca
 tgcggcccca tacagcctgc cacctgcaac tctaggaacc 16201 agagcccaca gctggaggggg
 atggcggagg gctgcttctg ccctgaggac cagatcctct 16261 tcaacgcaca catgggcatc tgcgtgcagg
 cctgcccctg cgtgggacct gatgggtttc 16321 ctaaatttcc cggggagcgg tgggtcagca actgccagtc
 ctgctgtgtg gacgagggtt 16381 cagtgtcggg gcaagtgaag cccctgccct gtgacgcca gggctagccc
 ccgctgtgca 16441 accgtcccgg cttcgtaacc gtgaccaggc cccggggcca gaacctctgc tgccccgaga
 16501 cggtgtgcgt gtgcaacaca accacctgcc cccagagcct gcctgtgtgc ccgccagggc 16561
 aggagtccat ctgcacccag gaggaggggc actgctgtcc caccttccgc tgcagacctc 16621 agctgtgttc
 gtacaatggc accttctacg ggggttggtgc aaccttcca ggccgcttc 16681 cctgccacat gtgtacctgc
 ctctctgggg acaccagga cccaacggtg caatgtcagg 16741 aggatgcctg caacaatact acctgtcccc
 agggctttga gtacaagaga gtggccgggc 16801 agtgctgtgg ggagtgcgtc cagaccgctc
 gcctcacgcc cgatggccag ccagtccagc 16861 tgaatgaaac ctgggtcaac agccatgtgg
 acaactgcac cgtgtacctc tgtgaggctg 16921 aggggtggagt ccatttgctg accccacagc ctgcatcctg
 cccagatgtg tccagctgca 16981 gggggagcct caggaaaacc ggctgctgct actcctgtga
 ggaggactcc tgtcaagtcc 17041 gcatcaacac gaccatcctg tggcaccagg gctgcgagac
 cgaggtcaac atcaccttct 17101 gcgagggtc ctgccccgga gcgtccaagt actcagcaga
 ggcccaggcc atgcagcacc 17161 agtgcacctg ctgccaggag aggcgggtcc acgaggagac
 ggtgcccttg cactgtccta 17221 acggctcagc catcctgcac acctacacc acgtggatga gtgtggctgc
 acgcccttct 17281 gtgtccctgc gcccatggct cccccacaca cccgtggctt cccggcccag gaggccactg
 17341 ctgtctgaga acgttctgcc tccatccca tgctctgtcc acctggagcc aggatgtgca 17401
 ttgtctgatc atgaaaacct tgggcctcct ctgcggagcc ccccgccctg tgtgtggcac 17461 cccgcgtcc
 gtgctcctgc tgcccacccc gtgggtgaaa ccggcccag aagggtgagg 17521 ggccagcagg
 accccttctg ggaggcgcc actcaggagt cctaccctgg gagagcctgt 17581 ggcccacctt ggcctgccc
 ctccctgatg tccctgggac gccctggaac aaactaagca 17641 tgtgcgggcc tatgtgtccc tgccacggcc
 ggagcgcccg cgcagcacgg attccagctg 17701 gccacgtccg gccgctgggg cagacaggct
 ggtccaggca aggccagctg ctgccaggaa 17761 gctgcgacag gcaaggcggc cgctgtcca
 tgcctgctgc agggtaactc agggctgagg 17821 tcgcaacggc caggctcagag aggggtcagc
 atcccaaagc cccctctgct caaccagcc 17881 cagttttgca aataaacct gagcattgag tacgtt

(167) In some embodiments of the methods of the disclosure, the wild type human MUC5B gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_002449.2):

(168) TABLE-US-00014 (SEQ ID NO: 12)

1	mgapsacrtl	vlalaamlvv
61	vfpslsplnp	ahngrvcstw
121	glvgsrpvvt	rvvikaqglv
181	alleldpkya	nqtclcgdf
241	ichrtllgpa	faechalvds
301	lnmqhqecgs	pctdtcsnpq
361	nttcsstcs	gglwqcqdlp
421		
481	vtlsldggdt	
541	vplmqvfvr	dpahqgqmcg
601	rnsfedpcsl	svenenyarh
661	clcaalssyv	hacaakgvql
721	tcsysfvpvd	gctcpagtf
781	lsclgaslqk	stgcaapmv
841	lvsdgsggci	aedcpcvhn
901	ghfitfdgdr	ysfegsceyi
961	syelilqegt	fkavargpgg
1021	rvcgclgnfd	dnaindfatr
1081	dstkyyeacv	ndacacdsgg

pqaetqgpve pswenaghtm dggaptsspt rrvsfvpvt
 gdfhyktfdg dvfrfpqlcn yvfsehcray yedfnvqlrr
 ngqreelpys rtgllveqsg dyikvsirlv
 ahnarltplq fgnlqkldgp
 lcrcptcpca
 cpgtcsvqgg ahistydekl ydlhgdcsvy lskkcadssf tvlaelrkcg ltdnencika
 airvqadggv flnsiytqlp lsaanitlft pssffivvqt glglqlqvql
 lcgfnfnqna ddftalsgvv eatgaafant wkaqaacana
 wcsrldpns afsrchsiin pkpfhsncmf dtncersed
 sdwrldgvctk ymqncpksqr yayvvdacqp tcrglseadv
 ndagacvpaq ecpcyahgtv lapgevvhde gavcsctggk
 ldcnssagt pgaecrsch tldvgcfsth cvsgcvcppg
 eatykpeti rvdntctcr nrrwecshrl clgtcvaygd
 thgtfrivte nipcgttgtt cskaklfve
 ayswdrktsv firhdydykg
 ctanpfrksw
 aqkqcsilh ptfaacrsqv
 dstkyyeacv
 ndacacdsgg
 dcecfctava
 ayaqachdag

1141 lcvsrwtpdt cplfcdfynp hggcewhyqp cgapclktcr npsghclvdl pglegcypkc 1201
ppsqqffned qmkcvaqcg ydkdgnnydv garvptaenc qscnctpsgi qcahsleact 1261
ctyedrtysy qdviynttdg lgaclicg sngtiirkav acpgtpattp ftftawvph 1321 sttspalpv
tvcvrevcrw sswynghrpe pglgggdfet fenlrqrgyq vcpvladiet 1381 raaqlpdmpl
eelgqqvdcd rmrqlmcans qqspplchdy elrvlcceyv pcgpspapgt 1441 spqpslsast
epavptptqt tatekttlwv tpsirstaal tsqtgsssgp vtvtpsapgt 1501 ttcqprcqw ewfdedypks
eqlggdvesy dkiraagghl cqpkdiecq aesfpnwla 1561 qvgqkvhcdv hfglvcrnwe
qegvfkmcyn yrirvlccsd dhcrgrattp pptteletat 1621 tttqalfst pqptsspglt rappastav
ptlsegltsr rytstlgtat tggtpptags 1681 teptvpgvat stlptrsalp gttgslgtwr psqpptlap
tmatsrarpt gtastasket 1741 lttslaplt selstsqat stprtettms plnttttsqg ttrcqpkcew
tewfdvdfpt 1801 sgvaggdmet feniraaggk mcwapksiec raenypevsi dqvgqvltes letglckne
1861 dqtrfnmcf nynrvlccd dyshcpstpa tsstatpsst pgttwiltkp tttattast 1921
gstattstl rtappkvlt ttattptvts skatpssspg tatalpalrs tattptatsv 1981 tpiptsslt
twtrlsqtt ptatmstatp sstpetahs tvltatatt gatgsvatps 2041 stpgtahtk vpttttft
atpssspgta ltpvwistt tpttrgstv tpssipgth 2101 tatvltttt tvatgsmatp ssstqstgtp
psltttatti tatgstnps stpgtppip 2161 vlttattpa atsnvtpss algtthtpv pntmatthgr
slppssphtv rtawtsatsg 2221 ilgtthitp stvsttlaa ttgtqhtp alssphpsr ttesppspg
tptgthtats 2281 rttatatsk trtstillps ptsapittv tmgcepqaw sewldysypm pgpsggdfd
2341 ysniraagga vceqplglec raqaqgvpl relgqvvecsl dfglvcrnr eqvgkfkmc 2401
nyeirvfccn yghcpstpat sstampsstp gttwiltelt ttatttestg statpsstpg 2461 ttwiltpst
tatvtvptgs tatasstqat agtphvstta tptvtsska tpfsspgtat 2521 alpaltat tptatsftai
pssslgtwt rlsqtttpta tmstatpsst petvhtstvl 2581 tttattgat gsvatpsst gtahtkvlt
tttftatp ssspgtatl pvwistttt 2641 ttrgstvps sipgtthtp vltttttva tgsmatpss
tqstgtpsl ttattitat 2701 gstnpsstp gtpippvlt ttattpaats stvtpssalg tthtpvpnt tathgrsl
2761 pssphvtrta wtsatsgtlg thitpstg tshtpaattg tqhstpls sphpsrtte 2821
sppspgtt ghtratsrt atatsktr stillpspts apittvmtg cepqawsew 2881 ldysympgp
sggdfdtysn iraaggavce qplglecraq aqgvplrel gqvvecsl 2941 glvcrnreq
gkfkmcfnie irvfccnygh cpstpsst atpsstpgt wiltqtaa 3001 tttattgsta ipsstpgt
ppkvltstat tptatsskat sssprtatt lpvltstatk 3061 statsftip sftlgtgtl peqttmat
mstihpsst ethtstvl tkatttrats 3121 smstpsstpg twiltelt aattaatgp tatpsstpgt
twiltpst atvtvptgst 3181 atasstrata gtlkvltsta tptvissra tpssspgtat alpaltat tptatsvtai
3241 pssslgtawt rlsqtttpta tmstatpsst petvhtstvl tttttrt gsvatpsst 3301 gtahtkvpt
tttftatp ssspgtalt pvwistttt ttrgstvps sipgttht 3361 vltttttva tgsmatpss
tqstgtpsl ttattitat gstnpsst gtpippvlt 3421 ttattpaats stvtpssalg tthtpvpnt
tathgrsl pssphvtrta wtsatsgilg 3481 thitpstv tshtpaatts tqhstpls sphpsrtte
sppspgtt ghtratsrt atatsktr stillpspts apittvmtg cepqawsew ldysympgp
sggdfdtysn 3601 iraaggavce qplglecraq aqgvplrel gqvvecsl glvcrnreq gkfkmcfnie
3661 irvfccnygh cpstpsst atpsstpgt wiltklta ttestgsta tpsstpgtw 3721 iltpstt
vtvptgst asstqatgt phvsttatt tvtsskatp sspgtalp 3781 alrstattpt atsftaipss
slgtwtrls qttptatms tatpsstpet ahtstvltt 3841 atttratsv atpsstpgta htktvptt
tgftvtpss pgtartppvw istttptts 3901 gstvtpssvp gthtptvlt tttvatgs matpssstq
sgtpslitt attitatgst 3961 tnpsstpgt pippvltta tpaatstv tpssalgth tppvpntat thgrslps
4021 phtvrtawts atsgltgth itpstgtsh tpaattgtt hstpalssph pssrttesp 4081
spgttptgt tatsrtata tpsktrstl lpssptsapi ttvvtgcep qawsewldy 4141 sympgpgg
dfdtysnira aggavceqpl glecraqa qvplgelqv vecslfdglv 4201 crnreqvgkf kmcfnieirv
fccnyghcps tpsstamp sstpgttwil telttatt 4261 astgstps stpgtappk vltspattpt
atsskatsss sprtattlv ltstatksta 4321 tsvtipsst lgtgtlpeq ttpvatmst ihpsstpet
htstvltkka ttratssts 4381 tpsstpgtw ilteltaat ttaatgptat psstpgtwti ltelttatt tastgatp
4441 sstpgttwil tepsttatv vptgstas sqatagtp vsttattptv tsskatpss 4501 pgtatpal
rstattptat sftaipsssl gtwtrlsqt tptatmsta tpsstpetv 4561 tstvtatat ttgatgsvat

psstpgtaht tkvptttttg ftatpssspg taltpvwis 4621 ttttpttpt tsgstvtps sipgtthtar
vltttttva tgsmapsss tqstgtpsl 4681 tttattitat gsttnpsst gttptvlt stattpaats
skatsssspr tattlpvlts 4741 tatstatsf tpipsstlwt twtvpqttt pmstmstiht sstpethts
tvltttatmt 4801 ratnstatps stlgttrilt eltttatta atgstatlss tpgttwile pstiatvmvp 4861
tgstatast lghtpkvv tmatmptat astvpssstv gtrtpavlp sslptfsvst 4921 vsssvltlr
ptgfpsshfs tpcfrafqg ffspegviyn kdragchfy avcnqhcdid 4981 rfqgactsp ppvssaplss
pspapgcdna iplrqnwtnw tlenctvarc vgdnrvvld 5041 pkpvanvtcv nkhlpikvsd
psqpcdfhye cecicsmwgg shvstfdgts ytfgrnctyv 5101 lmreiharf nlslyldnhy ctasataaaa
rcpralsihy ksmdivltvt mvhgkeegli 5161 lfdqipvssg fsknvglvsv lgttmrvti palgsvtfn
gqvfarlpy slfhntegq 5221 cgtctnnqrd dclqrdgta asckdmaktw lvpdsrkdgc
waptgtptta spaapvsst 5281 ttpcppqpl cdlmlsqvfa echnlvppgp ffnacisdhc rgrlevpcqs
leayaelcra 5341 rgvcswrga tggldlctcp ptkvykpcgp iqpatcnsrn qspqllegmae gcfcpedqil
5401 fnahmgicvq acpcvgpdgf pkfpgervws ncqscvdeg svsvqcklp cdaqgpppc 5461
nrpgfvtvr praenpcpe tvvcntttc pqlpvcppg qesictqeeq dccptfrcp 5521 qlcsyngtfy
gvgtfpgal pchmctclsg dtqdpvtqcq edacnnttcp qgfeykrvag 5581 qccgecvqta
cltpdgqpvt lnetwvnhv dntvylcea eggvhltpq pascpdvssc 5641 rgslrktgcc
ysceedscqv rinttilwhq gcetevnitf cegscpgask ysaeaqamqh 5701 qctccqerry heetvplhpc
ngsailhtyt hvdecgctpf cvpamapph trgfpaqeat 5761 av

(169) In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_198253.2, transcript variant 1):

(170) TABLE-US-00015 (SEQ ID NO: 13)

1	caggcagcgc	tgcgtcctgc
61	gccgcgcgct	ccccgctgcc
121	gctgccgctg	gccacgttcg
181	cggggacccg	gcggctttcc
241	cgcacggccg	cccccgccg
301	ggcccgagtg	ctgcagaggc
361	cgcgctgctg	gacggggccc
421	ctacctgcc	aacacgggtga
481	gcgccgcgtg	ggcgacgacg
541	ggtggctccc	agctgcgcct accaggtgtg
601	cactcaggcc	cggccccgc cacacgctag
661	ggcctggaac	catagcgtca gggaggccgg
721	gaggaggcgc	gggggagtg ccagccgaag
781	cgctgcccct	gagccggagc ggacgccct
841	gacgcgtgga	ccagtgacc gtggtttctg
901	agccacctct	ttggagggtg cgctctctgg cacgcgccac
961	ccagcaccac	gcgggcccc catccacatc gcggccacca
1021	tccccgggtg	tacgccgaga ccaagcactt cctctactcc tcaggcgaca
1081	gcggccctcc	ttctactca gctctctgag gccagcctg actggcgctc ggaggctcgt
1141	ggagaccatc	ttctgggtt ccaggccctg gatgccaggg actccccgca ggttgccccg 1201
1261	gcaagcacgc	gagctgcttg ggaaccacgc
1321	agcagccggt	gtctgtgcc
1381	ggacacagac	ccccgtcgcc
1441	gtacggcttc	gtgcgggcct gcctgcgccg
1501	gcacaacgaa	cgccgcttcc tcaggaacac caagaagttc
1561	caagctctcg	ctgcaggagc tgacgtggaa gatgagcgtg cgggactgcg
1621	caggagccca	gggggtggct gtgttccggc cgcagagcac cgtctgcgtg aggagatcct
1681	ggccaagttc	ctgcactggc tgatgagtgt gtacgtcgtc gagctgtcga ggtctttctt 1741
1801	gagcaagttg	

caaagcattg gaatcagaca gcacttgaag aggggtgcagc tgcgggagct 1861 gtcggaagca
 gaggtcagggc agcatcggga agccaggccc gccctgctga cgtccagact 1921 ccgcttcac ccgaagcctg
 acgggctgcg gccgatttg aacatggact acgtcgtggg 1981 agccagaacg ttccgcagag
 aaaagagggc cgagcgtctc acctcgaggg tgaaggcact 2041 gttcagcgtg ctcaactacg
 agcggggcgc ggcggggcgc ctctctgggc cctctgtgct 2101 gggcctggac gatatccaca
 gggcctggcg caccttcgtg ctgcgtgtgc gggcccagga 2161 cccgccgcct gagctgtact ttgtcaaggt
 ggatgtgacg ggcgcgtacg acaccatccc 2221 ccaggacagg ctacaggagg tcatcgccag
 catcatcaaa cccagaaca cgtactgcgt 2281 gcgtcggat gccgtgggtcc agaaggccgc ccatgggcac
 gtccgcaagg ccttcaagag 2341 ccacgtctct acctgacag acctccagcc gtacatgcga cagttcgtgg
 ctacactgca 2401 ggagaccagc ccgctgaggg atgccgtcgt catcgagcag agctcctccc tgaatgaggg
 2461 cagcagtggc ctcttcgacg tcttctacg cttcatgtgc caccacgccg tgcgcatcag 2521
 gggcaagtcc tacgtccagt gccaggggat cccgcagggc tccatcctct ccacgtgct 2581 ctgcagcctg
 tgctacggcg acatggagaa caagctgtt gcggggattc ggcgggacgg 2641 gctgctcctg cgtttggtg
 atgatttctt gttggtgaca cctcacctca cccacgcgaa 2701 aacctctc aggaccctgg tccgaggtgt
 ccctgagtat ggctgcgtgg tgaacttgcg 2761 gaagacagt gtgaactcc ctgtagaaga cgaggccctg
 ggtggcacgg cttttgtca 2821 gatgccggcc cacggcctat tcccctggg cggcctgctg ctggataccc
 ggaccctgga 2881 ggtgcagagc gactactcca gctatgccc gacctccatc agagccagtc tcacittcaa
 2941 ccgcggcttc aaggctggga ggaacatgcg tcgaaactc tttgggtct tgcggctgaa 3001
 gtgtcacagc ctgtttctgg atttcaggt gaacagctc cagacgggtgt gcaccaacat 3061 ctacaagatc
 ctctgctgc aggcgtacag gtttcacga tgtgtgctgc agtcccatt 3121 tcatcagcaa gtttgaaga
 accccacatt tttctgcgc gtcattctg acacggcctc 3181 cctctgctac tccatctga aagccaagaa
 cgcaggggat tgcgtggggg ccaagggcgc 3241 cgccggccct ctgccctcc aggccgtgca
 gtggctgtgc caccaagcat tctgtctaa 3301 gctgactga caccgtgtca cctacgtgcc actcctgggg
 tcatcagga cagcccagac 3361 gcagctgagt cggaagctcc cggggacgac gctgactgcc
 ctggaggccg cagccaaccc 3421 ggcaactgcc tcagacttca agaccatct ggactgatgg ccacccgccc
 acagccaggc 3481 cgagagcaga caccagcagc cctgtcacgc cgggctctac gtcccagga
 gggagggggc 3541 gcccacaccc agggccgcac cgctgggagt ctgaggcctg agtgagtgtt tggccgaggg
 3601 ctgcatgtcc ggctgaaggc tgagtgtccg gctgaggcct gagcgagtgt ccagccaagg 3661
 gctgagtgtc cagcacacct gccgtctca ctccccaca ggctggcgct cggtccacc 3721 ccagggccag
 ctttctca ccaggagccc ggctccact cccacatag gaatagtcca 3781 tcccagatt cgccattgtt
 caccctcgc cctgccctcc tttgcctcc acccccacca 3841 tccaggtgga gacctgaga aggaccctgg
 gagctctggg aatttgaggt gaccaaagg 3901 gtgccctgta cacaggcgag gacctgcac ctggatgggg
 gtccctgtgg gtcaaattgg 3961 ggggaggtgc tgtgggagta aaatactgaa tatatgagtt ttcagttt
 gaaaaaaa

(171) In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_937983.2, transcript variant 1):

(172) TABLE-US-00016 (SEQ ID NO: 14) 1 mpraprcrav rslrshyre vlplatfvrr
 lgpqgwrlvq rgdpaafral vaqlvcvpw 61 darpppaaps frqvsclkel varvlqrce
 rgaknvlafg falldgargg ppeaftsrv 121 sylpntvtda lrgsgawgll lrrvgddvll hllarcalfv
 lvapscayqv cgpplyqlga 181 atqarpppha sgprrrlgce rawnhsvrea gvplglpapg
 arrrggsasr slplkrpr 241 gaapeperp vqgswahpg rtrgpsdrgf cvvsparpae eatslegals
 gtrhshpsvg 301 rqhahagppst srpprpwdtp cppvyaetkh flyssgdkeq lrpsfllssl rpsltgarll
 361 vetiflsrp wmpgtprlp rlpqrywqmr plflllgnh aqcpygvllk thcplraavt 421
 paagvcarek pqgsvaapee edtdprllvq llrqhsspww vygfvracrl rlvppglwgs 481
 rhnerrflrn tkkfislghk aklsiqeltw kmsvrdcawl rrspgvvcvp aaehrlreei 541 lakflhwlm
 vyvvellrsf fyvtettfqk nrlffyrksv wsklqsigir qhlkrvqlre 601 lseaevrqhr earpalltsr
 lrfipkpdgl rpivnmdyvv gartfrekr aerltsrvka 661 lfsvinyera rrpqllgasv lglddihraw
 rtfvlrvraq dpppelyfvk vdvtagydti 721 pqdrltevia siikpntyc vrryavvqka ahghvrkafk
 shvstltdlq pymrqfvahl 781 qetsplrdav vieqssslne assglfdvfl rfmchhavri rgksyvqcqg

ipqgsilstl 841 lclsicygdme nklfagirrd glllrlvddf llvtphltha ktlrltivrg vpeygcvnvl
 901 rktvvnfpve dealggtafv qmpahglfpw cglildtrtl evqsdysya rtsirasltf 961
 nrgfkagrnm rrlkfgvrl kchslfldlq vnslqtvcn iykilllqay rfhacvlqlp 1021 fhqqvwknpt
 fflrvisdta slcysilkak nagmslgakg aagplpseav qwlchqafl 1081 kltrhrvtiv pllgslrtaq
 tqlsrklpgt titaleaaan palpsdfkti ld

(173) In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001193376.1, transcript variant 2):

(174) TABLE-US-00017 (SEQ ID NO: 15)

1	caggcagcgc	tgcgtcctgc	
61	gccgcgcgct	ccccgctgcc	
121	gctgccgctg	gccacgttcg	
181	cggggacccg	gcggctttcc	
241	cgcacggccg	ccccccgccg	
301	ggcccagatg	ctgcagaggc	
361	cgcgctgctg	gacggggccc	
421	ctacctgcc	aacacggtga	
481	gcgccgcgtg	ggcgacgacg	
541	ggtggctccc	agctgcgcct	accaggtgtg
601	cactcaggcc	cggccccgc	cacacgctag
661	ggcctggaac	catagcgtca	gggaggccgg
721	gaggaggcgc	gggggagcgt	ccagccgaag
781	cgctgccct	gagccggagc	ggacgccctg
841	gacgcgtgga	ccagtgacc	gtggtttctg
901	agccacctct	ttggagggtg	cgctctctgg
961	ccagcaccac	gcggggcccc	catccacatc
1021	tccccggtg	tacgccgaga	ccaagcatt
1081	gcggccctcc	ttctactca	gctctctgag
1141	ggagaccatc	ttctgggtt	ccaggccctg
1201	gatgccaggg	actccccgca	ggttgccccg
1261	gcaagtcccc	tacgggggtgc	tcctcaagac
1321	agcagccggt	gtctgtgccc	gggagaagcc
1381	ggacacagac	ccccgtcgcc	tggtgcagct
1441	gtacggcttc	gtgcggggcct	gcctgcgccg
1501	gcacaacgaa	cgccgcttcc	tcaggaacac
1561	caagctctcg	ctgcaggagc	tgacgtggaa
1621	caggagccca	gggggtggct	gtgttccggc
1681	ggccaagtcc	ctgcactggc	tgatgagtgt
1741	gtacgtcgtc	gagctgtctca	ggtctttctt
1801	gagcaagttg	caaagcattg	gaatcagaca
1861	gtcgggaagca	gaggtcaggc	agcatcgga
1921	ccgcttcac	ccccagcctg	acgggctgcg
1981	agccagaacg	ttccgcagag	aaaagagggc
2041	gttcagcgtg	ctcaactacg	agcgggcgcg
2101	gggcctggac	gataccaca	gggcctggcg
2161	cccgccgcct	gagctgtact	ttgtcaaggt
2221	ctcacggagg	tcacgccag	ggatgtgacg
2281	gcgctgggtat	gcccgtggcc	agaaggccgc
2341	acacgtctct	accttgacag	acctccagcc
2401	gtacatgcga	cagttcgtgg	ctcacctgca
2461	agctcctccc	tgaatgaggc	cagcagtggc
2521	tgccatcag		
2581	ctgcagcctg		
2641	gctgctcctg	cgtttggtgg	

atgatttctt gtggtgaca cctcaccta cccacgcgaa 2701 aaccttctc agctatgccc ggacctccat
cagagccagt ctcaccttca accgcggcct 2761 caaggctggg aggaacatgc gtcgaaact ctttggggtc
ttgcggctga agtgtcacag 2821 cctgtttctg gatttgcagg tgaacagcct ccagacgggtg tgcaccaaca
tctacaagat 2881 cctcctgctg caggcgtaca gggttcacgc atgtgtgctg cagctcccat ttcacagca
2941 agtttggaag aaccccacat ttttctgcg cgtcatctct gacacggcct ccctctgcta 3001
ctccatctg aaagccaaga acgcagggat gtcgctgggg gccaaaggcg ccgccggccc 3061
tctgccctcc gaggccgtgc agtggctgtg ccaccaagca ttctgtctca agctgactcg 3121 acaccgtgtc
acctacgtgc cactcctggg gtcactcagg acagcccaga cgcagctgag 3181 tcggaagctc
ccggggacga cgctgactgc cctggaggcc gcagccaacc cggcactgcc 3241 ctcagacttc
aagaccatcc tggactgatg gccacccgcc cacagccagg ccgagagcag 3301 acaccagcag
ccctgtcacg ccgggctcta cgtcccaggg agggaggggc ggcccacacc 3361 caggcccgca
ccgctgggag tctgaggcct gagtgagtgt ttggccgagg cctgcatgtc 3421 cggctgaagg ctgagtgtcc
ggctgaggcc tgagcgagtg tccagccaag ggctgagtgt 3481 ccagcacacc tgccgtcttc acttccccac
aggctggcgc tcggctccac cccagggccca 3541 gcttttctc accaggagcc cggcttcac tccccacata
ggaatagtcc atccccagat 3601 tcgccattgt tcaccctcg ccctgccctc cttgccttc cacccccacc
atccaggtgg 3661 agaccctgag aaggaccctg ggagctctgg gaatttgag tgaccaaagg tgtgccctgt
3721 acacaggcga ggaccctgca cctggatggg ggtccctgtg ggtcaaattg gggggaggtg 3781
ctgtgggagt aaaatactga atatagagt tttcagttt tgaaaaaaa

(175) In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001180305.1, transcript variant 2):

(176) TABLE-US-00018 (SEQ ID NO: 16) 1 mpraprcrav rsllrshyre vlplatfvrr
lgpqgwrivq rgdpaafral vaqlvcvpw 61 darpppaaps frqvsclkel varvlqrice
rgaknvlafg falldgargg ppeaftsvr 121 sylpntvtda lrgsgawgll lrrvgddvllv hllarcalfv
lvapscayqv cgpplyqlga 181 atqarpppha sgprrrlgce rawnhsvrea gvplglpapg
arrrggsasr slplpkrrr 241 gaapeperpt vgqgswahpg rtrgpsdrgf cvvsparpae eatslegals
gtrhshpsvg 301 rqhhagppst srpprpwdtp cppvyaetkh flyssgdkeq lrpsfllssl rpsltgarrl
361 vetiflsrp wmpgtprrlp rlpqrywqmr plflllgnh aqcpygvllk thcplraavt 421
paagvcarek pqgsvaapee edtdprrlvq llrqhsspww vygfvracrl rlvppglwgs 481
rhnerflrn tkkfislghk aklsiqeltw kmsvrdcawl rrspgvvcvp aaehrlreei 541 lakflhwlm
vyvvellrsf fyvtettfqk nrlffyrksv wsklqsigir qhlkrvqlre 601 lseaevrqhr earpalltsr
lrfipkpdgl rpivnmdyv vgartfrekr aerltsrvka 661 lfsvlnyera rrpqllgasv lglddihraw
rtfvlrvraq dpppelyfvk vdvtagydti 721 pqdrktevia siikpqntyc vrryavvqka ahghvrkafk
shvstltdlq pymrqfvahl 781 qetsplrdav vieqssslne assglfdvfl rfmchhavri rgksyvqcqg
ipqgsilstl 841 lclscygdme nklfagirrd glllrlvddf llvtphltha ktflsyarts irasltfnrg 901
fkagrnmrk lfgvlrlkch slfldlqvns lqtvcniyk illlqayrfh acvlqlpfhq 961 qvwknptffl
rvisdtasl ysilkaknag mslgakgaag plpseavqwl chqafllklt 1021 rhrtvyppll gslrtaqtql
srklpgttlt aleaaanpal psdfktild

(177) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_014883.3, transcript variant 1):

(178) TABLE-US-00019 (SEQ ID NO: 36) 1 atcaaatttc aactccaggc agtccttcca
gccatgtggg ttcagcggaa agagaagcaa 61 aaccactctt cctaaaatgt tagaagctgc tcttcgctta
ccttggggcc ttgcattgg 121 gagctgtttt tcacatcaaa gaatatgtgc tgaatggaat ttagtattt tgctgtcggt
181 ttaatatttt cgtctggtct tcctcagttc ttccagacgc ttctgagag aatgggggca 241 ggagctctag
ccatctgtca aagtaaagca gcggttcggc tgaaagaaga catgaaaaag 301 atagtggcag tgccattaaa
tgaacagaag gattttacct atcagaagtt atttgagtc 361 agtctccaag aactgaacg gcaggggctc
accgagaatg gcattccagc agtagtgtgg 421 aatatagtgg aatatttgac gcagcatgga cttaccaag
aaggctcttt tagggatgaat 481 ggtaacgtga aggtggtgga acaacttcga ctgaagtctg agagtggagt
gcccgtggag 541 ctcgggaagg acggtgatgt ctgctcagca gccagtctgt tgaagctgtt tctgagggag 601

ctgcctgaca gtctgatcac ctccagcgttg cagcctcgat ctttcaact ctttcaggat 661 ggcagaaatg
atgttcagga gagtagctta agagacttaa taaaagagct gccagacacc 721 cactactgcc tctcaagta
cctttgccag ttcttgacaa aagtagccaa gcatcatgtg 781 cagaatcgca tgaatgttca caatctcgcc
actgtatttg ggccaaattg ctttcatgtg 841 ccacctgggc ttgaaggcat gaaggaacag gacctgtgca
acaagataat ggctaaaatt 901 ctagaaaatt acaataccct gtttgaagta gagtatacag aaaatgatca
tctgagatgt 961 gaaaacctgg ctaggcttat catagtaaaa gaggtctatt ataagaactc cctgccccatc 1021
ctttaacaa gaggcttaga aagagacatg ccaaaaccac ctccaaaaac caagatccca 1081 aaatccagga
gtgagggatc tattcaggcc cacagagtac tgcaaccaga gctatctgat 1141 ggcatctctc agctcagctt
gcggctaagt tatagaaaag cctgcttgga agacatgaat 1201 tcagcagagg gtgctattag tgccaagtgtg
gtacccagtt cacaggaaga tgaaagacct 1261 ctgtcacctt tctatttgag tgctcatgta cccaagtca
gcaatgtgtc tgcaaccgga 1321 gaactcttag aaagaacat cccatcagct gtagaacaac atcttttga
tgtaataac 1381 tctggaggtc aaagttcaga ggactcagaa tctggaacac tatcagcatc ttctgccaca 1441
tctgccagac agcgcggccg ccagtccaag gaggcagatg aagttcgaca tgggagagac 1501 aagggactta
tcaacaaaga aaatactct tctgggttca accacctga tgattgtatt 1561 ttgaatactc aggaagtcca
aaaggtacac aaaaatactt ttggttgtgc tggagaaagg 1621 agcaagccta aacgtcagaa atccagtact
aaactttctg agcttcatga caatcaggac 1681 ggtcttctga atatggaaag tctcaattcc acacgatctc
atgagagaac tggacctgat 1741 gattttgaat ggatgtctga tgaaaggaaa ggaaatgaaa aagatgggtg
acacactcag 1801 cattttgaga gcccacaat gaagatccag gacatccca gcctatctga caccaaacag 1861
cagagaaatc aagatgccgg tgaccaggag gagagcttg tctccgaagt gcccagtcg 1921 gacctgactg
catttgttga tgaaaagaac tgggaagagc ctatccctgc tttctctcc 1981 tggcagcggg agaacagtga
ctctgatgaa gccacctct cgccgcaggc tgggcgcctg 2041 atccgtcagc tgctggacga agacagcgac
cccatgctct ctctcgggt ctacgcttat 2101 gggcagagca ggcaatacct ggatgacaca gaagtgcctc
cttccccacc aaactccat 2161 tctttcatga ggcggcgaag ctctctctg gggctctatg atgatgagca
agaggacctg 2221 acacctgccc agctcacacg aaggattcag agccttaaaa agaagatccg gaagttgaa
2281 gatagattcg aagaagagaa gaagtacaga ccttcccaca gtgacaaaagc agccaatccg 2341
gaggttctga aatggacaaa tgaccttgcc aaattccgga gacaacttaa agaatacaaaa 2401 ctaaagatat
ctgaagagga cctaactccc aggatgcggc agcgaagcaa cacactcccc 2461 aagagttttg gttcccaact
tgagaaagaa gatgagaaga agcaagagct ggtggataaa 2521 gcaataaagc ccagtgttga agccacattg
gaatctattc agaggaagct ccaggagaag 2581 cgagcggaaa gcagccgccc tgaggacatt aaggatatga
ccaaagacca gattgctaata 2641 gagaaagtgg ctctgcagaa agctctgtta tattatgaaa gcattcatgg
acggccggta 2701 acaaagaacg aacggcaggt gatgaagcca ctatacgaca ggtaccggct ggtcaaacag
2761 atctctccc gagctaacac catacccatc attggttccc cctccagcaa gcggagaagc 2821
cctttgctgc agccaattat cgagggcgaa actgcttct tcttaagga gataaaggaa 2881 gaagaggagg
ggtcagaaga cgatagcaat gtgaagccag acttcatggt cactctgaaa 2941 accgatttca gtgcacgatg
ctttctggac caattcgaag atgacgctga tggatttatt 3001 tccccaatgg atgataaaat accatcaaaa
tgacgccagg acacagggtt tcaaatctc 3061 catgctgcct caatacctga actcctggaa cacctccagg
aaatgagaga agaaaagaaa 3121 aggattcgaa agaaactcg ggattttgaa gacaactttt tcagacagaa
tggaagaaat 3181 gtccagaagg aagaccgcac tcctatggct gaagaataca gtgaataata gcacataaag 3241
gcgaaactga ggctcctgga ggtgctcatc agcaagagag aactgattc caagtccatg 3301 tgaggggcat
ggccaagcac aggggggttg cagctgcggt gagagttaac tgtccccaga 3361 gaaagtgcag ctctggaagg
cagccttggg gctggccctg caaagcatgc agccctctg 3421 cctctagacc atttggcatc ggctcctgtt
tccattgctt gccttagaaa ctggctggaa 3481 gaagacaatg tgacctgact taggcatttt gtaattggaa
agtcaagact gcagtatgtg 3541 cacatgcgca cgcgcatgca cgcacacaca cacacagtag tggagcttct
ctaactag 3601 cagagattaa tcaactacatt agacaacact catctacaga gaataacac tgttcttccc 3661
tgataactg agaaacaaga gaccattctc tgttaactg tgataaaaac aagctcagga 3721 ctttattcta
tagagcaaac ttgctgtgga gggccatgct ctcttggac ccagttaact 3781 gcaaactgac attggagccc
tatttctgct cgctgccatt ctagtacct ttccacagag 3841 ctgcgccttc ctacgtgtg tgaaagggtt
tcccctcag ccctcagga gatggaagct 3901 gcatctgccc acgatggcag tgacgtcatc atcttcagga
tgtttcttca ggacttctc 3961 agctgacaag gaattttggt ccctgcctag gaccgggtca tctgcagagg
acagagagat 4021 ggtaagcagc tgtatgaat ctgattttaa aaccagggtca tgggagaaga gcctggagat 4081

tctttctga acactgactg cacttaccag tctgatttta tctgtaaaca ccaagccagg 4141 ctgcatgct
 catggcaatc tgtttggggc tgtttgttg tggcactagc caaacataaa 4201 ggggcttaag tcagcctgca
 tacagaggat cggggagaga aggggcctgt gttctcagcc 4261 tctgagtac ttaccagagt ttaatTTTT
 taaaaaaat ctgcactaaa atccccaac 4321 tgacaggtaa atgtagcct cagagctcag cccaaggcag
 aatctaaatc acactatTTT 4381 cgagatcatg tataaaaaga aaaaaaaga gtcatgctgt gtggccaatt
 ataatTTTT 4441 tcaaagactt tgcacaaaa ctgtctatat tagacatTTT ggagggacca ggaaatgtaa 4501
 gacaccaaT cctccatctc ttcagtgtgc ctgatgtcac ctcattgatt gctgttactt 4561 ttttaactcc
 tgcgccaagg acagtgggtt ctgtgtccac cttgtgtct tgcgaggccg 4621 agcccaggca tctgctcgcc
 tgccacggct gaccagagaa ggtgcttcag gagctctgcc 4681 ttgacgacg tgttacagta tgaacacaca
 gcagaggcac cctcgtatgt ttgaaagt 4741 gccttctgaa agggcacagt ttaaggaaa agaaaaaga
 tgtaaaacta tactgacccg 4801 ttttcagttt taaagggtcg tgagaaactg gctgggtcaa tgggatttac
 agcaacattT 4861 tccattgctg aagtgagga gcagctctct tctgtcagct gaatgttaag gatggggaaa 4921
 aagaatgcct ttaagtttgc tctaatcgt atggaagctt gagctatgtg ttggaagtgc 4981 cctggtttta
 atccatacac aaagacggta cataatccta cagggttaaa tgtacataaa 5041 aatatagttt ggaattctt
 gctctactgt ttacattgca gattgctata atttcaagga 5101 gtgagattat aaataaaatg atgcacttta
 ggatgtttcc tattttgaa atctgaacat 5161 gaatcattca catgaccaa aattgtgtt ttttaaaat
 acatgtctag tctgtcctt 5221 aatagctctc taaataagc tatgatatta atcagatcat taccagttag ctttaaagc
 5281 acatttgtt aagactatgt ttttgaaaa atacgtaca gaatTTTT ttaagctaca 5341 aataaatgag
 atgctactaa ttgttttga atctgttgt tctgcaaag gtaaattaac 5401 taaagattta ttcaggaatc
 cccatttgaa tttgtatgat tcaataaaag aaacaccaa 5461 gtaagtata taaataaat tgtgtatgag
 atgttgtgtt ttccttga atttccacta 5521 actaactaac taactatat tcttcatgga atggagcca
 gaagaaatga gaggaagccc 5581 tttcacact agatcttatt tgaagaaatg tttgttagtc agtcagtcag
 tggtttctgg 5641 ctctgccgag ggagatgtgt tccccagcaa ccatttctgc agcccagaat ctcaaggcac 5701
 tagaggcggg gtcttaatta attggctca caaagacaaa atgctctgga ctgggattt 5761 tccttctgct
 tgttggaat atgtgttat taattagcac atgccaacaa aataaatgtc 5821 aagagtatt tcataagtgt
 aagtaaactt aagaattaa gactgcagac ttataattt 5881 ca

(179) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_055698.2, transcript variant 1):

(180) TABLE-US-00020 (SEQ ID NO: 37) 1 mgagalaicq skaavrlked mkkivavpln
 eqkdftyqkl fgvslqeler qgltenGpa 61 vvwNiveylt qhgltqeglf rvngnvkvve qlrlkfesgv
 pvelgkdgdv csaasllklf 121 lrelpdslit salqprfiql fqdgrndvqe sslrdlikel pdthycllky
 lcqfltkvak 181 hhvqnrmnvh nlatvfgpnc fhvppglegm keqdlcnkim akilenyntl feveytendh
 241 lrcenlarli ivkevykns lpilltrgle rdmpkpppkt kipksrsegs iqahrvlqpe 301 lsdgipqlsl
 rlsyrkacle dmnsaegais akvpssqed erlpsfpyls ahvpqvsnvS 361 atgellerti rsaveqhlfd
 vnnsaggqsse dsesgtlsas satsarqrrr qskeqdevrh 421 grdkglinke ntpsgfnhld dcilntqeve
 kvhknrtfgca gerskprqk sstklsehd 481 nqdglnmes lnstrshert gpddfewmsd erkgnedgg
 htqhfesptm kigehpslsd 541 tkqqrnqdag dqeesfvsev pqsdltalcd eknweepipa fsswqrensd
 sdeahlsppa 601 grlirqlde dsdpmlsprf yaygqsryl ddtvppspp nshsfmrrrs sslgsydeeq 661
 edltpaqltr riqlskkkir kfedrfeek kyrpshsdka anpevlkwtN dlakfrrqlk 721 esklkiseed
 ltpmrqrsn tlpksfgsql ekedekkkqel vdkaikpsve atlesiqrkl 781 qekraessrp edikdmtdq
 ianekvalqk allyyesihg rpvtknerqv mkplydryrl 841 vkqilsrant ipiigspssk rrspllqpII
 egetasffke ikeeeegsed dsnvkpdfmv 901 tlktdfsarc fldqfeddad gfispmddki psksqdtgl
 snlhaasipe llehlqemre 961 ekkrikkrl dfednffrqN grnvqkedrt pmaeeyseyk hikakrlle
 vliskrdtds 1021 ksm

(181) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001015045.2, transcript variant 2):

(182) TABLE-US-00021 (SEQ ID NO: 17) 1 attgaggagc agaaggagta ggggtcgggg
 gaggaggagg agcgccttta gtgctgcagc 61 agctgctgct ctgattggcc cgggtggtca gctgctccc

tggaacaaaa ggtcaaatg 121 gactgcagt taaatgtaga gaagcagccg ataatagc attgcctgaa
gaagtttga 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcgg 241
gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttact gtacagacga 301 ccaaacttg
cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg 361 gcttgtgaaa tcatgcctt
gcaaagttca caggaagatg aaagacctt gtcaccttc 421 tatttgatg ctcatgtacc ccaagtcagc
aatgtgtctg caaccggaga actcttagaa 481 agaaccatcc gatcagctgt agaacaacat cttttgatg
ttaataactc tggagggtcaa 541 agttcagagg actcagaatc tggaacacta tcagcatctt ctgccacatc
tgccagacag 601 cgccgccgcc agtccaagga gcaggatgaa gttcgacatg ggagagacaa gggacttatc 661
aacaaagaaa atactccttc tgggttcaac caccttgatg attgtattt gaatactcag 721 gaagtcgaaa
aggtacacaa aaatactttt gggtgtgctg gagaaaggag caagcctaaa 781 cgtcagaaat ccagtactaa
actttctgag cttcatgaca atcaggacgg tcttgtaat 841 atggaaagtc tcaattccac acgatctcat
gagagaactg gacctgatga ttttgaatgg 901 atgtctgatg aaaggaaagg aatgaaaaa gatggtggac
acactcagca tttgagagc 961 cccacaatga agatccagga gcatcccagc ctatctgaca ccaaacagca
gagaaatcaa 1021 gatgccggtg accaggagga gagctttgtc tccgaagtgc cccagtcgga cctgactgca
1081 ttgtgtgatg aaaagaactg ggaagagcct atccctgctt tctctcctg gcagcgggag 1141
aacagtgact ctgatgaagc ccacctctg ccgcaggctg ggcgcctgat ccgtcagctg 1201 ctggacgaag
acagcgaccc catgctctt cctcggttct acgcttatgg gcagagcagg 1261 caatacctgg atgacacaga
agtgcctct tccccacaa actcccattc ttcatgagg 1321 cggcgaagct cctctctggg gtcctatgat
gatgagcaag aggacctgac acctgcccag 1381 ctacacgaa ggattcagag ccttaaaaag aagatccgga
agtttgaaga tagattcgaa 1441 gaagagaaga agtacagacc ttccacagt gacaaagcag ccaatccgga
ggttctgaaa 1501 tggacaaatg acctgcaa attccggaga caacttaaag aatcaaaact aaagatatct 1561
gaagaggacc taactcccag gatgcggcag cgaagcaaca cactcccaa gagttttggt 1621 tcccaactg
agaaagaaga tgagaagaag caagagctgg tgataaagc aataaagccc 1681 agtgttgaag ccacattgga
atctattcag aggaagctcc aggagaagcg agcggaaagc 1741 agccgcctg aggacattaa ggatatgacc
aaagaccaga ttgctaata gaaagtggct 1801 ctgcagaaag ctctgttata ttatgaaagc attcatggac
ggccggtaac aaagaacgaa 1861 cggcagggtga tgaagccact atacgacagg taccggctgg taaacagat
cctctcccga 1921 gtaacacca taccatcat tggttcccc tccagcaagc ggagaagccc ttgtctgcag 1981
ccaattatcg agggcgaaac tgcttcttc ttcaaggaga taaaggaaga agaggagggg 2041 tcagaagacg
atagcaatgt gaagccagac ttcatggtca ctctgaaaac cgatttcagt 2101 gcacgatgct ttctggacca
attcgaagat gacgtgatg gatttattc cccaatggat 2161 gataaaatac catcaaaatg cagccaggac
acagggcttt caaatctca tgctgectca 2221 atacctgaac tctggaaca cctccaggaa atgagagaag
aaaagaaaag gattcgaaa 2281 aaacttcggg attttgaaga caacttttc agacagaatg gaagaaatgt
ccagaaggaa 2341 gaccgcactc ctatggctga agaatacagt gaataaagc acataaaggc gaaactgagg
2401 ctctggagg tgctcatcag caagagagac actgattcca agtccatgtg aggggcatgg 2461
ccaagcacag ggggctggca gctgcggtga gagtttactg tccccagaga aagtgcagct 2521 ctggaaggca
gccttggggc tggccctgca aagcatgcag cccttctgcc ttagaccat 2581 ttggcatcgg ctctgttc
cattgcctgc ctagaaact ggctggaaga agacaatgt 2641 acctgactta ggcattttgt aattggaaag
tcaagactgc agtatgtga catgcgcacg 2701 cgcagtcacg cacacacaca cacagtagtg gagcttct
aacactagca gagattaatc 2761 actacattag acaacactca tctacagaga atataactg ttctcctg
gataactgag 2821 aaacaagaga ccattctctg tctaactgtg ataaaaaaa gctcaggact ttattctata 2881
gagcaaaact gctgtggagg gccatgctt cttggaccc agttaactgc aaacgtgcat 2941 tggagcccta
ttgtctccg ctgccattt agtgacctt ccacagagct gcgcttct 3001 cacgtgtgtg aaaggtttc
ccctcagcc ctcaggtaga tggaagctgc atctgccac 3061 gatggcagtg cagtcacat cttcaggatg
ttcttcagg acttctcag ctgacaagga 3121 attttggtc ctgcctagga ccgggtcatc tgcagaggac
agagagatgg taagcagctg 3181 tatgaatgt gatttataa ccaggtcatg ggagaagagc ctggagattc
ttctgaac 3241 actgactgca ctaccagtc tgattttatc gtcaaacc aagccaggct agcatgctca 3301
tggcaatctg tttggggctg tttgtgtg gcactagcca aacataaagg ggcttaagtc 3361 agcctgcata
cagaggatcg gggagagaag gggcctgtgt tctcagctc ctgagtact 3421 accagagttt aatttttta
aaaaaatct gcactaaaat cccaaactg acaggtaa 3481 gtagccctca gagctcagcc caaggcagaa
tctaaatcac actatttctg agatcatgta 3541 taaaagaaa aaaaagaagt catgctgtgt ggccaattat

aattttttc aaagactttg 3601 tcacaaaact gcttatatta gacattttgg agggaccagg aaatgtaaga
caccaaattcc 3661 tccatctctt cagtgtgcct gatgtcacct catgatttgc tgttactttt ttaactcctg 3721
cgccaaggac agtgggttct gtgtccacct ttgtgctttg cgaggccgag cccaggcatc 3781 tgctcgctg
ccacggctga ccagagaagg tgcttcagga gctctgcctt agacgacgtg 3841 ttacagtatg aacacacagc
agaggcacc tcgtatgttt tgaaagttgc cttctgaaag 3901 ggacacagtt taaggaaaag aaaaagaatg
taaaactata ctgacccggtt ttcagtttta 3961 aagggtcggtg agaaactggc tggccaatg ggatttacag
caacattttc cattgctgaa 4021 gtgaggtagc agctctcttc tgtcagctga atgttaagga tggggaaaaa
gaatgccttt 4081 aagtttgctc ttaatcgtat ggaagcttga gctatgtgtt ggaagtgcc tggtttaaat 4141
ccatacacia agacgggtaca taatcctaca ggtttaaag tacataaaaa tatagtttgg 4201 aattctttgc
tctactgttt acattgcaga ttgctataat ttcaaggagt gagattataa 4261 ataaaatgat gcactttagg
atgtttccta ttttgaaat ctgaacatga atcattcaca 4321 tgacaaaaa ttgtgtttt taaaaatac
atgtctagtc tgtcctttaa tagtctctct 4381 aaataagcta tgatattaat cagatcatta ccagttagct
tttaaagcac atttgtttaa 4441 gactatgttt ttggaaaaat acgctacaga atttttttt aagctacaaa taaatgagat
4501 gctactaatt gttttggaat ctgttgtttc tgccaaagggt aaattaacta aagatttatt 4561 caggaatccc
catttgaatt tgtatgattc aataaaaaga aacaccaagt aagttatata 4621 aataaaattg tgtatgagat
gttggtttt cttttgtaat ttccactaac taactaacta 4681 acttatattc ttcattggaat ggagcccaga
agaaatgaga ggaagccctt ttcacactag 4741 atcttatttg aagaaatgtt tgtagtcag tcagtcagtg
gtttctggct ctgccgaggg 4801 agatgtgttc cccagcaacc atttctgcag cccagaatct caaggcacta
gaggcggtgt 4861 ctaattaat tggcttcaca aagacaaaat gctctggact gggatttttc ctttgctgtg 4921
ttgggaatat gtgtttatta attagcacat gccacaaaa taaatgtcaa gagttatttc 4981 ataagtgtaa
gtaaacttaa gaattaaaga gtgcagactt ataattttca

(183) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001015045.1, transcript variant 2):

(184) TABLE-US-00022 (SEQ ID NO: 18) 1 maceimplqs sqederplsp fylsahvpqv
snvsatgell ertirsaveq hlfdvnssgg 61 qssedsesgt lsassatsar qrrrskeqd evrhgrdkgl
inkentpsgf nhlddcilnt 121 qevkvhknt fgcagerskp krqkstkl elhdnqdgvl nmeslnstrs
hertgpddfe 181 wmsderkgne kdgghtqhfe sptmkigehp slsdtkqqrn qdagdqeef vsevpqsdlt
241 alcdeknwee pipafsswqr ensdsdeahl spqagrlir lldedsdpml sprfyaygqs 301
rqylldtevp pspnshsfm rrrsslsy ddeqedltpa qltrriqslk kkirkfedrf 361 eeekkyrph
sdkaanpevl kwtndlakfr rqlkeskli seedltprmr qrsntlpksf 421 gslekedek kqelvdkaik
psveatlesi qrklqekrae srpedikdm tkdqianekv 481 alqkallye sihgrpvtkn erqvmkplyd
ryrlvkqils rantipiigs psskrspil 541 qpiiiegetas ffkeikeeee gseddsnvkp dfmvtlktdf
sarcfldqfe ddadgfispm 601 ddkipksksq dtglsnlhaa sipellehlq emreekkir kklrldfednf
frqngrnvqk 661 edrtpmaeey seykhikakl rllvliskr dtdsksm

(185) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265578.1, transcript variant 3):

(186) TABLE-US-00023 (SEQ ID NO: 38) 1 attgaggagc agaaggagta ggggtcgggg
gaggaggagg agcgccttta gtgctgcagc 61 agctgctgct ctgattggcc cgggtggtca gctgctccc
tggaacaaaa ggtcaaaagt 121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa
gaagtttgga 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggt 241
gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttact gtacagacga 301 ccaaacttgc
cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg 361 gcttggtgaaa tcatgcctct
gcaaagtgt catgtacccc aagtcagcaa tgtgtctgca 421 accggagaac tcttagaaag aaccatccga
tcagctgtag aacaacatct tttgatgtt 481 aataactctg gaggtcaaag ttcagaggac tcagaatctg
gaacactatc agcatcttct 541 gccacatctg ccagacagcg ccgccgccag tccaaggagc aggatgaagt
tcgacatggg 601 agagacaagg gacttatcaa caaagaaaat actcctctg ggttcaacca cttgatgat 661
tgtattttga atactcagga agtcgaaaag gtacacaaaa atacttttgg ttgtgtgga 721 gaaaggagca
agcctaaacg tcagaaatcc agtactaaac ttctgagct tcatgacaat 781 caggacggtc ttgtgaatat

ggaaagtctc aattccacac gatctcatga gagaactgga 841 cctgatgatt ttgaatggat gtctgatgaa
aggaaaggaa atgaaaaaga tgggtggacac 901 actcagcatt ttgagagccc cacaatgaag atccaggagc
atcccagcct atctgacacc 961 aaacagcaga gaaatcaaga tgccggtgac caggaggaga gctttgtctc
cgaagtgcc 1021 cagtcggacc tgactgcatt gtgtgatgaa aagaactggg aagagcctat ccctgcttc 1081
tcctcctggc agcgggagaa cagtactct gatgaagccc acctctgcc gcaggctggg 1141 cgctgatcc
gtcagctgct ggacgaagac agcgaccca tgctctctcc tcggttctac 1201 gcttatgggc agagcaggca
atacctggat gacacagaag tgctctctc cccaccaa 1261 tccattctt tcatgaggcg gcgaagctcc
tctctggggt cctatgatga tgagcaagag 1321 gacctgacac ctgcccagct cacacgaagg attcagagcc
ttaaaaagaa gatccggaag 1381 ttgaagata gattcgaaga agagaagaag tacagacctt cccacagtga
caaagcagcc 1441 aatccggagg ttctgaaatg gacaaatgac ctggccaaat tccggagaca acttaaagaa 1501
tcaaaactaa agatatctga agaggaccta actcccagga tgcggcagcg aagcaacaca 1561 ctcccaaga
gttttggtc ccaacttgag aaagaagatg agaagaagca agagctggg 1621 gataaagcaa taaagccag
tgttgaagcc acattggaat ctattcagag gaagctccag 1681 gagaagcgag cggaagcag ccgacctgag
gacattaagg atatgacaa agaccagatt 1741 gctaatagaga aagtggctct gcagaaagct ctgttatatt
atgaaagcat tcatggacgg 1801 ccggtacaa agaacgaacg gcagggtgatg aagccactat acgacaggta
ccggtgggtc 1861 aaacagatcc tctccgagc taacaccata ccatcattg gttccccctc cagcaagcgg 1921
agaagccctt tgctgcagcc aattatcgag ggcgaaactg ctctctctt caaggagata 1981 aaggaagaag
aggaggggtc agaagacgat agcaatgtga agccagactt catggctact 2041 ctgaaaaccg atttcagtgc
acgatgctt ctggaccaat tcgaagatga cgctgatgga 2101 ttatttccc caatggatga taaaatacca
tcaaatgca gccaggacac agggcttca 2161 aatctccatg ctgcctcaat acctgaactc ctggaacacc
tccaggaaat gagagaagaa 2221 aagaaaagga ttcgaaagaa acttcgggat ttgaagaca acttttcag
acagaatgga 2281 agaaatgtcc agaaggaaga ccgcactct atggctgaag aatacagtga atataagcac
2341 ataaaggcga aactgaggct cctggagggtg ctcatcagca agagagacac tgattccaag 2401
tccatgtgag gggcatggcc aagcacaggg ggctggcagc tgcggtgaga gttactgtc 2461 cccagagaaa
gtgcagctct ggaaggcagc ctgggggtg gccctgcaa gcatgcagcc 2521 cttctgcctc tagaccatt
ggcatcggt cctgtttcca ttgcctgcct tagaaactgg 2581 ctggaagaag acaatgtgac ctgacttagg
cattttgtaa ttggaaagtc aagactgcag 2641 tatgtgcaca tgcgcacgcg catgcacgca cacacacaca
cagtagtgga gctttcctaa 2701 cactagcaga gattaatcac tacattagac aacactcatc tacagagaat
atacactgtt 2761 ctccctgga taactgagaa acaagagacc attctctgtc taactgtgat aaaaacaagc 2821
tcaggacttt attctataga gcaaactgc tgtggagggc catgctctcc ttggaccag 2881 ttaactgcaa
acgtgcattg gagccctatt tgctgccgt gccattctag tgaccttcc 2941 acagagctgc gccttctca
cgtgtgtgaa aggtttccc cttcagccct caggtagatg 3001 gaagctgcat ctgccacga tggcagtga
gtcatcatct tcaggatgtt tctcaggac 3061 tctcagct gacaaggaat ttggtccct gcctaggacc
gggtcatctg cagaggacag 3121 agagatgga agcagctga tgaatgctga tttaaaacc aggtcatggg
agaagagcct 3181 ggagattctt tctgaacac tgactgcact taccagtctg attttatcgt caaacaccaa 3241
gccaggctag catgctcatg gcaatctgtt tggggctgtt ttgtgtggc actagccaaa 3301 cataaagggg
cttaagtcag cctgcataca gaggatcggg gagagaaggg gcctgtgtc 3361 tcagcctctc gactacttac
cagagtttaa ttttttaa aaaaatctgc actaaaatcc 3421 ccaaactgac aggtaaatgt agccctcaga
gtcagccca aggcagaatc taaatcacac 3481 tatttcgag atcatgtata aaaagaaaaa aaagaagtca
tgctgtgtgg ccaattataa 3541 ttttttcaa agactttgc aaaaaactgt ctatattaga catttgagg
ggaccaggaa 3601 atgtaagaca ccaaatcctc catctcttca gtgtgcctga tgtcacctca tgatttgctg 3661
ttacttttt aactctgcg ccaaggacag tgggttctgt gtccacctt gtgcttgcg 3721 aggccgagcc
caggcatctg ctgcctgcc acggctgacc agagaaggtg cttcaggagc 3781 tctgccttag acgacgtgtt
acagtatgaa cacacagcag aggcacctc gtatgtttg 3841 aaagttgcct tctgaaaggg cacagttta
aggaaaagaa aaagaatgta aaactatact 3901 gacccgttt cagttttaa gggctgtgag aaactggctg
gtccaatggg atttacagca 3961 acattttcca ttgctgaagt gaggtagcag ctctcttctg tcagctgaat
gttaaggatg 4021 gggaaaaaga atgccttaa gtttgcctt aatcgatgg aagcttgagc tatgtgttg 4081
aagtgcctg gtttaatcc atacacaaag acggtacata atctacagg tttaaagta 4141 cataaaaaa
tagtttgaa ttcttgctc tactgtttac attgcagatt gctataatt 4201 caaggagtga gattataat
aaaatgatgc actttaggat gtttctatt ttgaaatct 4261 gaacatgaat cattcacatg accaaaaatt

gtgtttttt aaaaatacat gtctagtctg 4321 tcctttaata gctctcttaa ataagctatg atattaatca
gatcattacc agttagcttt 4381 taaagcacat ttgtttaaga ctatgtttt ggaaaaatac gctacagaat tttttttaa
4441 gctacaaata aatgagatgc tactaattgt ttggaatct gttgtttctg ccaaaggtaa 4501 attactaaa
gatttattca ggaatcccca ttgaatttg tatgattcaa taaaagaaaa 4561 caccaagtaa gttatataaa
ataaattgtg tatgagatgt tgtgtttcc ttgtaattt 4621 ccactaacta actaactaac ttatattctt
catggaatgg agcccagaag aatgagagg 4681 aagccctttt cactactagat cttatttgaa gaaatgtttg
ttagtcagtc agtcagtgg 4741 ttctggctct gccgaggag atgtgttccc cagcaaccat ttctgcagcc
cagaatctca 4801 aggcactaga ggcggtgtct taattaattg gcttcacaaa gacaaaatgc tctggactgg 4861
gatttttctt ttgctgtgtt gggaatatgt gtttattaat tagcacatgc caacaaaata 4921 aatgtcaaga
gttatttcat aagtgaagt aaacttaaga attaaagagt gcagacttat 4981 aattttca

(187) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252507.1, transcript variant 3):

(188) TABLE-US-00024 (SEQ ID NO: 39) 1 maceimplqs ahvpqvsnv atgellerti
rsaveqhlfd vnnsqgsse dsesgtlsas 61 satsarqrrr qskeqdevrh grdkglinke ntpsgfnhld
dcilntqeve kvhknrtfgca 121 gerskprqk sstklshld nqdglnmes lnstrshert gpddfewmsd
erkgnekdgg 181 htqhfesptm kigehpslsd tkqqrnqdag dquesfvsev pqsdltalcd eknweepipa
241 fsswqrensd sdeahlpqa grlirqlde dsdpmlsprf yayggsrql ddtevpssp 301
nshsfmrrrs sslgsydeeq edltpaqltr riqlskkir kfedrfeek kyrpshsdka 361 anpevlkwn
dlakfrqlk esklkiseed ltpmrqrsn tlpksfgsq ekedekkel 421 vdkaikpsve atlesiqrkl
qekraessrp edikdmtkdq ianekvalqk allyyesihg 481 rpvtknerqv mkplydryrl vkqilsrant
ipiispssk rrspllpqi egetasffke 541 ikeeeegsed dsnvkpdfmv tlktdfsarc fldqfeddad
gfispmdcki psksqdtgl 601 snlhaasipe llehlqemre ekkirklr dfednffrn grnvqkedrt
pmaeeyseyk 661 hikaklrle vliskrtds ksm

(189) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265579.1, transcript variant 4):

(190) TABLE-US-00025 (SEQ ID NO: 40) 1 attgaggagc agaaggagta ggggtcgggg
gaggaggagg agcgccttta gtgctgcagc 61 agtgcgtct ctgattggcc cgggtggtca gctgctccc
tggaacaaaa ggtcaaagt 121 gactgcagt taaatgtaga gaagcagccg ataaaatagc attgcctgaa
gaagtgtgga 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcgg 241
gcagcctcat gcccacaacc cagcttagcc actgtaagaa gacgttact gtacagacga 301 ccaaacttg
cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg 361 gcttgatgaaa tcatgcctct
gcaaagtca caggaagatg aaagacctt gtcaccttc 421 tatttgatg ctcatgtacc ccaagtcagc
aatgtgtctg caaccggaga actcttagaa 481 agaaccatcc gatcagctgt agaacaacat cttttgatg
ttaataactc tggaggtcaa 541 agttcagagg actcagaatc tggaacacta tcagcatctt ctgccacatc
tgccagacag 601 cgccgccgcc agtccaagga gcaggatgaa gttcgacatg ggagagacaa gggacttatc 661
aacaagaaa atactcttc tgggttcaac caccttgatg attgtattt gaatactcag 721 gaagtcgaaa
aggtacacaa aaatacttt gggtgtgctg gagaaaggag caagcctaaa 781 cgtcagaaat ccagtactaa
actttctgag cttcatgaca atcaggacgg tcttgtaat 841 atggaaaagc tcaattccac acgatctcat
gagagaactg gacctgatga tttgaatgg 901 atgtctgatg aaaggaaagg aatgaaaaa gatggtggac
acactcagca tttgagagc 961 ccacaatga agatccagga gcatcccagc ctatctgaca ccaaacagca
gagaaatcaa 1021 gatgccggtg accaggagga gagctttgtc tccgaagtgc ccagtcgga cctgactgca
1081 ttgtgtgatg aaaagaactg ggaagagcct atccctgct tctcctctg gcagcgggag 1141
aacagtgact ctgatgaagc ccacctctg ccgcaggctg ggcgcctgat ccgtcagctg 1201 ctggacgaag
acagcgaccc catgctctt cctcggttct acgcttatgg gcagagcagg 1261 caatactgg atgacacaga
agtgcctct tccccacaa actccattc ttcatgagg 1321 cggcgaagct cctctctggg gtcctatgat
gatgagcaag aggacctgac acctgcccag 1381 ctacacgaa ggattcagag ccttaaaaag aagatccgga
agtttgaaga tagattcgaa 1441 gaagagaaga agtacagacc ttccacagt gacaaagcag ccaatccgga
ggttctgaaa 1501 tggacaaatg acctgcaa attccggaga caactaaag aatcaaaact aaagatatct 1561

gaagaggacc taactcccag gatgcggcag cgaagcaaca cactcccca gagttttggt 1621 tcccaacttg
 agaaagaaga tgagaagaag caagagctgg tggataaagc aataaagccc 1681 agtgttgaag ccacattgga
 atctattcag aggaagctcc aggagaagcg agcggaaagc 1741 agccgcctg aggacattaa ggatatgacc
 aaagaccaga ttgctaata gaaagtggct 1801 ctgcagaaag ctctgttata ttatgaaagc attcatggac
 ggccggtaac aaagaacgaa 1861 cggcaggtga tgaagccact atacgacagg taccggctgg tcaaacagat
 cctctcccga 1921 gctaacacca taccatcat tgaagaagag gaggggtcag aagacgatag caatgtgaag
 1981 ccagacttca tggctactct gaaaaccgat ttcaagtgcac gatgctttct ggaccaattc 2041 gaagatgacg
 ctgatggatt tatttccca atggatgata aaataccatc aaaatgcagc 2101 caggacacag ggctttcaaa
 tctccatgct gcctcaatac ctgaactcct ggaacacctc 2161 caggaaatga gagaagaaaa gaaaaggatt
 cgaaagaaac ttccgggattt tgaagacaac 2221 ttttcagac agaatggaag aatgtccag aaggaagacc
 gcactcctat ggctgaagaa 2281 tacagtgaat ataagcacat aaaggcgaaa ctgaggctcc tggaggtgct
 catcagcaag 2341 agagacactg attccaagtc catgtgaggg gcatggccaa gcacaggggg ctggcagctg
 2401 cgggtgagagt ttactgtccc cagagaaagt gcagctctgg aaggcagcct tggggctggc 2461
 cctgcaaagc atgcagccct tctgcctcta gaccatttgg catcggctcc tgtttccatt 2521 gcctgcctta
 gaaactggct ggaagaagac aatgtgacct gacttaggca ttttgaatt 2581 ggaaagtcaa gactgcagta
 tgtgcacatg cgcacgcgca tgcacgcaca cacacacaca 2641 gtagtgagagc ttcctaaca ctagcagaga
 ttaatcacta cattagacaa cactcatcta 2701 cagagaatat acactgttct tccttgata actgagaaac
 aagagaccat tctctgcta 2761 actgtgataa aaacaagctc aggactttat tctatagagc aaacttgctg
 tggagggcca 2821 tgctctcctt ggaccaggtt aactgcaaac gtgcattgga gccctatttg ctgccgtgc 2881
 cattctagtg acctttccac agagctgcgc cttctcacg tgttgtaaag gttttccct 2941 tcagccctca
 ggtagatgga agctgcatct gccacgatg gcagtgcagt catcatctc 3001 aggatgttc ttcaggactt
 cctcagctga caaggaattt tggccctgc ctaggaccgg 3061 gtcacttgca gaggacagag agatggtaag
 cagctgtatg aatgctgatt ttaaaaccag 3121 gtcatgggag aagagcctgg agattcttc ctgaacactg
 actgcactta ccagtctgat 3181 tttatcgta aacaccaagc caggctagca tgctcatggc aatctgttg
 gggctgttt 3241 gttgtggcac tagccaaaca taaaggggct taagtcagcc tgcatacaga ggatcgggga 3301
 gagaaggggc ctgtgttctc agcctctga gtactacca gagtttaatt ttttaaaaa 3361 aaactgcac
 taaaatccc aaactgacag gtaaatgtag ccctcagagc tcagcccaag 3421 gcagaatcta aatcacacta
 ttttcgagat catgtataaa aagaaaaaaa agaagtcatt 3481 ctgtgtggcc aattataatt ttttcaaag
 actttgtcac aaaactgtct atattagaca 3541 ttttgagggg accaggaaat gtaagacacc aaatcctcca
 tctcttcagt gtgcctgatg 3601 tcacctcatg atttgctgtt actttttaa ctctgcgcc aaggacagtg gggtctgtgt
 3661 ccaccttgt gctttgcgag gccgagcca ggcatctgct cgctgccac ggctgaccag 3721
 agaaggtgct tcaggagctc tgccttagac gacgtgttac agtatgaaca cacagcagag 3781 gcaccctcgt
 atgttttgaa agttgccttc tgaaagggca cagttttaag gaaaagaaaa 3841 agaagttaa actatactga
 cccgtttca gtttaaaagg gtcgtgagaa actggctgg 3901 ccaatgggat ttacagcaac attttcatt
 gctgaagtga ggtagcagct ctctctgtc 3961 agctgaatgt taaggatggg gaaaaagaat gcctttaagt
 ttgctcttaa tcgtatggaa 4021 gcttgagcta tgtgttgaa gtgccctgg ttaatccat acacaaagac
 ggtacataat 4081 cctacaggtt taaatgtaca taaaaatata gtttgaatt cttgtctta ctgtttacat 4141
 tgcagattgc tataatttca aggagtga ttaataataa aatgatgcac ttaggatgt 4201 ttcctattt
 tgaaatctga acatgaatca ttcacatgac caaaaattgt gttttttaa 4261 aaatacatgt ctagtctgc
 cttaatagc tctcttaaat aagctatgat attaatacaga 4321 tcattaccag ttagcttta aagcacatt
 gttaagact atgttttgg aaaaatacgc 4381 tacagaattt tttttaagc taaaataaa tgagatgcta
 ctaattgtt tggaatctgt 4441 tgtttctgcc aaaggtaaat taactaaaga tttattcagg aatccccatt tgaattgta
 4501 tgattcaata aaagaaaaca ccaagtaagt tatataaat aaattgtgta tgagatgttg 4561 tgtttcctt
 tgtaatttcc actaactaac taactaact atattcttca tggaatggag 4621 ccagaagaa atgagaggaa
 gccctttca cactagatct tatttgaaga aatgtttgt 4681 agtcagtcag tcagtgtt ctggctctgc
 cgaggagat gtgttccca gcaaccatt 4741 ctgcagcca gaatctcaag gcactagagg cgggtgtcta
 attaattggc ttcacaaaga 4801 caaatgctc tggactggga ttttctt gctgtgttg gaatatgtgt ttattaata
 4861 gcacatgcca aaaaaataa tgtaagagt tattcataa gtgtaagtaa acttaagaat 4921 taaagagtgc
 agactataa tttca

(191) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene

of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252508.1, transcript variant 4):

(192) TABLE-US-00026 (SEQ ID NO: 41) 1 maceimplqs sqederplsp fylsahvpqv
snvsatgell ertirsaveq hlfdvnssgg 61 qssedsesgt lsassatsar qrrrqskeqd evrhgrdkgl
inkentpsgf nhlddcilnt 121 qevkvhknt fgcagerskp krqkstklsl elhndqdgvl nmeslnstrs
hertgpddfe 181 wmsderkgne kdgghtqhfe sptmkigehp slsdtkqqrn qdagdqeesf vsevpqsdlt
241 alcdeknwee pipafsswqr ensdsdeahl spqagrlirql lldedsdpml sprfyaygqs 301
rqylldtevp pspnshsfm rrrssslgsy ddeqedltpa qltrriqslk kkirkfedrf 361 eeekkyrpsh
sdkaanpevl kwtnldakfr rqlkesklki seedltprmr qrsntlpksf 421 gslekedek kqelvdkaik
psveatlesi qrklqekrae ssrpedikdm tkdqianekv 481 alqkallye sihgrpvtkn erqvmkplyd
ryrlvkqils rantipiiee eegseddsnv 541 kpdfmvtlkt dfsarcfldq feddadgfs pmddkipksc
sqdtglslnh aasipelleh 601 lqemreekkr irklklrdfed nffrqngrnv qkedrtpmae eyseykhika
klrllevlis 661 krtdtdsksm

(193) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265580.1, transcript variant 5):

(194) TABLE-US-00027 (SEQ ID NO: 42) 1 attgaggagc agaaggagta ggggtgcgggg
gaggaggagg agcgccttta gtgctgcagc 61 agctgctgct ctgattggcc cgggtggttca gctgcttccc
tggaacaaaa ggtcaaagt 121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa
gaagtttgga 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggg 241
gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttact gtacagacga 301 ccaaactgc
cgtggaagag acagttgtga gattcccttg caaattaca tacgagaatg 361 gcttgtagaa tcatgcctct
gcaaagactc ttagaaagaa ccatccgatc agctgtagaa 421 caacatcttt ttgatgttaa taactctgga
gggtcaaagt cagaggactc agaactgtga 481 aactatcag catcttctgc cacatctgcc agacagcgcc
gccgccagtc caaggagcag 541 gatgaagttc gacatgggag agacaaggga cttatcaaca aagaaaatac
tccttctggg 601 ttcaaccacc ttgatgattg tattttgaat actcaggaag tcgaaaagggt acacaaaaat 661
acttttggtt gtgctggaga aaggagcaag cctaaacgtc agaaatccag tactaaactt 721 tctgagcttc
atgacaatca ggacggctct gtgaatatgg aaagtctcaa ttccacacga 781 tctcatgaga gaactggacc
tgatgatttt gaatggatgt ctgatgaaag gaaaggaaat 841 gaaaaagatg gtggacacac tcagcatttt
gagagcccca caatgaagat ccaggagcat 901 ccagcctat ctgacaccaa acagcagaga aatcaagatg
ccggtgacca ggaggagagc 961 ttgtctccg aagtgtccca gtcggacctg actgcattgt gtgatgaaa
gaactgggaa 1021 gagcctatcc ctgcttctc ctctggcag cgggagaaca gtgactctga tgaagccac 1081
ctctcgccgc aggctgggag cctgatccgt cagctgctgg acgaagacag cgaccccatg 1141 ctctctctc
ggttctacgc ttatgggag agcaggcaat acctggatga cacagaagt 1201 cctcctccc cacaaactc
ccattcttc atgaggcggc gaagctctc tctgggggtcc 1261 tatgatgatg agcaagagga cctgacacct
gcccagctca cacgaaggat tcagagcctt 1321 aaaaagaaga tccggaagt tgaagataga ttgaagaag
agaagaagta cagaccttc 1381 cacagtga aagcagcaa tccggagggt ctgaaatgga caaatgacct
tgccaaattc 1441 cgagagacaac ttaaagaatc aaactaaag atatctgaag aggacctaac tcccaggatg 1501
cggcagcgaa gcaacacact cccaagagt ttggttccc aacttgagaa agaagatgag 1561 aagaagcaag
agctggtgga taaagcaata aagccagtg ttgaagccac attggaatct 1621 attcagagga agtccagga
gaagcgagcg gaaagcagcc gccctgagga cattaaggat 1681 atgaccaaag accagattgc taatgagaa
gtggctctgc agaaagctct gttatattat 1741 gaaagcattc atggacggcc ggtaacaaag aacgaacggc
aggtgatgaa gccactatac 1801 gacaggtacc ggctggtcaa acagatcctc tcccagagta acaccatacc
catcattggt 1861 tccccctcca gcaagcggag aagccctttg ctgcagcaa ttatcgaggg cgaaactgct 1921
tccttctca aggagataaa ggaagaagag gaggggtcag aagacgatag caatgtgaag 1981 ccagacttca
tggtcactct gaaaaccgat ttcagtgcac gatgcttct ggaccaattc 2041 gaagatgacg ctgatggatt
tatttccca atggatgata aaataccatc aaatgcagc 2101 caggacacag ggctttcaa tctccatgct
gcctcaatac ctgaactct ggaacacctc 2161 caggaaatga gagaagaaaa gaaaaggatt cgaaagaaac
ttcgggattt tgaagacaac 2221 ttttcagac agaattggaag aaatgtccag aaggaagacc gcactcctat
ggctgaagaa 2281 tacagtgaat ataagcat aaaggcgaaa ctgaggctcc tggaggtgct catcagcaag

2341 agagacactg attccaagtc catgtgaggg gcatggccaa gcacaggggg ctggcagctg 2401
 cgggtgagagt ttactgtccc cagagaaagt gcagctctgg aaggcagcct tggggctggc 2461 cctgcaaagc
 atgcagccct tctgcctcta gaccatttgg catcggctcc tgtttccatt 2521 gcctgcctta gaaactggct
 ggaagaagac aatgtgacct gacttaggca ttttgaatt 2581 ggaaagtcaa gactgcagta tgtgcacatg
 cgcacgcgca tgcacgcaca cacacacaca 2641 gtagtggagc tttctaaca ctagcagaga ttaatcacta
 cattagacaa cactcatcta 2701 cagagaatat acactgttct tccttgata actgagaaac aagagacat
 tctctgtcta 2761 actgtgataa aaacaagctc aggactttat tctatagagc aaacttgctg tggagggcca 2821
 tgctctcctt ggaccagtt aactgcaaac gtgcattgga gccctatttg ctgccgctgc 2881 cattctagt
 acctttccac agagctgcgc cttcctcagc tgtgtgaaag gttttccct 2941 tcagccctca ggtagatgga
 agctgcatct gccacgatg gcagtgcagt catcatcttc 3001 aggatgttct ttcaggactt cctcagctga
 caaggaattt tggtcctgc ctaggaccgg 3061 gtcactgca gaggacagag agatggtaag cagctgtatg
 aatgctgatt ttaaaccag 3121 gtcatgggag aagagcctgg agattcttct ctgaacactg actgcactta
 ccagtctgat 3181 tttatcgta aacaccaagc caggctagca tgctcatggc aatctgtttg gggctgtttt 3241
 gttgtggcac tagccaaaca taaaggggct taagtcagcc tgcatacaga ggatcgggga 3301 gagaaggggc
 ctgtgttctc agcctcctga gtacttacca gagttaatt ttttaaaaa 3361 aaatctgcac taaaatcccc
 aaactgacag gtaaatgtag ccctcagagc tcagcccaag 3421 gcagaatcta aatcacacta tttcgagat
 catgtataaa aagaaaaaaa agaagtcag 3481 ctgtgtggcc aattataatt ttttcaaag acttgtcac
 aaaactgtct atattagaca 3541 ttttgaggg accaggaaat gtaagacacc aaatcctcca tctcttcagt
 gtgcctgatg 3601 tcacctcatg atttgcgtt actttttta ctctgcgcc aaggacagt ggttctgtgt 3661
 ccaccttgt gcttgcgag gccgagccca ggcactgtct cgctgccac ggctgaccag 3721 agaaggtgt
 tcaggagctc tgccttagac gacgtgttac agtatgaaca cacagcagag 3781 gcaccctcgt atgtttgaa
 agttgccttc tgaaagggca cagttttaag gaaaagaaaa 3841 agaatgtaa actatactga cccgtttca
 gttttaaagg gtcgtgagaa actggctggt 3901 ccaatgggat ttacagcaac attttccatt gctgaagtga
 ggtagcagct ctctctgtc 3961 agctgaatgt taaggatggg gaaaaagaat gcctttaagt ttgctctaa
 tcgatggaa 4021 gcttgagcta tgtgttgaa gtgccctggt ttaatccat acacaaagac ggtacataat 4081
 cctacagggt taaatgtaca taaaaatata gtttgaatt ctttgcctc ctgtttacat 4141 tgcagattgc
 tataatttca aggagtgaaga ttataataa aatgatgcac ttaggatgt 4201 ttctatttt tgaaatctga
 acatgaatca ttcacatgac caaaaattgt gttttttta 4261 aaatacatgt ctagtctgtc cttaatagc
 tctctaaat aagctatgat attaatcaga 4321 tcattaccag ttagcttta aagcacattt gttaagact
 atgttttgg aaaaatacgc 4381 tacagaattt tttttaagc taaaataaa tgagatgcta ctaattgttt tggaatctgt
 4441 tgtttctgcc aaaggtaat taactaaaga tttattcagg aatccccatt tgaatttgta 4501 tgattcaata
 aaagaaaaca ccaagtaagt tatataaat aaattgtgta tgagatgtg 4561 tgttttcctt tgaatttcc
 actaactaac taactaactt atattcttca tggaatggag 4621 ccagaagaa atgagaggaa gcccttttca
 cactagatct tatttgaaga aatgtttgt 4681 agtcagtcag tcagtgtt ctggctctgc cgaggagat
 gtgttccca gcaaccatt 4741 ctgcagcca gaatctcaag gcactagagg cggtgtctta attaattggc
 ttcacaaaga 4801 caaatgctc tggactggga ttttcttt gctgtgttg gaatatgtt ttattaatta 4861
 gcacatgcc aaaaaataaa tgtcaagagt tattcataa gtgtaagtaa acttaagaat 4921 taaagagtgc
 agactataa tttca

(195) In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252509.1, transcript variant 5):

(196) TABLE-US-00028 (SEQ ID NO: 43) 1 maceimplqr lletirsav eqhlfdvnns
 ggqssedses gtlssassats arqrrrske 61 qdevrhgrdk glinkentps gfnhlddcil ntqevkvkh
 ntfgcagers kpkqrksstk 121 lselhdnqdg lvnmeslnst rshertgpdd fewmsderkg nekdgghtqh
 fesptmkique 181 hpslsdtkqq rinqdagdee sfvsevpqsd ltalcdeknw eepipafssw qrensdidea 241
 hlspqagrli rqlldedsdp mlsprfyayg qsrqylldte vppsppnshs fmrrrsslsg 301 syddeqedlt
 paqltrriqs lkkkirkfed rfeekkyrp shsdkaanpe vlkwtnlak 361 frqlkeskl kiseedltpr
 mrqrsntlpk sfgsqleked ekkqelvdka ikpsveatle 421 siqrklqekr aessrpelik dmtkdqiane
 kvalqkally yesihgrpvt knerqvmkpl 481 ydryrlvkqi lsrantipii gspsskrsp llqpiiaget
 asffkeikee eegseddsnv 541 kpdfmvtlkt dfsarcfldq feddadgfs pmddkipskc sqdtglsnlh

aasipelleh 601 lqemreekkr irkklrdfed nffrqngrnv qkedrtprmae eyseykhika klrlllevlis 661
krtdtdsksm

(197) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_004415.3, transcript variant 1):

(198) TABLE-US-00029 (SEQ ID NO: 44) 1 aagaaaccgg ccaggtgtgg cctagggcgc
cagtgccagc ggggaggaga ctgctccgc 61 cgccgaccaa caccaacacc cagctccgac gcagctcctc
tgcgccttg ccgcctccg 121 agccacagct ttctcccg cctgcccc ggcccgtcgc cgtctccgcg
ctcgcagcgg 181 cctcgggagg gccaggtag cgagcagcga cctcgcgagc ctccgcact cccgcccggg 241
tccccggcgc tccgcctatc ctggcccc tccgtttct ccgcgccggc ccgcctcgct 301 tatgcctcgg
cgctgagccg ctctccgat tgcccgcga catgagctgc aacggaggct 361 cccaccgcgc gatcaacact
ctgggccgca tgatccgcgc cgagtctggc ccggacctgc 421 gctacgaggt gaccagcggc ggccgggggca
ccagcaggat gtactattct cggcgccggc 481 tgataccga ccagaactcg gacggctact gtcaaacagg
cacgatgtcc aggcaccaga 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg
cgagcagagc 601 tcactgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat 661
tggatgagtg tttgcccag gccaatgacc aaatggaaat cctcgacagc ttgatcagag 721 agatgcggca
gatgggccag ccctgtgatg ctaccagaa aaggcttct cagctccaag 781 agcaaatgcg agcccttat
aaagccatca gtgtccctcg agtccgcagg gccagctcca 841 aggggtgggg aggtactact tgcagagtg
gctctggctg ggatgagttc accaaacatg 901 tcaccagtga atgttgggg tggatgaggc agcaaagggc
ggagatggac atggtggcct 961 ggggtgtgga cctggcctca gtggagcagc acattaacag ccaccggggc
atccacaact 1021 ccactggcga ctactgctg cagctggaca aaatcaaagc cgacctgcgc gagaaatctg 1081
cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtcctt gagaggatgg 1141 atcacctgcg
acagctgcag aacatcattc aggccacgtc caggagatc atgtggatca 1201 atgactgcga ggaggaggag
ctgctgtacg actggagcga caagaacacc aacatcgctc 1261 agaaacagga ggccttctcc atacgcatga
gtcaactgga agttaaagaa aaagagctca 1321 ataagctgaa acaagaaagt gaccaactg tcctcaatca
gcatccagct tcagacaaaa 1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag
atcaccaagt 1441 gcattgatgt tcactgaaa gaaaatgctg cctacttca gtttttgaa gaggcgcagt 1501
ctactgaagc atacctgaag gggctccagg actccatcag gaagaagtac ccctgcgaca 1561 agaacatgcc
cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga 1621 aaatccttga atacaagcgt
cagggtgcaga acttggtaaa caagtctaag aagattgtac 1681 agctgaagcc tcgtaacca gactacagaa
gcaataaacc cattattctc agagctctc 1741 gtgactacaa acaagatcag aaaatcgtc ataaggggga
tgagtgtatc ctgaaggaca 1801 acaacgagcg cagcaagtgg tacgtgacgg gcccgggagg cgttgacatg
ctgttccct 1861 ctgtggggct gatcatccct cctccgaacc cactggcgt ggacctctc tgcaagattg 1921
agcagtacta cgaagccatc ttggctctgt ggaaccagct ctacatcaac atgaagagcc 1981 tgggtcctg
gcactactgc atgattgaca tagagaagat cagggccatg acaatcgcca 2041 agctgaaaac aatgcggcag
gaagattaca tgaagacgat agccgacctt gagttacatt 2101 accaagagtt catcagaaat agccaaggct
cagagatgtt tggagatgat gacaagcgga 2161 aaatacagtc tcagttcacc gatgccaga agcattacca
gaccctggtc attcagctcc 2221 ctggctatcc ccagcaccag acagtacca caactgaaat cactcatcat
ggaacctgcc 2281 aagatgtcaa ccataataaa gtaattgaaa ccaacagaga aaatgacaag caagaaacat 2341
ggatgctgat ggagctgcag aagattcgca ggcagataga gcactgcgag ggcaggatga 2401 ctctcaaaaa
cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta 2461 acgagcttaa gattgtgcag
aatgattcac aagcaattgc tgaggtctc aaccagctta 2521 aagatatgct tgccaacttc agaggttctg
aaaagtactg ctatttacag aatgaagtat 2581 ttggactatt tcagaaactg gaaaatatca atggtgttac
agatggctac ttaaatagct 2641 tatgcacagt aagggcactg ctccaggcta ttctcaaac agaagacatg
ttaaagggtt 2701 atgaagccag gctcactgag gaggaaactg tctgctgga cctggataaa gtggaagctt 2761
accgctgtgg actgaagaaa ataaaaatg acttgaactt gaagaagtcg ttgtggcca 2821 ctatgaagac
agaactacag aaagcccagc agatccactc tcagacttca cagcagtatc 2881 cactttatga tctggacttg
ggcaagtctg gtgaaaaagt cacacagctg acagaccgct 2941 ggcaaaggat agataaacag atcgacttta
ggttatggga cctggagaaa caaatcaagc 3001 aattgaggaa ttatcgtgat aactatcagg ctttctgcaa
gtggctctat gatgctaaac 3061 gccgccagga ttcttagaa tccatgaaat ttggagattc caacacagtc

atgcggtttt 3121 tgaatgagca gaagaacttg cacagtgaaa tatctggcaa acgagacaaa tcagaggaag 3181
 tacaaaaaat tgctgaactt tgcgccaatt caattaagga ttatgagctc cagctggcct 3241 catacacctc
 aggactggaa actctgctga acatacctat caagaggacc atgattcagt 3301 ccccttctgg ggtgattctg
 caagaggctg cagatgttca tgctcggtag attgaactac 3361 ttacaagatc tggagactat tacaggttct
 taagtgagat gctgaagagt ttggaagatc 3421 tgaagctgaa aaataccaag atcgaagttt tggaagagga
 gctcagactg gcccagagatg 3481 ccaactcgga aaactgtaat aagaacaaat tcctggatca gaacctgcag
 aaataccagg 3541 cagagtgttc ccagttcaaa gcgaagcttg cgagcctgga ggagctgaag agacaggctg
 3601 agctggatgg gaagtcggct aagcaaaatc tagacaagtg ctacggccaa ataaaagaac 3661
 tcaatgagaa gatcacccga ctgacttatg agattgaaga tgaaaagaga agaagaaaat 3721 ctgtggaaga
 cagatttgac caacagaaga atgactatga ccaactgcag aaagcaaggc 3781 aatgtgaaaa ggagaacctt
 ggttggcaga aattagagtc tgagaaagcc atcaaggaga 3841 aggagtacga gattgaaagg ttgagggttc
 tactgcagga agaaggcacc cggaagagag 3901 aatatgaaaa tgagctggca aaggtaagaa accactataa
 tgaggagatg agtaatttaa 3961 ggaacaagta tgaaacagag attaacatta cgaagaccac catcaaggag
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 4501 ataccagtgg ctaccgggct cagatagaca atctacccg agaaaacagg agcttatctg 4561
 aagaaataaa gaggctgaag aacactctaa cccagaccac agagaatctc aggagggttg 4621 aagaagacat
 ccaacagcaa aaggccactg gctctgaggt gtctcagagg aaacagcagc 4681 tggaggttga gctgagacaa
 gtcactcaga tgcgaacaga ggagagcgta agatataagc 4741 aatctcttga tgatgctgcc aaaaccatcc
 aggataaaaa caaggagata gaaagggttaa 4801 aacaactgat cgacaaagaa acaaatgacc ggaaatgcct
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 ccagggtgaa acagcgctg gagtgtgaga aacagcaaat tcagaatgac ctgaatcagt 5941 ggaagactca
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 agtcttagga gtgagatcga aagactccaa gcagagatca 6061 agagaattga agagagggtgc aggcgtaagc
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 gttgctggaa gcccaagcag ctactggctt tatagtggat cctgttagca 7261 acttgagggt accagtggag
 gaagcctaca agagaggtct ggtgggcatt gagtcaaag 7321 agaagctcct gtctgcagaa cgagctgtca
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 aaagggccac ggtattcgct 7441 tattagaagc acagatcgca accggggggga tcattgacct aaaggagagc
 catcgtttac 7501 cagttgacat agcatataag aggggctatt tcaatgagga actcagttag attctctcag 7561
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 agaaagatgc attaaggatg aggaaacagg gctctgtctt ctgcctctga 7681 aagaaaagaa gaaacagggtg
 cagacatcac aaaagaatac cctcaggaag cgtagagtgg 7741 tcatagttaga cccagaaacc aataaagaaa
 tgtctgttca ggaggcctac aagaagggcc 7801 taattgatta tgaaccttc aaagaactgt gtgagcagga
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 gggttcatag atggccgcgc cgcacagagg ctgcaagaca 8641 ccagcagcta tgccaaaatc ctgacctgcc
 ccaaaacca attaaaaata tcctataagg 8701 atgccataaa tcgctccatg gtagaagata tcaactgggt
 gcgcttctg gaagccgcct 8761 ccgtgtcgtc caagggctta cccagccctt acaacatgtc ttcggctccg
 gggctccgct 8821 ccggctcccg ctcgggatct cgctccggat ctgctccgg gtcccgcagt gggctccgga
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 gcactagtag tcagttggga gtggttgcta taccttgact tcatttatat 9001 gaatttccac ttattaaat
 aatagaaaag aaaatcccg tgcttgcatg agagtgatg 9061 gacattctat gcttacagaa aatataacca
 tgattgaaat caaatagtaa aggtgttct 9121 ggcttttat ctcttagct catctaaat aagcagtaca
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 ggagataaaa attaaatgga tcaactgat ttagtcatt ctgcttctca tctaaatatt 9361 tccatattct
 gtattaggag aaaattaccc tcccagcacc agccccctc tcaaaccctc 9421 aacccaaaac caagcatttt
 ggaatgagtc tccttagtt tcagagtgtg gattgtataa 9481 cccatatact ctgcagta cttgtttgtg
 ttggtattaa ttgactgtg catgacagcg 9541 gcaatcttt ctttgggtcaa agttttctgt ttattttgct tgtcatattc
 gatgtacttt 9601 aaggtgtctt tatgaagttt gctattctgg caataaactt ttagactttt gaagtgtttg 9661
 tgttttaatt taatatgtt ataagcatgt ataaacatt agcatatttt tatcataggt 9721 ctaaaaatat ttgtttacta
 aatacctgtg aagaaatacc attaaaaaac tatttggttc 9781 tgaattctta ctagaaaaaa aa

(199) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_004406.2, transcript variant 1):

(200) TABLE-US-00030 (SEQ ID NO: 45) 1 mscnggshpr intlgrmira esgpdlryev

tsggggtsrm yysrrgvitd qnsdgycqtg 61 tmsrhqnqnt iqellqncsd clmraelivq pelkygdgiq
ltrsreldec faqandqmei 121 ldsliremrrq mgqpccdayqk rllqlqeqmr alykaisvpr vrrasskggg
gytcqsgsgw 181 deftkhvtse clgwmrqrra emdmvawgvd lasveqhins hrgihnsigd yrwqldkika
241 dlreksaiyq leeeyenllk asfermdhlr qlqniiqats reimwindce eeellydwsd 301
kntniaqkqe afsirmsqle vkekelnlk qesdqlvinq hpasdkieay mdtlqtqsw 361 ilqitkciv
hlkenaayfq ffeeaqstea ylkglqdsir kkypcdknmp lqhleqike 421 lekerekile ykrqvqnln
kskkivqlkp rnpdyrsnpk iilralcdyk qdqkivhkgd 481 ecilkdnnr skwyvtgpgg vdmlypsvg
iipppnplav dlsckieqyy eailalwnql 541 yinmkslvsw hycmidieki ramtiaklt mrqedymkti
adelhyqef irnsgsemf 601 gdddkrkiqs qftdaqkhyq tlvqlpgyp qhqtvttei thhgctqdn
hmkvietnre 661 ndkqetwmlm elqkirrqie hcegrmtlkn lpladqgssh hitvkinelk svqndsggia 721
evlnqlkdml anfrgsekyc ylnqevfglf qkleningvt dgylslctv rallqailqt 781 edmlkvyear
lteetvclld ldkveayrcg lkkikndlnl kksllatmkt elqkaqqihs 841 qtsqqyplyd ldlgkfgek
tqltdrwqri dkqidfrlwd lekqikqlrn yrdnyqafck 901 wlydakrrqd slesmkfgds ntvmrflneq
knlhseisgk rdkseevqki aelcansikd 961 yelqlasyts gletlnipi krtmiqspsg vilqeaadvh
aryielltrs gdyrrflsem 1021 lksledlklk ntkievlee lrlardanse ncnknkfldq nlqkyqaecs
qfkaklasle 1081 elkrqaeldg ksakqnlkdc ygqikelnek itrlyeied ekrrrksved rfdqqkndyd 1141
qlqkarqcek enlgwqkles ekaikekeye ierlrllqe egtrkreyen elakvrnhyn 1201 eemsnlrnky
eteinitktt ikeismqked dsknlrnqld rlsrenrdlk deivrlndsi 1261 lqateqrra eenalqqkac
gseimqkkqh leielkqvmq qrsednarhk qsleeaakti 1321 qdknkeierl kaefqeeakr rweyenelsk
vrnnydeei slknqfetei nitktihql 1381 tmqkeedtsq yraqidnlr enrlseeik rlnltlqt
enlrveedi qqqkatgsev 1441 srrkqqleve lrqvtqrte esvryksld daaktiqdn keierlkqli
dketndrkcl 1501 edenarlqry qydlqkanss atetinklkv qeqeltrli dyervsgert vkddqtrfq 1561
nslkelqlqk qkveelnrl krtasedsck rkkleeleg mrrslkeqai kitnitqle 1621 qasivkkre
ddlrqqravl dghlrekqr qeelrlsse vealrrllq eqesvkqahl 1681 nehfqkaie dksrslnesk
ieierlqsl enitkehml eeelnrlle yddlrrgrse 1741 adsdknatit elrsqlqisn nrtlelqgli
ndlqrerenl rqeiefqkq aleasnriq 1801 sknqctqvq eresllvik vleqdkarlq rledelnrak
stleaetrvk qrlecekqqi 1861 qndlnqwtq ysreeairk ieserekser eknlrseie rlqaeikrie
ercrrkleds 1921 tretqsqlet ersrygreid klrrpygsh retqtecewt vdtkslvfdg lrkkvtamql 1981
yecqlidkt ldkllkgkks veevaseiqp flrgagsiag asapkekys lveakrkli 2041 spestmille
aqaatggiid phrnekltd saiardlidf ddrqqiyaa kaitgfddpf 2101 sgktvsvsea ikknlidret
gmrleaqia sggvdpvns vflpkdvala rgldirdlyr 2161 slndprdsqk nfvdpvttkk vsyvqlkerc
riephtgll lsvqkrmsf qgirqpvtvt 2221 elvdsgilrp stvnelesgq isydevgeri kdfllqgssci
agiynettkq klgiyeamki 2281 glvrpgtale lleaqaatgf ivdpvsnrl pveaykrgl vgiefkekll
saeravtgin 2341 dpetgniisl fqamnkelie kghgirlea qiatggiidp keshrlpvd aykrqynee 2401
lseilsdpsd dtkgffdpnt eenltylqlk ercikedetg lcllplkek kqvqtsqknt 2461 lrkrrvivd
petnkemsvq eaykkglidy etfelceqe ceweititg sdgstrvvlv 2521 drktgsqydi qdaidkglvd
rkffdqyrsg slstqfadm islkngvgt ssmgsgvsdd 2581 vfsssrhesv skistissv nltirssfs
dtleesspia aifdenlek isitegiern 2641 ivdsitqrl leaqaatggi ihpttgqkls lqdaysqgi
dqdmatrikp aqkafifeg 2701 vkgkkkmsaa eavkekwlpy eagqrflefq yltggldpe vhgristea
irkgidgra 2761 aqlqdtssy akiltcpkt lkisykdain rsmveditgl rleasvss kglpsynms 2821
sapgsrsgsr sgsrsgsrsg srsgrsrsf datgnssysy sysfsssig h

(201) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001008844.2, transcript variant 2):

(202) TABLE-US-00031 (SEQ ID NO: 19) 1 aagaaaccgg ccaggtgtgg cctaggcgcc
cagtgccagc ggggaggaga ctgcctccgc 61 cgccgaccaa caccaacacc cagctccgac gcagctcctc
tgcgcccttg ccgccctccg 121 agccacagct ttctcccgcc tctgcccc ggcccgctgc cgtctccgcg
ctgcagcgg 181 cctcgggagg gccaggtag cgagcagcga cctcgcgagc ctccgcact cccgcccgt 241
tccccggccg tccgcctatc cttggcccc tccgcttct ccgcgcggc ccgcctcgct 301 tatgcctcgg
cgctgagccg ctctccgat tgccgcccga catgagctgc aacggaggct 361 ccacccgcg gatcaacact

ctgggcccga tgaatccgcgc cgagtctggc ccggacctgc 421 gctacgaggt gaccagcggc ggccgggggca
ccagcaggat gtactattct cggcgcggcg 481 tgatcaccga ccagaactcg gacggctact gtcaaacggg
cacgatgtcc aggcaccaga 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg
cgagcagagc 601 tcactgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat 661
tggatgagtg ttttcccag gccaatgacc aaatggaaat cctcgacagc ttgatcagag 721 agatgcggca
gatgggcccag ccctgtgatg cttaccagaa aaggcttctt cagctccaag 781 agcaaagtcg agccctttat
aaagccatca gtgtccctcg agtccgcagg gccagctcca 841 aggggtggtgg aggctacact tgtcagagtg
gctctggctg ggatgagttc accaaacatg 901 tcaccagtga atgtttgggg tggatgaggc agcaaagggc
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cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtccttt gagaggatgg 1141 atcacctgcg
acagctgcag aacatcattc aggccacgtc caggagatc atgtggatca 1201 atgactgcga ggaggaggag
ctgctgtacg actggagcga caagaacacc aacatcgctc 1261 agaaacagga ggccttctcc atacgcatga
gtcaactgga agttaaagaa aaagagctca 1321 ataagctgaa acaagaaagt gaccaacttg tcctaatca
gcatccagct tcagacaaaa 1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag
atcaccaagt 1441 gcattgatgt tcactgaaa gaaaatgctg cctactttca gtttttgaa gaggcgcagt 1501
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cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga 1621 aaatccttga atacaagcgt
caggtgcaga acttggtaaa caagtctaag aagattgtac 1681 agctgaagcc tcgtaacca gactacagaa
gcaataaacc cattattctc agagctctct 1741 gtgactacaa acaagatcag aaaatcgtgc ataaggggga
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cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta 2461 acgagcttaa gagtgtgcag
aatgattcac aagcaattgc tgaggttctc aaccagctta 2521 aagatatgct tgccaacttc agaggttctg
aaaagtactg ctatttacag aatgaagtat 2581 ttgactatt tcagaaactg gaaaatatca atgggtgtac
agatggctac ttaaatagct 2641 tatgcacagt aagggcactg ctccaggcta ttctccaaac agaagacatg
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gagagagaag aacagtctta ggagttagat cgaaagactc caagcagaga 4261 tcaagagaat tgaagagagg
tgcaggcgta agctggagga ttctaccagg gagacacagt 4321 cacagttaga aacagaacgc tcccgatatac
agagggagat tgataaactc agacagcgc 4381 catatgggtc ccatcgagag acccagactg agtgtgagt
gaccgttgac acctccaagc 4441 tgggtgttga tgggctgagg aagaaggtga cagcaatgca gctctatgag
tgtcagctga 4501 tcgacaaaac aacctggac aaactattga aggggaagaa gtcagtggaa gaagttgctt 4561
ctgaaatcca gccattcctt cggggtgcag gatctatgc tggagcatct gcttctcta 4621 aggaaaaata
ctctttgta gaggccaaga gaaagaaatt aatcagccca gaatccacag 4681 tcatgcttct ggaggcccag
gcagctacag gtggtataat tgatcccat cggaatgaga 4741 agctgactgt cgacagtgc atagctcggg
acctcattga cttcgatgac cgtcagcaga 4801 tatatgcagc agaaaaagct atcactgggt ttgatgatcc
attttcaggc aagacagtat 4861 ctgtttcaga agccatcaag aaaaatttga ttgatagaga aaccggaatg
cgctgctgg 4921 aagcccagat tgcttcaggg ggtgtagtag acctgtgaa cagtgtctt ttgcaaaaag 4981
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gaacgagctg tcaactggta taatgatcct gaaacaggaa 5581 acatcatctc ttgttcaa gccatgaata
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cccaaaggag agccatcgtt 5701 taccagtga catagcatat aagaggggct atttcaatga ggaactcagt
gagatttct 5761 cagatccaag tgatgatacc aaaggatttt ttgaccccaa cactgaagaa aatcttacct 5821
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agtaagtaag attccacca tatccagcgt caggaattta accataagga 6361 gcagctcttt ttcagacacc
ctggaagaat cgagcccat tgcagccatc ttgacacag 6421 aaaacctgga gaaaatctcc attacagaag
gtatagagcg gggcatcgtt gacagcatca 6481 cgggtcagag gcttctggag gctcaggcct gcacaggtgg
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gacatggcca 6601 ccaggctgaa gcctgctcag aaagccttca taggcttcga ggggtggaag ggaaagaaga
6661 agatgtcagc agcagaggca gtgaaagaaa aatggctccc gtatgaggct ggccagcgt 6721
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atcctgacct gccccaaaac caaattaaat atactctata 6901 aggatgccat aaatcgctcc atggtagaag
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gtcttcggct ccgggggtccc 7021 gctccggctc ccgctcggga tctcgtccg gatctcgtc cgggtcccgc
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cactttatta aataatagaa aagaaaatcc cgggtgctgc agtagagtga 7261 taggacattc tatgcttaca
gaaaatatag ccatgattga aatcaaatag taaaggctgt 7321 tctggctttt tatcttcta gctcatcta

aataagcagt acacttggat gcagtgcgtc 7381 tgaagtgcta atcagttgta acaatagcac aaatcgaact
taggatttgt ttcttctctt 7441 ctgtgtttcg attttgatc aattctttaa ttttgaagc ctataatata gttttctatt 7501
cttgagata aaaattaaat ggatcactga tattttagtc attctgcttc tcatctaaat 7561 atttccatat
tctgtattag gagaaaatta ccctcccagc accagccccc ctctcaaacc 7621 cccaacccaa aaccaagcat
tttgaatga gtctccttta gtttcagagt gtggattgta 7681 taacccatat actcttcgat gtacttggtt
ggtttggat taatttgact gtgcatgaca 7741 gcggcaatct tttctttgtt caaagttttc tgtttatttt gcttgcata
ttcgaatgac 7801 tttaaggtgt ctttatgaag tttgctattc tggcaataaa ctttagact tttgaagtgt 7861
ttgtgttta attaatatg ttataagca tgtataaaca ttagcatat tttatcata 7921 ggtctaaaaa tatttgttta
ctaaatacct gtgaagaaat accattaaaa aactatttgg 7981 ttctgaattc ttactagaaa aaaaa

(203) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001008844.1, transcript variant 2):

(204) TABLE-US-00032 (SEQ ID NO: 20) 1 mscnggshpr intlgrmira esgpdlyev
tsggggtsrm yysrrgvitd qnsdgyctg 61 tmsrhqnqnt iqellqncsd clmraelivq pelkygdgiq
ltrsreldec faqandqmei 121 ldsliremrq mgqpcdayqk rllqlqeqmr alykaisvpr vrrasskggg
gytcqsgsgw 181 deftkhvtse clgwmrqra emdmvawgv lasveqhins hrgihnsigd yrwqldkika
241 dlreksaiyq leeyenllk asfermdhlr qlqniiqats reimwindce eeellydwsd 301
kntniaqkqe afsirmsqle vkekelnlk qesdqlvinq hpasdkieay mdtlqtqsw 361 ilqitkcidv
hlkenaayfq ffeaaqstea ylkglqdsir kkypcdknmp lqhleeqike 421 lekerekile ykrqvqnln
kskivqlkp rnpdyrsnpk iilralcdyk qdqkivhkgd 481 ecilkdnnr skwyvtgpgg vdmlypsvgi
iipppnplav dlsckieyy eailalwnql 541 yinmkslvsw hycmidieki ramtiaklkt mrqedymkti
adlhyqef irnsggsemf 601 gdddkrkiqs qftdaqkhyq tivilpgyp qhqtvtttei thhgctqdn
hkvietnre 661 ndkqetwmlm elqkirrqie hcegrmtlkn lpladqgssh hitvkinelk svqndsgaia 721
evinqlkdml anfrgsekyc ylnqvfglf qkleningvt dgylnslctv rallqailqt 781 edmlkvyear
lteetvclld ldkveayrcg lkkikndlnl kksllatmkt elqkaqqihs 841 qtsqqyplyd ldlgkfgek
tqltdrwqri dkqidfrlwd lekqikqlrn yrdnyqafck 901 wlydakrrqd slesmkfgds ntmrflneq
knlhseisgk rdkseevqki aelcansikd 961 yelqlasyts gletlnipi krtmiqspsg vilqeaadvh
aryelltrs gdyrrflsem 1021 lksledlklk ntkievlee lrlardanse ncnknkfldq nlqkyqaecs
qfkaklasle 1081 elkrqaeldg ksakqnlkdc ygqikelnek itrlyeied ekrkrsved rfdqqkndyd 1141
qlqkarqcek enlgwqkles ekaikekeye ierlrllqe egtrkreyen elakasnriq 1201 esknqctqv
qeresllvki kvleqdkarl qrledelnra kstleatrv kqrlecekq 1261 iqndlnqwt qysrkeear
kieserekse reknsrsei erlqaeikri eercrrkled 1321 stretsqle tersrygrei dklrqrpygs
hretqtecew tvdtsklvfd glrkvtamq 1381 lyecqlidkt tldkllkgkk sveevaseiq pflrgagsia
gasaspkey slveakrkl 1441 ispestvml eaqaatggii dphrneklv dsaiardlid fddrqqiya
ekaitgfddp 1501 fsgktvsvse aiknldre tgmrlleaqi asggvdpvn svflpkdval arglirdly 1561
rslndprdsq knfvdptkk kvsyvqlker criephtgll llsvqkrms fqgirqpvtv 1621 telvdsgilr
pstvnelesg qisydevger ikdflqgssc iagiynettk qlgiyeamk 1681 iglvrpgtal elleaqaatg
fivdpvsnlr lpveeaykrg lvgiefkekl lsaeravtgy 1741 ndpetgniis lfqamnkeli ekghgirle
aqiatggiid pkeshrlpvd iaykrgyfe 1801 elseilsdps ddtkgffdpn teenitylql kercikdeet
glcllplkek kkqvqtsqkn 1861 tlrkrrviv dpetnkemsv qeaykkgld yetfkelceq ecweeitit
gsdgstrvvl 1921 vdrktgsqyd iqdaidkglv drkffdqys gslstqfad mislknvgvt ssmgsgvsv
1981 dvfsssrhes vskistissv rnltrsssf sdtleesspi aaifdenle kisitegier 2041 givdsitgqr
lleaqactgg iihpttgqkl slqdavsqgv idqdmtrlk paqkafigfe 2101 gvkgkkkmsa aeavkekwp
yeagqrflef qyltgglvdp evhgristee airkgfidgr 2161 aaqlqdtss yakiltcpkt klkisykdai
nrsmveditg lrlleaasvs skglpspynm 2221 ssapgsrgs rsgsrgsrs gsrsgsrrgs fdatgnssys
ysysfsssi gh

(205) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001319034.1, transcript variant 3):

(206) TABLE-US-00033 (SEQ ID NO: 46) 1 aagaaaccgg ccaggtgtgg cctaggcgcc

cagtgccagc ggggaggaga ctgctccgc 61 cgccgacca caaccaacacc cagctccgac gcagctcctc
tgcgcccttg ccgccctccg 121 agccacagct ttctcccg cctgcccc ggcccgctgc cgtctccgcg
ctcgagcgg 181 cctcgggagg gccaggtag cgagcagcga cctcgcgagc ctccgcact cccgccccgt 241
tccccggccg tccgcctatc ctggcccc tccgtttct ccgcgccggc ccgctcgct 301 tatgcctcgg
cgctgagccg ctctcccgat tgcccgcga catgagctgc aacggaggct 361 cccaccgcg gatcaacact
ctgggcccga tgatccgcgc cgagtctggc ccggacctgc 421 gctacaggt gaccagcggc ggcgggggca
ccagcaggat gtactattct cggcgccggc 481 tgatcaccga ccagaactcg gacggctact gtcaaacgg
cacgatgtcc aggcaccaga 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg
cgagcagagc 601 tcactgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat 661
tggatgagt tttgcccag gccaatgacc aatggaaat cctcgacagc ttgatcagag 721 agatcgggca
gatgggccag ccctgtgatg ctaccagaa aaggcttct cagctccaag 781 agcaaatgcg agcccttat
aaagccatca gtgtccctcg agtcgcagc gccagctcca 841 aggggtggtg aggtacact tgcagagt
gctctggctg ggatgagttc accaaacatg 901 tcaccagtga atgttgggg tggatgaggc agcaaagggc
ggagatggac atgggtggcct 961 ggggtgtgga cctggcctca gtggagcagc acattaacag ccaccggggc
atccacaact 1021 ccacggcgca ctacgctgg cagctggaca aaatcaaagc cgacctgcgc gagaaatctg 1081
cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtcctt gagaggatgg 1141 atcacctgcg
acagctgcag aacatcattc aggccacgct caggagatc atgtggatca 1201 atgactgcga ggaggaggag
ctgctgtac actggagcga caagaacacc aacatcgtc 1261 agaaacagga ggcttctcc atacgcatga
gtcaactgga agttaaagaa aaagagctca 1321 ataagctgaa acaagaaagt gaccaactg tcctaatca
gcatccagct tcagacaaaa 1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag
atcaccaagt 1441 gcattgatg tcactgaaa gaaaatgctg cctacttca gtttttgaa gaggcgagt 1501
ctactgaagc atacctgaag gggctccagg actccatcag gaagaagtac ccctgcgaca 1561 agaacatgcc
cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga 1621 aaatcctga atacaagcgt
caggtgcaga acttggtaaa caagtctaag aagattgtac 1681 agctgaagcc tcgtaacca gactacagaa
gcaataaacc cattattctc agagctctc 1741 gtgactacaa acaagatcag aaaatcgtc ataaggggga
tgagtgtatc ctgaaggaca 1801 acaacgagc cagcaagtgg tacgtgacgg gcccgggagg cgttgacatg
ctgttccct 1861 ctgtggggct gatcatcct cctccgaacc cactggccgt ggacctctc tgcaagattg 1921
agcagtacta cgaagccatc ttggctctgt ggaaccagct ctacatcaac atgaagagcc 1981 tgggtcctg
gcactactgc atgattgaca tagagaagat cagggccatg acaatcgcca 2041 agctgaaaac aatgcggcag
gaagattaca tgaagacgat agccgacctt gagttacatt 2101 accaagagtt catcagaaat agccaaggct
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gaccctggtc attcagctcc 2221 ctggctatcc ccagcaccag acagtacca caactgaaat cactcatcat
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cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta 2461 acgagcttaa gagtgtgcag
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accgctgtgg actgaagaaa ataaaaaatg acttgaactt gaagaagtcg ttgtggcca 2821 ctatgaagac
agaactacag aaagcccagc agatccactc tcagacttca cagcagtatc 2881 cactttatga tctggacttg
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caagaggctg cagatgttca tgctcggtac attgaactac 3361 ttacaagatc tggagactat tacaggttct
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cagtatgata 6601 ttcaagatgc tattgacaag ggccttgtg acaggaagtt cttgatcag taccgatccg 6661
gcagcctcag cctcactcaa ttgctgaca tgatctctt gaaaaatggg gtcggcacca 6721 gcagcagcat
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tccagcgtca ggaatttaac cataaggagc agctctttt 6841 cagacacct ggaagaatc agccccattg
cagccatctt tgacacagaa aacctggaga 6901 aaatctccat tacagaaggt atagagcggg gcatcgttga
cagcatcacg ggtcagaggc 6961 ttctggaggc tcaggcctgc acaggtggca tcatccacc aaccagggc

cagaagctgt 7021 cacttcagga cgagctctcc cagggtgtga ttgaccaaga catggccacc aggctgaagc
 7081 ctgctcagaa agccttcata ggcttcgagg gtgtgaaggg aaagaagaag atgtcagcag 7141
 cagaggcagt gaaagaaaaa tggctcccgt atgaggctgg ccagcgcttc ctggagtcc 7201 agtacctcac
 gggaggtctt gttgaccgg aagtgcattg gaggataagc accgaagaag 7261 ccatccggaa ggggttcata
 gatggccgcg ccgcacagag gctgcaagac accagcagct 7321 atgccaaaat cctgacctgc cccaaaacca
 aattaaaaat atcctataag gatgccataa 7381 atcgctccat ggtagaagat atcactgggc tgcgccttct
 ggaagccgcc tccgtgtcgt 7441 ccaagggctt acccagccct tacaacatgt cttcggtccc ggggtcccgc
 tccggctccc 7501 gctcgggatc tcgctccgga tctcgctccg ggtcccgcag tgggtcccgg agaggaagct
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 gtcagttggg agtgggtgct atacctgac ttcatttata tgaattcca 7681 ctttattaaa taatagaaaa
 gaaaatcccg gtgcttgag tagagtata ggacattcta 7741 tgcttacaga aaatatagcc atgattgaaa
 tcaaatagta aaggctgttc tggctttta 7801 tcttcttagc tcattctaaa taagcagtagc acttgatgc
 agtgcgtctg aagtgcata 7861 cagttgtaac aatagcacia atcgaactta ggattgttt cttctcttct gtgttcgat
 7921 tttgatcaa tctttaatt ttggaagcct ataatacagt tttctattct tggagataaa 7981 aattaaatgg
 atcactgata ttttagtcat tctgcttctc atctaaatat ttccatattc 8041 tgtattagga gaaaattacc
 ctcccagcac cagccccct ctcaaaccac caaccacaaa 8101 ccaagcattt tggaatgagt ctctttagt
 ttcagagtgt ggattgtata acccatatac 8161 tcttcgatgt actgtttgg ttggtatta attgactgt
 gcatgacagc ggcaatctt 8221 tctttgtca aagtttctg tttatttgc ttgcatatt cgatgtactt taaggtgtct
 8281 ttatgaagtt tgctattctg gcaataaact tttagactt tgaagtgtt gtgtttaat 8341 ttaatattg
 tataagcatg tataaacatt tagcatatt ttatcatagg tctaaaaata 8401 tttgtttact aaatacctgt
 gaagaaatac cattaataaa ctatttggtt ctgaattct 8461 actagaaaaa aaa

(207) In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001305963.1, transcript variant 3):

(208) TABLE-US-00034 (SEQ ID NO: 47) 1 mscnggshpr intlgrmira esgpdlyev
 tsggggtsrm yysrrgvitd qnsdgyctg 61 tmsrhqnqnt iqellqncsd clmraelivq
 pelkygdgiq ltrsreldec faqandqmei 121 ldsliremrq mgqpcdayqk rllqlqeqmr alykaisvpr
 vrrasskggg gytcsqsgsw 181 deftkhvtse clgwmrqrra emdmvawgvd lasveqhins
 hrgihnsigd yrwqldkika 241 dlreksaiyq leeeyenllk asfermdhrl qlqniiqats reimwindce
 eeellydwsd 301 kntniaqkqe afsirmsqle vkekelnlk qesdqlvlnq hpasdkieay mdlqtqwsd
 361 ilqitkcidv hlkenaayfq ffeeaqstea ylkglqdsir kkypcdknmp lqhlleqike 421
 lekerekile ykrqvqnlnv kskkivqlkp rnpdyrsnpk iilralcdyk qdqkivhkgd 481 ecilkdnner
 skwyvtgpgg vdmlypsvgl iipppnplav dlsckieqyy eailalwnql 541 yinmkslvsw
 hycmidieki ramtiaklt mrqedymkti adlehyqef irnsggsemf 601 gdddkrkiqs
 qftdaqkhyq tlviqlpgyp qhqtvttei thhgctqdv hnkviennre 661 ndkqetwmlm elqkirrqie
 hcegrmtlkn lpladqgssh hitvkinelk svqndsqaia 721 evlnqlkdml anfrgsekyc ylqnevfglf
 qkleningvt dgylnslctv rallqailqt 781 edmlkvyer lteeetvcl dlvkveayrcg lkkikndl
 kksllatmkt elqkaqqihs 841 qtsqqyplyd ldlgkfgekv tqldtrwqri dkqidfrlwd lekqikqlrn
 yrdnyqafck 901 wlydakrrqd slesmkfgds ntmrflneq knlhseisgk rdkseevqki aelcansikd
 961 yelqlasyts gletlnipi krtmiqspg vilqeaadvh arielltrs gdyrflsem 1021 lksledlklk
 ntkievleee lrlardanse ncnknkfldq nlqkyqaecs qfkaklasle 1081 elkrqaeldg ksakqnlkdc
 ygqikelnek itrlyeied ekrrksved rfdqqkndyd 1141 qlqkarqcek enlgwqkles ekaikekeye
 ierlrllqe egtrkreyen elakvrnhyn 1201 eemsnlnky eteinitkt ikeismqked dsknlrnqld
 rlsrenrdlk deivrlndsi 1261 lqateqrra eenalqqkac gseimqkkqh leielkqvmq qrsednarhk
 qsleaaakti 1321 qdknkeierl kaefqeeakr rweyenelsk asnriqesk qctqvqere slvkikvle 1381
 qdkarlqrle delnrakstl eaetrvkqrl ecekqqiqnd lnqwktyqr keeairkies 1441 erekserekn
 slrseierlq aeikrieerc rnkledstre tqsqleters ryqreidklr 1501 qrpqgshret qtecewtvdt
 sklvdglrk kvtamqlyec qlidkttldk llkgkksvee 1561 vaseiqpflr gagsiagasa spekyslve
 akrkklipe stvmlleaqa atggiidphr 1621 nekltvdsai arldidddr qqiyaaekai tgfdpfsqgk
 tvsyseaikk nlidretgmr 1681 lleaqiasgg vvdpvnsvfl pkdvalargl idrdlyrsln dprdsqknfv

dpvtkkkvsy 1741 vqlkercrie phtglillsv qkrmsfqi rqpvtvtelv dsgilrpstv nelesgqisy 1801
devgerikdf lqgssciagi ynettkqklg iyeamkiglv rpgtalelle aqaatgfivd 1861 pvsnlrlpve
eaykrglvgi efkekllsae ravtgyndpe tgniislfqa mnkeliekgh 1921 girlleaia tggidpkes
hrlpvdiayk rgyfneelse ilsdpddtk gffdpnteen 1981 ltylqlkerc ikdeetglcl lplkekkkqv
qtsqkntlrk rrvvivdpet nkemsvqeay 2041 kkgldiyetf kelceqecew eetitgsdg strvvlvdrk
tgsqydiqda idkglvdrkf 2101 fdqyrsgsls ltqfadmisl kngvgtsssm gsgvsddvfs ssrhesvski
stissvrnlt 2161 irssfsdtl eesspiaaif dtenlekisi tegiergivd sitgqrilea qactggiihp 2221
ttgqklslqd aysqgvidqd matrlkpaqk afigfegvkg kkkmsaaev kekwlpveag 2281 qrflefqylt
gglvdpevhg risteeairk gfidgraaqr lqdtssyaki ltcpktklki 2341 sykdainrsm veditglrl
eaasysskgl pspynmssap gsrsgsrgs rsgsrgsrs 2401 gsrrgsfdat gnsssysysys fssssigh
(209) In some embodiments of the methods of the disclosure, the wild type human AZGP1 gene of
the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:
NM_001185.3):

(210) TABLE-US-00035 (SEQ ID NO: 21) 1 ccattggcct gtagattcac ctcccctggg
cagggcccca ggaccagga taatatctgt 61 gcctcctgcc cagaaccctc caagcagaca caatgtaag
aatggtgcct gtctgtctgt 121 ctctgtctgt gcttctgggt cctgtgtcc cccaggagaa ccaagatggt
cgttactctc 181 tgacctatat ctacactggg ctgtccaagc atgttgaaga cgtccccgcg ttcaggccc 241
tggctcact caatgacctc cagttcttta gatacaacag taaagacagg aagtctcagc 301 ccatgggact
ctggagacag gtggaaggaa tggaggattg gaagcaggac agccaacttc 361 agaaggccag
ggaggacatc tttatggaga ccctgaaaga catcgtggag tattacaacg 421 acagtaacgg gtctcacgta
ttgcagggaa ggtttggttg tgagatcgag aataacagaa 481 gcagcggagc attctggaaa tattactatg
atggaagga ctacattgaa ttcaacaaag 541 aaatcccagc ctgggtcccc ttcgaccag cagcccagat
aaccaagcag aagtgggagg 601 cagaaccagt ctacgtgcag cgggccaagg cttaacctgga
ggaggagtgc cctgcgactc 661 tgcggaaata cctgaaatac agcaaaaata tcttgaccg gcaagatcct
ccctctgtgg 721 tggtcaccag ccaccaggcc ccaggagaaa agaagaaact gaagtgcctg gcctacgact
781 tctaccagg gaaaattgat gtgcactgga ctcgggccgg cgaggtgcag gagcctgagt 841
tacggggaga tgttctcac aatggaaatg gcactacca gtctgggtg gtggtggcag 901 tgcccccgca
ggacacagcc ccctactcct gccacgtgca gcacagcagc ctggcccagc 961 ccctcgtggt gccctgggag
gccagctagg aagcaagggt tggaggcaat gtgggatctc 1021 agaccagta gctgcccttc ctgcctgatg
tgggagctga accacagaaa tcacagtcaa 1081 tggatccaca aggcctgagg agcagtgtgg ggggacagac
aggaggtgga ttggagacc 1141 gaagactggg atgcctgtct tgagtagact tggacccaaa aaatcatctc
accttgagcc 1201 cacccccacc ccattgtcta atctgtagaa gctaataaat aatcatcct ccttgccatg 1261
cataaaaaaa aaaaaaaa

(211) In some embodiments of the methods of the disclosure, the wild type human AZGP1 gene of
the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:
NP_001176.1):

(212) TABLE-US-00036 (SEQ ID NO: 22) 1 mvrmpvlls llllgpavp qenqdgrysl
tyiytglskh vedvpafqal gslndlqffr 61 ynskdrksqp mglwrqvegm edwkqdsqql
karedifmet lkdiveyynd sngshvlqgr 121 fgceiennrs sgafwkyyyd gkdyiefnke ipawvpfdpa
aqitkqkwea epvyvqraka 181 yleeecpatl rkylkyskni ldrqdppsvv vtshqapgek kklkclaydf
ypgkidvhwt 241 ragevqepel rgdvlhngng tyqswvvav ppqdtapysc hvqhsslaqp lvvpweas

(213) In some embodiments of the methods of the disclosure, the wild type human OBFC1 gene of
the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:
NM_024928):

(214) TABLE-US-00037 (SEQ ID NO: 23) 1 aaatgcgctg gcggggagac cgggggttggt
ccctggcggg gcagggggcg ggctcaggcc 61 ggaactccag agacgacctc agccaactgc
tctgcgccg ggccggggctg tcgccgccag 121 cggtcccgag cgccggaagg gccaggtctc
agggtcctg gagctgcagg cggcgggagg 181 ggctacaaat gcttgactca gtgatgcaga accttcaga
gttagctgga agccacagcc 241 ctgcctcttg atgcagcctg gatccagccg gtgtgaagag gagaccctt
ccctcttggt 301 gggtttgat cctgtgttc tagccttgc aaaactctac atcagggata tcttgacat 361

gaaggagtc cgccagggtc caggtgtatt ttgtacaat ggacatccaa taaaacaggt 421 agatgtcttg
 ggaactgtca ttggagttag agaaagagat gctttctaca gttatggagt 481 ggatgacagc actggagtta
 taaactgcat ctgctggaaa aagtgaata ctgagtctgt 541 atcagctgct ccaagtgcag caagagagct
 cagcttaacc tcacaactta agaagctaca 601 agagaccatt gagcagaaaa caaagataga gatcggggac
 acgatccgag tcagaggcag 661 tatccgcaca tacagagaag agcgagagat tcatgccacc acttactata
 aagtggacga 721 cccagtgtgg aacattcaaa ttgcaaggat gcttgagctg cccactatct acaggaaagt
 781 ttatgaccag ccttttcaca gctcagccct agagaaagaa gaggcactaa gcaatccagg 841
 cgccctggac ctccccagtc tcacgagttt gctgagttaa aaagccaaag aattctcat 901 ggagaacaga
 gtgcagagct ttaccagca ggagctggaa atgggtggagt cttgtctgc 961 ccttgccaat cagcctgtga
 ttcacagtgc ctctccgac caagtgaatt ttaagaagga 1021 caccacttcc aaggcaattc atagtatatt
 taagaatgct atacaactgc tgcaggaaaa 1081 aggacttgtt ttccagaaag atgatggttt tgataaccta
 tactatgtaa ccagagaaga 1141 caaagacctg cacagaaaga tccaccggat cattcagcag gactgccaga
 aaccaaata 1201 catggagaag ggctgtcact tcctgcacat cttggcctgt gctcgcctga gcatccgcc 1261
 gggcctgagc gaggtgtgtc tgcagcaagt tctggagctc ctggaggacc agagtacat 1321 tgtcagcaca
 atggagcact actacacagc gttctgagca gagacacgca gaccagctga 1381 ggaggacaaa gataaggtgg
 cattcaccac caggctctga ctttcagcat catgcagggg 1441 cttatctgtc tggaggcagt tacctcataa
 taaactataa aatatagtca tcttggaat 1501 gggatttggc ataaatgttg ttggctcctt tctgtccact
 atgtccttgg tgtacaatga 1561 ctttgcctc agccatgaca caacaagaa accctcctg ttgagctcct
 ggctggactg 1621 tgcgttgttc gcagagcaga atggggagga aacagtgttg gcagctaac tgatgtgtg 1681
 ggttggagtc tcttccatgg caaagggaca ccacagggtg gtgaacattc aggaactgag 1741 gggcatatgg
 cctgatcaca cagttctaag ctttcaaaa cttcagggtt tcagagacct 1801 tctgtgggc ctctcttgc
 ggctaagaac cggtttaggg gagtagttct ccctggatga 1861 gtgcttacag ttctgtggc tcagttacca
 gcagtggggg tgagacctgg gtcgatgtc 1921 ttacaggcc tgcccagaga tgggaataaa cagggatcca
 cagcgtgact atgtgttgt 1981 catttcctt ttattcctt gggaatcgaa aggtgtcca gtacattcc ctgcactac
 2041 agaggtgcat gactaaatac attgtccctc gatgccctg aagatcacgg aggcagtcag 2101
 ccaattgcct ggcaggttgt agatgttatt ttcagggttg ccgctgagtg tgcaggatgt 2161 gctgacacca
 tccagacaaa gactcggat gtgcccagac aggtgatgga gtcagtctt 2221 tgctcagaat gacaaggtaa
 aggaaaaaca tctgaggtat gttgtaggcc tgttctgaca 2281 gcaaaatgac aatccagcc agcaaaaata
 aagtgtggag aaagatttg agttaattac 2341 agtcattca cagaaggcac tgccttcgtc tgctgcattt
 gctcttgatg tgataagctc 2401 ttcgtggctc agctggagat ccttaggcc tggagagttg ctctctctc
 cgtggaaaca 2461 ggacagtctt tatacgaga agtccgctgc agctcgatac gtcaggctga gagctagaac 2521
 cagtagattg cctcctgtca tagactttg taatgatgca aaccttgc gatttctaac 2581 agtgattatg
 tagtggctgc cctgcatctt ctctgtgtac agaagggtcc ctagcataga 2641 gtctgcctgg aatgatgtcc
 tgggcagttc ttccttgagg tcagcagctg ttccacgttg 2701 aatgcatctg attagtggg ctgcccagga
 aggagttagc aatcagaagg taaaagggc 2761 ataccctgc ctatagcaac tctgctcta ggggtttatc
 tcaaggagat ggctacacaa 2821 gtgtgaaagg atggtgcac aaggtgttca ttgctgtata atctagaatt
 ctatattggg 2881 gaaaatacct ataggaaaa agttaattac ggttcttggg cacaatgaaa tactatgcag 2941
 ctatgaaaaa aatgatgaaa gcagacagac agtgttgcca tggcacactg tcctagtag 3001 attagtggg
 aagtagatag agttatagat ctgtttctat agtataacac cattatctac 3061 agtccctgt gtgtatgat
 atatccgtag agagagtgt tatttctgca tggaggtctt 3121 tataaatgta gcacatgtac atatatat
 atatacacac acacagtcga ccactccctt 3181 ctctggaag tactttccgc gtttggtt caggacacca
 agctctctgg ttgctcctc 3241 tcaggttctt ttgtcagt ctctgcctcc ctgaggactc agtcccagac
 ctcttttcta 3301 tctggcttgc tctactgggt gtctccagca gccacatgga ttataccatc tacatgctgt 3361
 ctaacacctc agtttaaacc cagaatgggc ctctccctg aactgcagac ccctatatc 3421 agtttgctac
 tgacatctcc acttaggtct ctaatggaca tctcagattt cacaggccca 3481 aagccaggct cccaattact
 cctgacccca ggcttgcctc tgatagtac atgaggcagc 3541 caaatgccta ggcagagagg ggagggtccc
 aatgaaacc ccacgttcaa gcaaaatca 3601 gcctgaaggc taaaagacca gattgctgtt cctggatgaa
 acccaccagc cagagtggga 3661 acttctgttc ctgtttgcc acccttccc aattgttct tctgaatac
 gccttaacca 3721 atcgaatgtt gcctttcca gtaataccta cagcctgccc ctcccccat tctgagccca 3781
 taaaagacc cagactcccc catattaagg ggaacttct gccttgggt agggggacca 3841 cccccagtc

tctctctgt tgaaaactgt ttcatactc aataaaactc ccagctttgc 3901 tcaacttcc actgtcagca
 cattctcatt cttctttggg gctgggcaag aactcaacca 3961 gtgtggaagc catacttggc ccaggcgggt
 gaagtgggcg ggccgtctcc tgcagcaggt 4021 agcatgggtca agcgaggccc aggtgggccc tcaccagcca
 gaggtccctg gcttgcaaag 4081 tgaccgagaa aaaaatcctg tgccactcct ttggaaaatg tccctgattc
 aggaagaggt 4141 agtccatcc agttgctcaa accaaatcca ttggcttctt tctttctatc atacctcaca 4201
 tccaatctgt ctgcaagtct tttggctcta cttcagaat atctccagaa tcttaactgc 4261 ttcacctcc
 tccccggcct cctcagtcct ctctgcttcc gccctggccc ctcttgggct 4321 gttcacagca cagcagctgt
 tgccacctg ttaatgtcc cactctcta cagccttcgg 4381 tcttgcccca ggtaggagcc tgaggctgca
 cagaggtcag cacggccccg cttacctgc 4441 cctcccagcc cagccgcacg ggccttgac acatgcctcg
 gcatattcct gccttagggc 4501 tgggtgctct gctatttct ctcccaggt aacctatga agtgcctccc
 tctgccctct 4561 ttccagcctt tacttgagtg tcacctctc agtgaggcct gccctcattc ctcttcgct 4621
 gtttgaacc catctctgt ccccttccc agaactcct ttctacttc gttttcttc 4681 acagtactg
 atactgccta acacactcca tggtttcta cttgccctgt ttattattt 4741 ccccaatag acagaatgtt
 ccatgatggc agaattctct gtttgttct cttccatgtc 4801 ccagcacct agaacagtgc ctgacgcac
 tcctaagcaa tacgaccaat aagtatgtgt 4861 ctggctgcct tccggctgcc agtgtctgcc tcttctag
 gggcagtggg tgcgggggtg 4921 ctttctaca tgtcttagta ggctgtgcag gctggaagtg ctcagaagt
 acacccccag 4981 ggagcagcct cagccaacag caccttggt gtaaagccc cagctccctc gccctcaggt
 5041 aagcattgct gaggcacacg ttccatactc tttccacag ttctccgtg ggactgagca 5101
 ccacccagcc acccacagga gcagctaacc tgataaccac cagcctcacc ctccctgcct 5161 tacttccccg
 ctccccttta ccacatgtg acctccaga tgcatttctt gtttccggt 5221 ctctgtctca ggattggctc
 ctggatgaac acaaactaac actatgttca caaatatatt 5281 tgggaaatgc tggatgaata attatacaca
 tcagacagat tactagaaat tctcacaaa 5341 gggatgcaca tgttacctt gcattggtgag atctcaggtg
 ctttttacc cacatagcta 5401 tctttggca ttttataat tagcaagtgc tcaacttcc actgtcagta cattctcatt
 5461 cttcttgggc gctggacaag aattcaaccg gtgtgtaagc cagactcggc ccgggcagtc 5521
 tcaactcct gactccttat ataatttcta caaaaattat aaagctattt cccactcccc 5581 accccacatt
 catgtaacct gaagcatgag taaaccaaga atgaggtagg cctctgtctt 5641 ctaagcaaca tcagaactct
 aagaacatga gggactctta gaaaactctc tggagctaac 5701 cacagctggg tcaactgtca tgtactgaag
 accagccaga gggttcccct gaaaaggagg 5761 gaaactgagc aaacattctc cagttctctt agtgtgcaca
 tgtttcagga ggtgtgaacc 5821 ccacatgtag cttgtgtagg caagaagaca aatagtcta ctgtctggc
 aaggatttgt 5881 ttgaagagcc atgattatgc ccatatggt agccaccagt gctccccatc cctgtaagac 5941
 acttcttct cattatttct tctctgatg gtgtgccagg atgtggcca agagaagcca 6001 agtggaagaa
 aggctgttca gtgacaagga acctaaact tagtgccaag gactgaaacc 6061 aagtaaactt gtaatttcc
 atgatggaaa catctacact ttctcattag tggcctctac 6121 agcagttgcc ccaaagaagc gtctcattgt
 tttttacta catttatgtg aagcatacag 6181 gcaaactcag aaagactgtg ataaggctcg ccagagatgc
 ctgcacaggt gctgggggaa 6241 aagcaggacc atctgaagg gagatgggtg ctgtggacaa agaactctgc
 agtgggttct 6301 atttgcata tttctgctgg tggaggctgt aatgtgagc tcaaactccc acataagtga 6361
 gttttcattg taatccagaa tgtttttaa tcacctact tctattgaac ttgcactatc 6421 atctgttaac
 ctctactgta ttattaaat aaacctgaat aggtaaatca cagtacagca 6481 aaa

(215) In some embodiments of the methods of the disclosure, the wild type human OBFC1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_079204.2):

(216) TABLE-US-00038 (SEQ ID NO: 24) 1 mqpgrsree etpsllwld pvflafakly
 irdildmkes rqpvgvflyn ghpikqvdl 61 gtvigvrerd afysygvdds tgvincicwk klntesvsaa
 psaarelsit sqlklqeti 121 eqtkieigd tirvrsirt yreereihat tykvddpvw niqiarmlel
 ptiyrkvydq 181 pfhssaleke ealsnpgald lpsltsllse kakeflmenr vqsfyqqele mvesllslan 241
 qpvihsassd qvnfkddts kaihsifkna iqllekglv fqkddgfdnl yyvtredkdl 301 hrkihriiqq
 dcqkpnhmek gchflhilac arlsirppls eavlqqvlel ledqsdvst 361 mehytaf

(217) In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_015205.2, transcript variant 1):

(218) TABLE-US-00039 (SEQ ID NO: 25) 1 gcggccgcac tagtaccgg
gagcccatgg gcgcgccgag ccgggcgcgg gggcgctgaa 61 cggcgagcgg ggagcgccg
gaggagccat ggactgcagc ctctgtcgga cgctctgca 121 cagatactgt gcaggagaag agaattgggt
ggacagcagg accatctacg tgggacacag 181 ggagccacct ccgggcgcag aggctacat
cccacagaga taccagaca acaggatcgt 241 ctctccaag tacacatttt ggaactttat acccaagaat
ttattgaac aattcagaag 301 agtagccaac ttttattcc ttatcatatt tctggtgcag ttgattattg
atacccccac 361 aagtccagt acaagcggac ttccactctt cttgtcatt actgtgacgg ctatcaaaca 421
gggttatgaa gactggcttc gacataaagc agacaatgcc atgaaccagt gtctgttca 481 tttcattcag
cacggcaagc tcgttcggaa acaaagtcga aagctgcgag ttggggacat 541 tgtcatggtt aaggaggacg
agacctttcc ctgcgacttg atcttcttt ccagcaaccg 601 gggagatggg acgtgccacg tcaccaccgc
cagcttggat ggagaatcca gccataaaac 661 gcattacgcg gtccaggaca ccaaaggctt ccacacagag
gaggatatcg gcggacttca 721 cgccaccatc gactgtgagc agccccagcc cgacctctac aagtctgtgg
gtcgcataca 781 cgtttacagt gacctgaatg acccgtgggt gaggccctta gcatcgaaa acctgctgt
841 tagaggagct aactgaaga aactgagaa aatcttgggt gtggctattt acacgggaat 901
ggaaaccaag atggcattaa attatcaatc aaaatctcag aagcgtatcg ccgtggaaaa 961 atcgatgaat
gcgttctca ttgtgtatct ctgcattctg atcagcaaag ccctgataaa 1021 cactgtgctg aaatacatgt
ggcagagtga gccctttcgg gatgagccgt ggtataatca 1081 gaaaacggag tcggaaaggc agaggaatct
gttctcaag gcattcacgg acttctggc 1141 ctctatggc ctcttaact acatcatccc tgtgtccatg
tacgtcacgg tcgagatgca 1201 gaagttctc ggctcttact tcatcacctg ggacgaagac atgttgacg
aggagactgg 1261 cgaggggcct ctggtgaaca cgtcggaact caatgaagag ctgggacagg tggagtacat
1321 cttcacagac aagaccggca ccctcacgga aaacaacatg gattcaagg agtgtgcat 1381
cgaaggccat gtctacgtgc ccacgtcat ctgcaacggg caggtcctcc cagagtcgtc 1441 agaatcgac
atgattgact cgtccccag cgtcaacggg agggagcgcg aggagctgt 1501 ttccgggcc ctctgtctct
gccacaccgt ccagggtgaaa gacgatgaca gcgtagacgg 1561 cccaggaaa tcgccggacg gggggaaatc
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tacctaaggc tgaaggacaa 1681 ttacatggag atattaaaca gggagaacca catcgaaagg ttgaattgc
tggaaatttt 1741 gagttttgac tcagtcagaa ggagaatgag tgtaattgta aaatctgcta caggagaaat 1801
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cgagccagag tggagcgtaa cgcagtggag gggctccgaa cttgtgtgt 1921 tgctataaa aggctgatcc
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cagtctccac acagtaacat ttaactgata attcattaat 8581 cagcttgaa aaattaaatt gtaattaa
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tctgctcag tttcttata aagaaactca cacaa

(219) In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NP_056020.2, transcript variant 1):

(220) TABLE-US-00040 (SEQ ID NO: 26) 1 mdcslvrtl hrycageenw
vdsrtiyvgh repppgaeay ipqrypndri vsskytfnf 61 ipknlfefr rvanfyflii flvqliidtp
tspvtsglpl ffvitvtaik qgyedwlrhk 121 adnamnqcpv hfiqhglvr kqsrklrvgd ivmvkedetf
pcdliflssn rgdgtchvtt 181 asldgesshk thyavqdtkg fhteediggl hatieceppq pdlykfvgr
nvysdlnp 241 vrplgsenll lrgatlknte kifgvaiytg metkmalnyq sksqkrsave ksmnafliv
301 lciliskali ntlvkymwqs epfrdepwyn qkteserqm lflkaftdfl afmvlfnyii 361
pvsmvvtvem qkflgsyfit wdmdfdeet gegplvntsd lneelgqvey iftdktgtlt 421
ennmefkecc ieghvvyphv icngqvlpes sgidmidssp svngreerel ffralclht 481
vqvkdtdsvd gprkspdgk scvyisspd evalvegvr lgftyrlkd nymeilnren 541 hierfellei
lsfsvrrm svivksatge iylfckgads sifprviegk vdqirarver 601 naveglrtlc vaykrliee
yegickllqa akvalqdrek klaeayeqie kdltilgata 661 vedrlqekaa dtiealqkag ikvwvltgdk
metaaatcy cklfrntql lelttkrie 721 qslhdvlfel sktvrhsgs ltrdnlscls admqdyglii
dgaalslimk predgssgny 781 relfleics csavlccrma plqkaqivkl ikfskehpit laigdgandv
smileahvgi 841 gvigkegrqa arnsdyaipk fkhllkmlv hghfyirys elvqyfykn vcfifpqfly
901 qffcgfsqqt lydtayltly nisftslpil lyslmeqhv idvlkrdptl yrdvaknall 961 rwrvfiiwtl
lgldalvff fgayfvfent tvtsngqifg nwtfgtlvft vmvftvltkl 1021 aldthywtwi nhfviwgsll
fyvvfslwv gviwplnyq rmyyvfqiml ssgpawlaiv 1081 llvtisllpd vlkkvlcrql wptatervqt
ksqclsveqs tifmlsqts slsf

(221) In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:

(222) TABLE-US-00041 (SEQ ID NO: 48) 1 gcggccgcac tagtaccccg
gagcccatgg gcgcgccgag ccgggcgcg gggcgctgaa 61 cggcgagcgg ggagcggccc
gaggagccat ggactgcagc ctctgctgga cgctcgtgca 121 cagatactgt gcaggagaag agaattgggt
ggacagcagg accatctacg tgggacacag 181 ggagccacct ccgggcgag aggcctacat
cccacagaga taccagaca acaggatcgt 241 ctctccaag tacacattt ggaactttat accaagaat
ttatttgaac aattcagaag 301 agtagccaac ttttattcc ttatcatatt tctggtgcag ttgattattg
atacaccac 361 aagtccagt acaagcggac ttccactctt cttgtcatt actgtgacgg ctatcaaaca 421
gggttatgaa gactggcttc gacataaagc agacaatgcc atgaaccagt gtcctgttca 481 ttcattcag
cacggcaagc tcttcggaa acaaagtcga aagctgcgag ttggggacat 541 tgtcatggtt aaggaggacg
agaccttcc ctgcgacttg atcttcttt ccagcaaccg 601 gggagatggg acgtgccacg tcaccaccgc
cagcttggat ggagaatcca gccataaaac 661 gcattacgag gtccaggaca ccaaaggctt ccacacagag
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cacagtgggt 5221 gggataaaat aaggtgacaa acatcacacc aaagatgagg gtacgagca actggcttga
5281 gcagacagaa cggggaagac tccactctgt cccgaggggc cagccgcagg cgtccccagg 5341
gccaccctgc cctgaggctc ttgtgtggc gccctggctt ggcagccctg cccacgtgc 5401 ccccgcaaac
aatgggtgt gctgtttac agccctttt aggaaccaa tatgggcata 5461 aatgtaacac ctgtacggg
ggcagattct ctgtatgtc agttaacaaa ttattgtaa 5521 tgtatttt tagaaatctt aaaattgcct
ttgactgaa gtatttcat agctgtttat 5581 atctcttta ttcatatt taacatactg tctaattta aaaataggt
ttaaagctt 5641 tcattttta gttatgaaa tttggccac ttacattta gattctgtg agagtttga 5701
ctgaatgtc caatctctga tgaatgcga tttcagatt tgattttat ctctacacac 5761 acctctctt ttcttggtat
ttctggtggc agtgattagt tgaacagcac atttaaggca 5821 cgataattg ctacacttt tctttacaat
ttgtgcaat tcatctgt tctatgtt 5881 cattgttaat tgccatcct cagcctaaa aatagaagat
tctcacgtga aggttagta 5941 agttgggtcc cagctctgcc tgtgtggaga tagtcacat gtacctga
caacaagttt 6001 tagtgtgaaa gtcactaac tttacacac tccaaacgt cttttaaaa attgctggg 6061
aaattattaa atgaatgtc ctgatgatt gaaatagaca aggggcacga gataaaaaag 6121 aaaaggatga
gaagatcctc agtgaatgac gttgcagggt cttcatgaa tttccacct 6181 cgcagtagt agtattact
tgccttaaac taacttgaa gcaagtaat tcaacttga 6241 gcactttgt gagtttgaa aaacttatt
tgtgtctga caggtaata aattatcaat 6301 ttgtaattca gcatgttggt cagagacacg gtcactgatt
cacaccagt cctgccaca 6361 gaccgtctc gacacgaca gtgggcctgc tgcattgac acaccagtc
cctgccacag 6421 accgtctcag acacgcacag tgggcctgct gcatgattca caccagtcc ctgccacaga 6481
ccgtctcaga cagcacagt gggcctgct catgcgtgtt acctggctt tggctccacg 6541 ctactcata
gccatgtcca catgggggtc tgcacacagg atcactcaca tatgtacatg 6601 taccaccac aaacgtgcaa
gctctgcac acatgcatc acacaaacgt gtacacaagt 6661 gtgagctct acacgcatc acacacacac
gtgtacatgc accaaagcat gtgtgacct 6721 cagacatgca gaacatgcac gtgtacacat accacagaca
cgcgtgtgca tgctctaca 6781 caatacatat gcacatat tgaacagcgt aagtctctac acacggacgt

gtgatacaca 6841 catgcatgta caggtaagca cacatgtaca agctcctaca ggcttgctct cacacacgtg 6901
 tatgcacagc agagagacgt atgagcttct actgcacaca tgcacacaca cagcacacg 6961 tacattcact
 acaaacgtgc agcctcctgc acacgtgcac attcatgtgt acaccacaaa 7021 tgagttccca gacgtgtaaa
 cacacgtgca cacatcgtaac acatgtgagc tccacacagc 7081 acacacagat gcacatggac acaccccaaa
 cacgcacagg ctctacaca catgcacaca 7141 cgtgtacacc acaaacgagc tcccagacat gtaaacacac
 gtctcccaca cgtgagctcc 7201 cacacgtaca catgcacatg tacgcaccac aaacacatgc gcaggctcct
 gcaggcgtga 7261 atacacacat gcacacacat atacacacat gtgccacaaa caagtgcaca ctgtcctggt 7321
 gtcctgcact gcctcctgcc tcttgctga ggggcccctg tgagaggcct ctggatgggc 7381 atgggaagat
 gggctcccctg gccccagcc catgcctccc tgggatgaag agtccccctc 7441 ctggcagaat gtctgggctt
 tgcagagcag gccccggggg tgaagtcgca gcttactta 7501 caccagctgc tctgtgagca aggcttggtg
 ccctggacaa ggccctccc cttagggag 7561 gtccagcctc gcaagctgaa acctccccctc ggctcagccc
 tataccaggc ggccacagca 7621 ggactggcca caccacgcc gcacctcctc cgtgcacgcg tcggagcacg
 gccagccttc 7681 cgccacgagc cagctgggaa gggccgcggc cgcctaaagc cccagtcaac ccagcctgtg
 7741 tctgagcaga cagggcgaac aagcaggcca caccgtctcg agggaggagg ccagatgcgg 7801
 ccagcgtctc caacagggtg accatccgct cggcttgctg agcgtttaa caaatgtta 7861 gacaggctgt
 ggggactccc ctgagttgag ccttgccag gggctccgtg ctgtcgcggg 7921 aaacctccag ccttgttctt
 caaaccactc agctcatgtg tttgactg actagtactg 7981 aataatacaa ccactcttat ttaatgttag
 tattatttat ttgacaactc agtgtctaac 8041 agcttgatat gcaggctcct gcctcctaca tttcttagg
 aagttacca tttgtaactt 8101 taaaaacagg aaaaatatca gttggcaaat gcaatctttt tttttttaa gctaaagggtg
 8161 ggtgaactgg aatgaaaatc tttctgatgt tgtgtctata agcagccttg atgggatatg 8221 ttagaagtgt
 catgaaagtg tgattctact tttgcagaaa aatctaaaga tcaattata 8281 tagctttatt tttacttta
 tcaaagtata cagaatttta atatgcatac attgtgtctg 8341 acttaaaatt ataagtctg cgtcaccatt
 taaatgtct gttcattatg taatgtaata 8401 aaagaaggct tcaaaaatg tatttaacat gaatggatc
 catagttgtc atcatcataa 8461 atactggagt ttattttta attattaaac atagtaggtg cattaacata aatcagctc
 8521 cacacagtaa catttaactg ataattcatt aatcagctt gaaaaattaa attgttaatt 8581 aaaccaatct
 aacatttcag taaagtttat tttgatgct tctgtttta acttttatt 8641 ctgtagataa actgactgga taatattata
 ttgactttt ctctagatta tctaagcagg 8701 agacctgaat ctgcttgcaa taaagaataa aagtctgctt
 cagtttctt ataaagaaac 8761 tcacacaa

(223) In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_115565.3, transcript variant 2):

(224) TABLE-US-00042 (SEQ ID NO: 49) 1 mdcslvrtl hrycageenw
 vdsrtiyvgh repppgaeay ipqrypndri vsskytfnf 61 ipknlfefqr rvanfyflii flvqliidtp
 tspvtsglpl ffvitvtaik qgyedwlrhk 121 adnamnqcpv hfiqhglvr kqsrklrvgd ivmvkedetf
 pcdlilfssn rgdgtchvtt 181 asldgesshk thyavqdtkg fhteediggl hatieceppq pdlykfvgri
 nvysdlnpvp 241 vrplgsenll lrgatlknte kifgvaiytg metkmalnyq sksqkrsave ksmnaflivv
 301 lciliskali ntlvkymwqs epfrdepwyn qkteserqrn lflkaftdfl afmvlfnyii 361
 pvsmvvtvem qkflgsyfit wdmdmfdeet gegplvntsd lneelgqvey iftdktgtlt 421
 ennmeffecc ieghvypvhv icngqvlpes sgidmidssp svngrereel ffralclcht 481
 vqvkdddsud gprkspdgk scvyisspd evalvegvr lgftyrlkd nymeilnren 541 hierfellei
 lsfdsvrrm svivksatge iylfckgads sifprviegk vdqirarver 601 naveglrtlc vaykrliee
 yegickllqa akvalqdrek klaeayeqie kdltilgata 661 vedrlqekaa dtiealqkag ikvwvltgdk
 metaaatcya cklfrntql lelttkrie 721 qslhdvlfel sktvlrhsgs ltrdnlscls admqdyglii
 dgaalslimk predgssgny 781 relfleics csavlccrma plqkaqivkl ikfskehpit laigdgandv
 smileahvgi 841 gvigkegrqa arnsdyaipk fkhllkmlv hghfyirisi elvqyffyn vcfifpqfly
 901 qffcgfsqqt lydtayltly nisftslpil lyslmeqhv idvlkrdptl yrdvaknall 961 rwrvfiiwlt
 lgldalvff fgayfvfent tvtsngqifg nwtfgtlvft vmvftvtlkl 1021 aldthywtwi nhfviwgsll
 fyvvfslwv gviwpflnyq rmyyvfqiml ssgpawlaiv 1081 llvtisllpd vlkkvlcrql wptatervqn
 gcaqprdrds eftplaslqs pgyqstcpsa 1141 awysshssqv tlaawkekvs tepppilggs hhhcssipsh
 scprsvrgml v

(225) In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_002225.3, transcript variant 1):

(226) TABLE-US-00043 (SEQ ID NO: 50)

1	tttccgcagt	taggggctgc	tatttcaacg				
cagggagata	aaaagaaaaa	aacacttgct	61	cttctacccc	gctaaaaaca	ctcatcctag	ggagcacgcc
agcatttgca	gcgttcgggg	121	cagggccact	cggcctgcgg	ccgttgcaact	ggctggaagc	tggcaggcga
tcacggttga	181	ttggctcggg	tgcggtccaa	gggcagcaac	gccttcggcg	ggccgcctag	ggtgattggc
241	tgctgcagcc	caccccctag	ccggtttggt	gggcggcgaa	gcctggattg	gtggagctaa	301
gagctggctc	agtttcagcg	ctggctcttc	gtgcatggca	gagatggcga	ctgcgactcg	361	gctgctgggg
tggcgtgtgg	cagctggag	gctgcggccg	ccgcttgccg	gcttcgtttc	421	ccagcgggcc	cactcgcttt
tgcccgtgga	cgatgcaatc	aatgggctaa	gcgaggagca	481	gaggcagctt	cgtagacca	tggctaagtt
ccttcaggag	cacctggccc	ccaaggccca	541	ggagatcgat	cgcagcaatg	agttcaagaa	cctgcgagaa
ttttggaagc	agctggggaa	601	cctggggcgta	ttgggcatca	cagcccctgt	tcagtatggc	ggctccggcc
tgggctacct	661	ggagcatgtg	ctggtgatgg	aggagatatc	ccgagcttcc	ggagcagtgg	ggctcagtta
721	cggtgccac	tccaacctct	gcataacca	gctgtacgc	aatgggaatg	aggcccagaa	781
agagaagtat	ctcccgaagc	tgatcagtgg	tgagtacatc	ggagccctgg	ccatgagtga	841	gcccataatga
ggctctgatg	ttgtctctat	gaagctcaaa	gcggaataa	aaggaaatca	901	ctacatcctg	aatggcaaca
agttctggat	cactaatggc	cctgatgctg	acgtcctgat	961	tgtctatgcc	aagacagatc	tggctgctgt
gccagcttct	cggggcatca	cagccttcat	1021	tgtggagaag	ggtatgcctg	gctttagcac	ctctaagaag
ctggacaagc	tggggatgag	1081	gggctctaac	acctgtgagc	taatctttga	agactgcaag	attcctgctg
ccaacatcct	1141	gggccatgag	aataagggtg	tctacgtgct	gatgagtggg	ctggacctgg	agcggctggt
1201	gctggccggg	gggcctcttg	ggctcatgca	agcggctctg	gaccacacca	ttccctacct	1261
gcactgagg	gaagcctttg	gccagaagat	cggccacttc	cagttgatgc	aggggaagat	1321	ggctgacatg
tacacccgcc	tcatggcgtg	tcggcagtat	gtctacaatg	tcgccaaggc	1381	ctgcgatgag	ggccattgca
ctgctaagga	ctgtgcaggt	gtgattcttt	actcagctga	1441	gtgtgccaca	caggtagccc	tggacggcat
tcagtgtttt	ggtggcaatg	gctacatcaa	1501	tgactttccc	atgggcccgt	ttcttcgaga	tgccaagctg
tatgagatag	gggctgggac	1561	cagcagagtg	aggcggctgg	tcacggcgag	agccttcaat	gcagacttcc
actagtctctg	1621	agacccttcg	cccccttttc	ctgcacctag	tggcctttct	tgggaagtag	agatgtggcg
1681	gctttccac	cctgcccaca	gcaggccctc	ctgcccagct	gctcttgta	gccctctggc	1741
ctctggatga	ggttgagttc	tccacaacag	ctcccaagca	tcattgggct	cgcagccggg	1801	cctgtgccac
ggctagtgtt	gtgtgattta	aaatggactc	agcaggaagc	atattgtctg	1861	gggattgttg	ggacagggtt
tggtgactct	gtgcccttgc	tctctaactt	ctgagccac	1921	ctcccagggt	aggcacctgg	gggcatgcag
gtgccacct	cccagggtag	gcacctgggg	1981	gcattgcaggt	accacacctt	ttctcttggg	tgaggctctg
gcaaggagat	ctctctgctc	2041	aagcacagca	gaatcatggc	ccctctccat	gaattggaac	ttggtacagg
ttaagtatcc	2101	ctaactctga	aatctgaaac	acttgttggt	ccaagcattt	tggataaggc	aaattcaact
2161	ttcagtctct	ttctggggg	aaaaaaataa	taaacctagc	ctagccaggc	gtggtggctc	2221
atgcttgtaa	tcccagcact	tcaggaggct	gagatgggtg	gatcacctga	ggtcaggagt	2281	tcaagaccag
cctggccaac	atgtggaac	ctgcctcaa	ctaaaaatag	aaaaaaatta	2341	gttgggcatg	gtggtgggca
cctgtaatcc	cagctacttc	aggaggctga	ggcaggagaa	2401	ttacttgaac	ccaggaggcg	gacgttgtag
tgagccgagc	ttgtgccatt	gcactccagc	2461	ctgggcgaca	agagcaaaac	tcttcaaaaa	acaaaaacaa
acaaaaaac	cctggccctt	2521	gtttcttcca	gtttctagag	gtatcagctc	ctagcagctt	atgaacacat
atgcttgctt	2581	ggccaggcaa	ggtggtgtgt	gcctgtaatc	ccagcacttt	gggaggccaa	ggcagggtga
2641	tcacttgtag	tcaggagttc	aagaccagcc	tgccaacgt	ggtgaaacct	catctctact	2701
aaaaatacaa	aaattagcca	ggggtggtgg	tgcacgtctg	taatccagc	tactcaggag	2761	gctgaggcag
gagaatcact	tgaacccggg	agggtgaggt	tgcaatgagc	caatatgaca	2821	ccgctgcagt	ccagcctggg
ccatagagtg	agactctgtc	tcaaaaaagg	aaagaaaaat	2881	aggctgggca	cagtgactca	tgctgtaat
cccaacactt	tgggaggccg	aggcagggtg	2941	atcacgaggt	caggagtcca	agaccagcct	ggccaagatg
gtaaaacctc	gtctctacta	3001	aaaatacaaa	aattagccag	gtgtggtggc	aggctcctgt	aatcccagct
actcaggagg	3061	ctgaggcaga	gaattgcttg	aacccgggag	gcagagtgtg	cagttagcca	agatcacacc
3121	actgcactcc	agcttgagc	acagagcgag	actctgtctc	aaaaaataat	aggccaggca	3181
tgggtggtca	acgtctgtaa	tcccagcact					

ttgggaggcc gaggcgggca gatcacaagg 3241 tcaggagttc gagaccagcc tgacgaccaa catggtgaaa
 cctcgtctct actaaaaata 3301 caaaaattag ccaggcctgg tggcacgcgc ctgtaatccc agttacacag
 aagactgagg 3361 caggagaatc gcttgaacgc aggaggcaga ggttgacagga gctgagatcg cgccattgca
 3421 ctccagcctg ggcaacagag tgagactctg tctcaaaaaa taataataaa ataatgaac 3481
 acacatgctg ctgagtccgc aggggggggca gacgagagga cagcgtgctt ttgtgtactg 3541 ttggaagact
 ggctcctcct gtacagcacc tctgagccct tgtgcaccgc cctgccacgg 3601 gcaccatcca gtcctggccc
 tgtgaccacc cacagctgac tgggcagcag gcacaggccc 3661 taccgagca ggccggagtt ggctcgcag
 actccagctg aggctgcctg tgtacattc 3721 tccagatacc ctatggctaa tttgttata actgcacagt
 ggctgctgcc attttgtatt 3781 aaatatattg tgaaacaaac ctatctgggg agaagcaatc tacttgcccgc
 tgcttctgt 3841 ctggatccag cttgtgtcct tggagagtgg ctggcccagg tctattcct gtcctccagc 3901
 ccgttcttc atgagggaca ggaaggtaaa atcagccctt aggagagagg ttcagcctc 3961 ctttcccag
 atctccagct gagttttaa ggaagcaggg agcccagagt gctaagtct 4021 tacagccaga aggaagctta
 tagatttctg aaaaccgccc cttgtttt aaaaagatca 4081 acacaattg actttctca ggtaaaacg
 aactagaatc cagatctgct catggcaaaa 4141 atgggggtgt tctgagaatt ccagcttgg gccgcactgt
 acagcagtct ggatagagt 4201 tgatctgaga agggaatggg tctgggtgt tccaccctt ccgagtcca
 aaaagaggga 4261 actggtttc ttggttctca gccagcagc acctatcctg gctcttggtc ctggcctgca 4321
 gccagtgtc gttcctagcc tgaggctga gacaggtggg gttggctcct caccaacccc 4381 agttccgtcc
 catcctgagg gcaagatcct gggctcatag gcagtcctt tcttctt 4441 gtctgtctc ctgctatgt
 ggagatgaat gtgactaaaa gggccatctt gctggcttaa 4501 tgtgtggctg gagagaccag cctggagaca
 atgtggcaaa atggggcgct tcatccagtc 4561 tgtctaagcc ctgtcgactt ggggaggtga tttcttct
 ggttctatat gtgaagcaaa 4621 ataaatgtt taaaattaa agcaaaaaa acaaaatgaa ccatgaaaaa aaa
 (227) In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2

gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession
 number: transcript variant 1):

(228) TABLE-US-00044 (SEQ ID NO: 51) 1 maematatrl lgwrvaswrl rpplagfvsq
 rahsllpvdd ainglseeqr qlrqtamakfl 61 qehlapkage idrsnefkn refwkqlgnl gvlgitapvq
 ygsglgyle hvlvmeeisr 121 asgavglisyg ahsnlcinql vrnagneaqke kylpklisge yigalamsep
 nagsdvsmk 181 lkaekkgnhy ilngnkfwit ngpdadlviv yaktdlaavp asrgitafiv ekgmpgfsts 241
 kkldklgmrg sntcelifed ckipaanihg henkgvyvlm sgldlerlrvl aggpplgmqa 301 vldhtipylh
 vreafgqkig hfqlmqgkma dmytrlmacr qyvynvakac deghtakdc 361 agvilysaec atqvaldgiq
 cfgngyind fpmgrflrda klyeigagts evrrlvigra 421 fnadfh

(229) In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2
 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession
 number: NM_001159508.1, transcript variant 2):

(230) TABLE-US-00045 (SEQ ID NO: 27) 1 ttcccgagc taggggctgc tatttcaacg
 caggagata aaaagaaaaa aacacttgct 61 ctctacccc gtaaaaaaca ctatcctag ggagcacgcc
 agcatttgca gcgttcgggg 121 cagggccact cggcctgcgg ccgttgact ggctggaagc tggcaggcga
 tcacggttga 181 ttggctcggg tgcggtccaa gggcagcaac gccttcggcg ggccgcctag ggtgattggc
 241 tgctgcagcc cacccttag ccggtttggt gggcggcgaa gcctggattg gtggagctaa 301
 gagctggctc agtttcagcg ctggctcttc gtgcatggca gagatggcga ctgcgactcg 361 gctgctgggg
 tggcgtgtgg cgagctggag gctgcggccg ccgcttgccg gcttcgtttc 421 ccagcgggcc cactcgctt
 tgcccgtgga cgatgcaatc aatgggctaa gcgaggagca 481 gaggcaggaa ttttgaagc agctggggaa
 cctgggcgta ttgggcatca cagcccctgt 541 tcagtatggc ggctccggcc tgggctacct ggagcatgtg
 ctggtgatgg aggagatatt 601 ccgagcttcc ggagcagtgg ggctcagtta cggtgcccac tccaacctct
 gcatcaacca 661 gctgtacgc aatgggaatg agggccagaa agagaagtat ctcccgaagc tgatcagtgg
 721 tgagtacatc ggagccctgg ccatgagtga gcccaatgca ggctctgatg ttgtctctat 781
 gaagctcaaa gcggaaaaga aaggaaatca ctacatcctg aatggcaaca agttctggat 841 cactaatggc
 cctgatgctg acgtctgat tgtctatgcc aagacagatc tggctgctgt 901 gccagcttct cggggcatca
 cagccttcat tgtggagaag ggtatgcctg gctttagcac 961 ctctaagaag ctggacaagc tggggatgag
 gggctctaac acctgtgagc taatcttga 1021 agactgcaag attcctgctg ccaacatcct gggccatgag

aataagggtg tctactgtct1081 gatgagtggg ctggacctgg agcggctggg gctggccggg gggcctcttg
ggctcatgca 1141 agcggctctg gaccacacca ttccctacct gcacgtgagg gaagcctttg gccagaagat 1201
cggccacttc cagttgatgc aggggaagat ggctgacatg tacacccgcc tcatggcgtg 1261 tcggcagtat
gtctacaatg tcgccaaggc ctgcatgag ggccattgca ctgctaagga 1321 ctgtgcaggt gtgattcttt
actcagctga gtgtgccaca caggtagccc tggacggcat 1381 tcagtgtttt ggtggcaatg gctacatcaa
tgactttccc atgggccgct ttctcgaga 1441 tgccaagctg tatgagatag gggctgggac cagcgagggtg
aggcggctgg tcatcggcag 1501 agccttcaat gcagactttc actagtctg agacccttcg ccccttttc
ctgcacctag 1561 tggcctttct tgggaagtag agatgtggcg gcttcccac cctgccaca gcaggccctc 1621
ctgccagct gctctgtca gccctctggc ctctggatga ggttgagttc tccacaacag 1681 ctcccaagca
tcatgggcct cgcagccggg cctgtgccac ggctagtgtt gtgtgattta 1741 aatggactc agcaggaagc
atattgtctg gggattgtg ggacaggttt tggtagctct 1801 gtgcccttgc tctctaactt ctgagccac
ctcccagggt aggcacctgg gggcatgcag 1861 gtgccacct cccagggtag gcacctgggg gcattgcaggt
accacctct ttctctggg 1921 tgaggctctg gcaaggagat ctctctgctc aagcacagca gaatcatggc
ccctctccat 1981 gaattggaac ttggtacagg ttaagtatcc ctaatcctga aatctgaaac acttgtggtt 2041
ccaagcattt tggataaggc aaattcaact ttcagtctct ttctggggg aaaaaataa 2101 taaacctagc
ctagccaggc gtggtggctc atgcttgtaa tcccagcact tcaggaggct 2161 gagatgggtg gatcacctga
ggtcaggagt tcaagaccag cctggccaac atgtggaaac 2221 ctgcctcaa ctaaaaatag aaaaaatta
gttgggcatg gtggtgggca cctgtaatcc 2281 cagctacttc aggaggctga ggcaggagaa ttacttgaac
ccaggaggcg gacgttgag 2341 tgagccgagc ttgtgccatt gcactccagc ctgggcgaca agagcaaaac
tctcaaaaa 2401 acaaaacaaa acaaaaaaac cctggccctt gtttctcca gttctagag gtatcagctc 2461
ctagcagctt atgaacacat atgcttgctt ggccaggcaa ggtggtgtgt gcctgtaatc 2521 ccagcacttt
gggaggccaa ggcagggtga tcaattgcag tcaggagtgc aagaccagcc 2581 tgtccaacgt ggtgaaaccc
catctctact aaaaatacaa aaattagcca ggggtggtgg 2641 tgcacgtctg taatcccagc tactcaggag
gctgaggcag gagaatcact tgaaccggg 2701 aggtggaggt tgcaatgagc caatatgaca ccgtgcagt
ccagcctggg ccatagagt 2761 agactctgtc taaaaaagg aaagaaaaat aggctgggca cagtgactca
tgctgtaat 2821 cccaacactt tgggaggccg aggcagggtg atcacagggt caggagtcca agaccagcct 2881
ggccaagatg gtaaacctc gtcttacta aaaatacaa aattagccag gtgtggtggc 2941 aggctcctgt
aatcccagct actcaggagg ctgaggcaga gaattgctt aaccggggag 3001 gcagagtttg cagtgcagca
agatcacacc actgcactcc agcttgagc acagagcgag 3061 actctgtctc aaaaaataat aggccaggca
tggtggctca acgtctgtaa tcccagcact 3121 ttgggaggcc gaggcgggca gatcacaagg tcaggagtgc
gagaccagcc tgacgaccaa 3181 catggtgaaa cctcgtctct actaaaaata caaaaattag ccaggcctgg
tggcacgcgc 3241 ctgtaatccc agttacacag aagactgagg caggagaatc gcttgaacgc aggaggcaga
3301 ggttgagga gctgagatcg cgccattgca ctccagcctg ggcaacagag tgagactctg 3361
tctcaaaaaa taataataa ataatgaac acacatgctg ctgagtcgc agggggggca 3421 gagcagagga
cagcgtgctt ttgtgtactg ttggaagact ggctcctct gtacagcacc 3481 tctgagccct tgtgaccgc
cctgccacgg gcaccatcca gtctggccg tgtgaccacc 3541 cacagctgac tgggcagcag gcacaggccc
taccgagca ggccggagt ggctgcag 3601 actccagctg aggctgcctg gtacatttc tccagatacc
ctatggctaa tttgttata 3661 actgcacagt ggctgtgcc atttgtatt aatatattg tgaacaaac
ctatctgggg 3721 agaagcaatc tacttgccgc tgcttctgt ctggatccag cttgtgtct tggagagtgg 3781
ctggcccagg tctattcct gtctccagc ccgttcttc atgagggaca ggaaggtaaa 3841 atcagccctt
aggagagagg tctagcctc ctttcccag atctcccagt gaggtttaa 3901 ggaagcaggg agcccagagt
gctaagtct tacagccaga aggaagctta tagatttctg 3961 aaaaccgcc cttgtttt aaaaagatca
acacaattg actttctaa ggtcaaaacg 4021 aactagaatc cagatctgct catggcaaaa atgggggtgt
tctgagaatt ccagctttgg 4081 gccgactgt acagcagtct ggatagagt tgatctgaga agggaatggg
tctgggtgt 4141 tccaccctt ccgagttcca aaaagaggga actggtttc ttggttctca gccagcagc 4201
acatatcctg gctcttggtc ctggcctgca gccaaagtgt gttcctagcc tgaggcttga 4261 gacagggtggg
gttggctcct caccaacccc agttccgtcc catcctgagg gcaagatcct 4321 gggctcatag gcagtcctt
tacttctt gtctgtctc ctgctatgtt ggagatgaat 4381 gtgactaaaa gggccatctt gctggctta
tgtgtggctg gagagaccag cctggagaca 4441 atgtggcaaa atggggcgct tcatccagtc tgtctaagcc
ctgtcgactt ggggaggtga 4501 tttcttctt gtttctatat gtgaagcaaa ataatgttt taaaattaa

agcaaaaaaa 4561 acaaaatgaa ccatg

(231) In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NP_001152980.1, transcript variant 2):

(232) TABLE-US-00046 (SEQ ID NO: 28) 1 maematatrl lgwrvaswrl rpplagfvsq
rahslpvdd ainglseeqr qefwkqlgnl 61 gvlgitapvq yggsglgyle hvlvmeeisr asgavglisyg
ahsnlcinql vrngneaqke 121 kylpkklisge yigalamsep nagsdvvsmk lkaekkgnh y ilngnkfwit
ngpdadvliv 181 yaktdlaavp asrgitafiv ekgmpgfsts kkldklgmrg sntcelifed ckippaanilg 241
henkgvyvlm sgldlerlvl aggpplgmqa vldhtipylh vreafigqkig hfqlmqgkma 301 dmytrlmacr
qyvynvakac deghtakdc agvilysaec atqvaldgiq cfigngyind 361 fpmgrflrda klyeigagts
evrrlvigra fnadfh

(233) In some embodiments of the methods of the disclosure, the wild type human DPP9 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_139159.4):

(234) TABLE-US-00047 (SEQ ID NO: 29) 1 caacttccgg gtcaaagggtg cctgagccgg
cgggtccctt gtgtccgccg cggctgtcgt 61 cccccgtcc cgccacttcc ggggtcgcag tcccgggcat
ggagccgcga ccgtgaggcg 121 ccgctggacc cgggacgacc tgcccagtcc ggccgcccgc
ccacgtcccg gtctgtgtcc 181 cacgcctgca gctggaatgg aggtctctg gacccttag aaggcacccc
tgccctcctg 241 aggtcagctg agcggttaat gcggaagggt aagaaactgc gcctggacaa ggagaacacc
301 ggaagtggga gaagcttctc gctgaattcc gagggggctg agaggatggc caccaccggg 361
acccaacgg cgcaccgagg cgacgcagcc gccacagatg accgggccgc ccgcttcag 421
gtgcagaagc actcgtggga cgggctccgg agcatcatcc acggcagccg caagtactcg 481 ggctcattg
tcaacaaggc gccccacgac ttccagttg tgcagaagac ggatgagtct 541 gggccccact cccaccgcct
ctactacctg ggaatgccat atggcagccg agagaactcc 601 ctctctact ctgagattcc caagaaggtc
cggaaagagg ctctgtctg cctgtctctg 661 aagcagatgc tggatcatt ccaggccacg cccaccatg
gggtctactc tcgggaggag 721 gagtctgta gggagcgga acgcctgggg gtcttcggca tcacctcta
cgacttcac 781 agcgagagtg gcctcttct cttccaggcc agcaacagcc tcttccactg ccgcgacggc
841 ggcaagaacg gttcatggt gtcccctatg aaaccgctgg aaatcaagac ccagtgtca 901
gggccccgga tggaccccaa aatctgcct gccgacctg ctttctctc cttcatcaat 961 aacagcgacc
tgtgggtggc caacatcgag acaggcgagg agcggcggct gaccttctgc 1021 caccaagggt tatccaatgt
cctggatgac cccaagtctg cgggtgtggc caccttctgc 1081 atacaggaag agttcgaccg cttcactggg
tactggtggt gccccacagc ctctgggaa 1141 gggtcagagg gcctcaagac gctgcgaatc ctgtatgagg
aagtcatga gtccgagggt 1201 gaggtcattc acgtcccctc tcttgcgta gaagaaagga agacggactc
gtatcggtag 1261 cccaggacag gcagcaagaa tccaagatt gccttgaac tggctgagtt ccagactgac 1321
agccaggggc agatcgtctc gaccaggag aaggagctgg tgcagccctt cagctcgctg 1381 tccccgaagg
tggagtacat cgccagggcc ggggtggacc gggatggcaa atacgcctgg 1441 gccatgttcc tggaccggcc
ccagcagtgg ctccagctcg tctctctcc cccggccctg 1501 tcatcccga gcacagagaa tgaggagcag
cggctagcct ctgccagagc tgtccccagg 1561 aatgtccagc cgtatgtggt gtacgaggag gtcaccaacg
tctggatcaa tgttcatgac 1621 atcttctatc cttccccca atcagaggga gaggacgagc tctgcttct
ccgcgccaat 1681 gaatgcaaga ccggcttctg ccattgtac aaagtcaccg ccgttttaa atcccagggc 1741
tacgattgga gtgagccctt cagccccggg gaagatgaat ttaagtgcc cattaaggaa 1801 gagattgtc
tgaccagcgg tgaatgggag gtttggcga ggcacggctc caagatctgg 1861 gtcaatgagg agaccaagct
ggtgtacttc cagggcacca aggacacgcc gctggagcac 1921 cacctctacg tggtcagcta tgaggcggcc
ggcgagatcg tacgctcac cagccccggc 1981 ttctccata gctgctccat gagccagaac ttgacatgt
tcgtcagcca ctacagcagc 2041 gtgagcacgc cgccctgctg gcacgtctac aagctgagcg gccccacga
cgacccctg 2101 cacaagcagc cccgcttctg ggctagcatg atggaggcag ccagctgccc cccggattat
2161 gttctccag agatcttcca ttccacacg cgctcggatg tgcggctcta cggcatgac 2221 tacaagcccc
acgcttgca gccagggaag aagcacccca ccgtctctt tgtatatgga 2281 ggccccagg tgcagctggt
gaataactcc ttcaaaggca tcaagtactt gcggctcaac 2341 aactggcct ccctgggcta cgccgtgggt
gtgattgacg gcaggggctc ctgtcagcga 2401 gggcttcggt tcgaaggggc cctgaaaaac caaatgggcc

aggtggagat cgaggaccag 2461 gtggagggcc tgcagttcgt ggccgagaag tatggcttca tcgacctgag
 ccgagttgcc 2521 atccatggct ggtcctacgg gggcttctc tcgctcatgg ggctaacca caagccccag 2581
 gtgttcaagg tggccatcgc ggggtccccg gtcaccgtct ggatggccta cgacacaggg 2641 tacactgagc
 gctacatgga cgtccctgag aacaaccagc acggctatga ggcggggtcc 2701 gtggccctgc acgtggagaa
 gctgccaat gagcccaacc gcttgcttat cctccacggc 2761 ttcttgacg aaaacgtgca cttttccac
 acaaaactcc tcgtctcca actgatccga 2821 gcagggaaac cttaccagct ccagatctac cccaacgaga
 gacacagtat tcgctgcccc 2881 gagtcggcg agcactatga agtcacgttg ctgcacttcc tacaggaata
 cctctgagcc 2941 tgcccaccgg gagccgccac atcacagcac aagtggctgc agcctccgcg gggaaccagg
 3001 cgggagggac tgagtggccc gcgggcccca gtgaggcact ttgtcccgcc cagcgtggc 3061
 cagccccgag gagccgctgc cttaccgcc cgcagcctt ttatctttt ttaaagctc 3121 ttgggttta
 tgtccgctgc ttcttggtg ccgagacaga gagatgggtg tctcgggcca 3181 gccctctc tccccgctt
 ctgggaggag gaggtcacac gctgatggc actggagagg 3241 ccagaagaga ctgaggagg cgggctgcct
 tccgctggg gctccctgtg acctctcagt 3301 cccctggccc ggccagccac cgtccccagc acccaagcat
 gcaattgcct gtccccccg 3361 gccagcctcc ccaactgat gtttggtt tgttggggg gatattttc ataattatt
 3421 aaaagacagg ccggcgcgcg tggctcacgt ctgtaatccc agcactttg gaggctgagg 3481
 cgggcggatc acctgaggtt gggagttcaa gaccagcctg gccaacatgg ggaaaccccg 3541 tcttactaa
 aaatacaaaa aattagccgg gtgtggtggc gctgcctat aatcccagct 3601 actcgggagg ctgaggcagg
 agaatcgctt gaaccggga ggtggaggtt gcggtgagcc 3661 aagatcgac cattgcactc cagcctgggc
 aacaagagcg aaactctgtc taaaaataa 3721 taaaaataa aagacagaaa gcaaggggtg cctaaatcta
 gacttggggt ccacaccggg 3781 cagcggggtt gcaaccagc acctggtagg ctccatttct tccaagccc
 gagcagaggg 3841 tcatcgggc cccacaggag aagcggccag ggcccgcggg gggcaccacc tgtggacagc
 3901 cctctgtcc ccaagcttc aggcaggcac tgaacgcac cgaactcca cgctctgtg 3961
 gtcagtggcg gctgtccct cccagccca gccgcccagc cacatgtgtc tgcctgacc 4021 gtacacacca
 ggggttccgg ggttgggagc tgaaccatcc ccacctcagg gttatattc 4081 cctctccct tccctcccg
 ccaagagctc tgccaggggc gggcaaaaaa aaaagtaaaa 4141 agaaaagaaa aaaaaaaaaa agaaacaaac
 cacctctaca tattatgaa agaaaatatt 4201 ttgtcgatt cttattctt tataattatg cgtggaagaa
 gtagacacat taaacgattc 4261 cagttggaaa aaaaaaaaaa aaaaaa

(235) In some embodiments of the methods of the disclosure, the wild type human DPP9 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_631898.3):

(236) TABLE-US-00048 (SEQ ID NO: 30) 1 mrkvkklrld kentgswrsf slnsegaerm
 attgptadr gdaaatddpa arfqvqkhs 61 dglrsiihgs rkysglivnk aphdfqfvqk tdesgphshr
 lyylgmpygs renslysei 121 pkkvkeall llswkqmlhd fqatphhgv sreeellrer krlgvfgits
 ydfhesglf 181 lfqasnlfh crdggkngfm vspmkpleik tqcsgprmdp kicpadpaff sfinnsdwv 241
 anietgeerr ltfchqglsl vlddpksagv atfviqeedf rftgywwcpt aswegseglk 301 tlilyeevd
 esevevihvp spaleerktd syryprtgs npiaklae fqt dsqgkiv 361 stqekelvqp fsslfpkvey
 iaragwtrdg kyawamfldr pqqwlqlvll ppalfipste 421 neeqrlasar avprnvqpyv vyeevtnvwi
 nvhdifypfp qsegedelcf lranecktgtf 481 chlykvtavl ksqgydwsep fspgedefkc pikeeialts
 gewevlarhg skiwvneetk 541 lvyfqgtdt plehhlyvvs yeaageivrl ttpgfshscs msqnfdmfvs
 hyssystppc 601 vhvyklsgpd ddplhkqprf wasmmeaasc ppdyvppeif hfhtsdvrl ygmiykphal
 661 qpgkkhptvl fvyggpqvql vnnsfkgiky lrlntlaslg yavvvidgrg scqrgrlrfeg 721
 alknqmgqve iedqveglqf vaekygidl srvaihgwsy ggflslmgli hkpqvfkvai 781 agapvtvwma
 ydtgyterym dvpennqhgy eagsvalhve klpnepnrl ilhgfldenv 841 hffhtnflvs qliragkpyq
 lqiypnerhs ircpesgehy evtlhflqe yl

(237) In some embodiments of the methods of the disclosure, the wild type human SIGLEC14 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001098612.1):

(238) TABLE-US-00049 (SEQ ID NO: 31) 1 actcaccctc cggcttctg tcggggctt
 ctgacccca cccacgttt ggacattgg 61 agcatttct tccctgacag ccggacctgg gactgggctg
 gggccctggc ggatggagac 121 atgctgcccc tgctgctgt gccctgtg tgggggggggt ccctgcagga

gaagccagtgc 181 tacgagctgc aagtgcagaa gtcggtagac gtgcaggagg gcctgtgcgt cctgtgccc
 241 tgctccttct cttacccctg gagatcctgg tattcctctc cccactcta cgtctactgg 301 ttccgggacg
 gggagatccc atactacgct gaggtgtggt ccacaaacaa cccagacaga 361 agagtgaagc cagagaccca
 gggccgattc cgcctccttg gggatgtcca gaagaagaac 421 tgctccctga gcatcggaga tgccagaatg
 gaggacacgg gaagctattt cttccgcgtg 481 gagagaggaa gggatgtaa atatagctac caacagaata
 agctgaactt ggaggtgaca 541 gccctgatag agaaaccga catccacttt ctggagcctc tggagtccgg
 ccgccccaca 601 aggctgagct gcagccttc aggatcctgt gaagcgggac cacctctcac attctcctgg
 661 acgggggaatg ccctcagccc cctggacccc gagaccacc gctctcggga gtcaccctc 721
 acccccaggc ccgaggacca tggcaccaac ctcacctgtc aggtgaaacg ccaaggagct 781 caggtgacca
 cggagagaac tgtccagctc aatgtctct atgtccaca gaacctcgcc 841 atcagcatct tcttcagaaa
 tggcacaggc acagccctgc ggatcctgag caatggcatg 901 tcggtgccca tccaggaggg ccagtcctg
 ttctcgcct gcacagtga cagcaacccc 961 cctgcctcac tgagctggtt ccgggaggga aaagccctca
 atccttcca gacctaatg 1021 tctgggacc tggagctgcc taacatagga gctagagagg gaggggaatt
 cacctgccgg 1081 gttcagcatc cgctgggctc ccagcacctg tcctcatcc ttctgtgca gagaagctcc 1141
 tcttctgca tatgtgaac tgagaaacag cagggtcct ggcccctcgt cctcacctg 1201 atcagggggg
 ctctcatggg ggctggctc ctctcacct atggcctcac ctggatctac 1261 tataccaggt gtggaggccc
 ccagcagagc agggctgaga ggctggctg agcccctccc 1321 gctcaagaca gaactgaggt gtggacactt
 agccctgtgg gacacatgca ggacatcact 1381 gtcagcttct ttctggaagc tcacatcca ctgactacc
 ctcttttct tctgcccc 1441 tacccttct acttattccc ctctgctgt gagtcttgcc ccaccacacc
 tgcaccccc 1501 tctgcacccc atccctctc cacctgccct tctcttccct ctccatccac catctccagc 1561
 cctgtgaagg gaatgtactt tcggtcttat accccatta cccattacc aaaagtacc 1621 tttttttt tttttttt
 ttgagacaga gtctactct gttgcacagg ctggagtca 1681 gtggcacaat ctccgttcac tgcaacctcc
 acctctgggg ttcaagcaat tctctgcct 1741 cagcctcct agtagctggg attacaggtg cctgccacca
 catccagta attttttt 1801 ttgtatgtt agtagatg gggtttacc atgttgcca ggtctgaac tctgacctc
 1861 aagcaatcca ctgcattggc ctccaaagt gctggcatta caggtatgag ccaccgtgcc 1921
 tggctgcaa aagttacct ctaaacactt gaatttctg tctctcagc ttcctatcc 1981 atataggcac
 agagaggcag cattgtttt ccagttaaaa ctctacctca ttgtgattat 2041 tatccaatac aattgttaca
 aaataagtaa aacttttat aaacaataca acataactga 2101 tttactctt taa

(239) In some embodiments of the methods of the disclosure, the wild type human SIGLEC14 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001092082.1):

(240) TABLE-US-00050 (SEQ ID NO: 32) 1 mlpllllpll wggslqekpv yelqvqksvt
 vqeglcvlvp csfsypwrs wsspplyvyw 61 frdgeipyya evvatnnpdr rvkpetqgrf rllgdvqkkn
 cslsigdarm edtgsyffrv 121 ergrdvkysy qqnklnevt aliekpdi hf leplesgrpt rlscslpgsc
 eagppltfsw 181 tgnalspldp ettrsseltl tprpedhgtl ltcqvkrqga qvttertvl nvsyapqnla 241
 isiffnrgtg talrilsngm svpiqegqsl flactvdsn paslswfreg kalnpsqtsm 301 sgtlelpnig
 areggetcr vqhplgsqhl sfilsvqrss sscicvtekq qgswplvltl 361 irgalmgagf llygtltwi y
 ytrcgppqqs raerpg

(241) In some embodiments of the methods of the disclosure, the wild type human ADM2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001253845.1):

(242) TABLE-US-00051 (SEQ ID NO: 33) 1 cgcccacgcc cggcgccccg
 accgcggagg actccccgag cccgcgccgc catggcccgg 61 atcccagcgg ccgccttggg
 ttgcatcagc ctctctgcc tgcagctccc tggctcgtg 121 tcccgcagcc tgggaggggga cccgcgaccc
 gtcaaaccaca gggagcccc agcccggagc 181 cttccagca gcctgcagcc caggcaccac
 gcaccccgac ctgtggtctg gaagcttcac 241 cgggccctcc aggcacagag ggggtgccggc ctggcccctg
 ttatgggtca gcctctccgg 301 gatggtggcc gccaacactc gggccccga agacactcgg gccccgcag
 gacccaagcc 361 cagctctgc gactgggctg tgtgctgggc acctgccagg tgcagaatct cagccaccgc
 421 ctgtggcaac tcatgggacc ggccggccgg caggactcag ctctgtgga cccagcagc 481
 cccacagct atggctgagg tggggccggg ccacaccct gccatccca gccagggtgc 541 tgtgccccg

tcagagctg cagctgagcc ccactgaag ccagctcct cggagctgca 601 gacagcaggt cctgcagcaa
caatacctgc acggctttgc acacgtaaac ctaggctggt 661 ctacacgcag tgctggtacg tcaaggagcc
taaacaccct gaaattgtga cccctgggg 721 gacagctgcc agacacagct ggcggcagca ccagatgcta
agcgcttcag agaggaggtg 781 tctgccaga gatgtggagc agaagctggg ccctgaacac acggggccat
gtctggacga 841 gcaggggaga gaggctgaac tggccagaag tggcccctcc gctgctggtc cagtcaact
901 gaagcccggc cttgtgcctg ggctgttct gctctcatgc acaaccagcc cttccactg 961
cctgcctgtg ggacaggagg gggagcgtgg gatgctgtag cccccggggt tgggcaaggg 1021 aaggatggtg
gccctccaga ggtcatgaag ggacctctgt ggctccagct gccaaccctg 1081 gagcccagac cgaggtggcc
atggagactc cacctggatc ccctgtagga ggccaggag 1141 gggaactcag cagttcagga gccacccaa
accattctgg gacagggaca cccctttcta 1201 cccagggca gggcagggtt ggggtggggca agatcccca
gcccgactag accacactca 1261 cctgaagggg gtgagacct tgttggcagc cagacaaggg tggggctcca
caggcagcac 1321 aggcgccccca ccaccacca gtttggggac ccagtgggac caggtgcggg ggcagagggt
1381 gacttacaa gagccaggga gggcagcca ggccaagt acagcaagaa caagaaccac 1441
tgccggcgtg cacagactg gtgtgtgtcc ttcctgggg ggacggggga ctacatgtg 1501 cctgccactg
gagcctctca accgtccagc agaacacggg gttcagaaag ggctcctt 1561 gctatttagc gaactag
catttaattt acaaatgtt gtaggggtca cctctcggc 1621 catccacga gggtcgcat gatcccca
actctagagg ccgcagcaga gctcaggaca 1681 tccccaca gagcttccc ctacttct acctccaagg
gggagggtcc tggaagcgcc 1741 ccccaggcg ccgcccctgt gcttgcctcc cgagctcagg gattgccgag
tccacgtaac 1801 tgacctgtac tccacaggc cctgtgggaa cggctccaggc tggctcctcc ctgtggaggc 1861
ctcctgacac tgagagatgt actaggattg cagcaaagg gtgcagggtg atgggcccga 1921 cagcaggca
gtcaaggcca gtccttggg agaagcactg ggtcagggtga ggtctgagga 1981 cagcaggcct tcctagggg
aaggagctgg gagtgccaag gcccagggt cacaggaggc 2041 gtggctgctg agaggctgca
gggtggaggg gcctcggcct cagagtcag tgccctgtga 2101 ccactgaagg gtgtcagcag agcacacggc
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gtcaggagct cagcgtccgc tactcagccc 2221 agccaaaacc ctcccagacg tctcctctcc tgcctgggca
aagtccagct tggcaccccg 2281 tctggggcct gcctgtggtc agggccaagt gttcctct ccaggaaagc
cttaccctc 2341 ctatgcct gtagtcagga ggccgctgc tgaaccctc cgtgtcct cggtgcaa 2401
atcagacca cctgacacca tcacgcggag gccagcagc acctgcacc attccagct 2461 gctctggcca
aaatctccgc tcggccaggc cccgtggctc acactgtaa tctagcaca 2521 ttgggaggcc aaggcaggca
catcacctga gttcaggagt tcaagaccag cctggccaac 2581 atggtgaaat cccgtctcta ctaaaacag
aaaattatcc gggcgtggtg gcacatgact 2641 gtaatcccag ctactcagga ggctgaggca ggaggatcac
ttgaacctgg gaggcggagg 2701 ttgcagttag ctgagattgc gccattgcac tccagcctgg gcaacaagag
caaaattctg 2761 cctcaaaaaa aaaaatagta ataatacaa aattagctgg gcgtggtggc acatgccagt 2821
aattccatct actcgggagg ctgaggcagg agaatcgtct aagcccggga ggtggaggtt 2881 gcagtgagcc
cagatggcgc tgctgcactc aagcttgat gacagagcaa gactccgtt 2941 caaaaaaaaa aaacctctc
tctcttca caccttctc tgaatccac ccggtccac 3001 ctctgaacc tatccagaca ctttctctg
accaggcac cacctgctt cggggcgatg 3061 gccgtagct ctcccaggc acctgtctgc atccctctg
ccagtgcag ctgagcacgt 3121 gacctaccg tgttgggaca cgtgaggata cagccttgac cccaggggc
tgacattcta 3181 gggggagata gaaggagaca aacgtagaag gtagaataag tgggtggtg agtggcaggg
3241 agtgctgagt gccacaggaa gtcagacaag gaaggagagt gtggggcagg tgccgttaa 3301
atggggggcg ctggggtctc ctacagttg ctctcagct cagctgtgcc aggatctgt 3361 tgagttagt
cagctgcca cagccctctt gctgaccc tgaagcccag aactctgatc 3421 ttcacagccc taggtatggc
cccagcacc cactgccctc tctctgccc cagccgactg 3481 ctgttcccag attccttg ccacgtcca
agacgccagc tctgccgagg gactttgtt 3541 ctacgggtg cctccatgcc tgcaggggcc atgcatggga
agttgcgtg gcggcctggg 3601 tgttggcgt tccgtgctg ctccaactc ccgtgaggcc ccttcccag
agcctgacac 3661 actctgtggc cgaactctag gcaggtgcc ctgagtcct tctcgacga ggcctgacc 3721
catccccatc ctgctgggc ccggcagcc cggtgttagc aagaatctc taaatcagt 3781 tatggagaat
taccaccct cgatatctga tccattct catctccac cttgatctc 3841 ataccctgc cggcctctg
caagatctc attgagccac tccagtgaga atccccctac 3901 cctgaaggc cgcctaaca attcccatc
cgctgaccc tccaacgcca tcaatctca 3961 gctgtggtg ttgaactcg aggtgagctc ctctaccac

tctctgaat aaagcttttc 4021 tcacattttt aaaaaaa aaaaa

(243) In some embodiments of the methods of the disclosure, the wild type human ADM2 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001240774.1):

(244) TABLE-US-00052 (SEQ ID NO: 34) 1 mariptalg cislclqlp gslsrslggd
prpvkprepp arspssslqp rhpaprpvvw 61 klhralqaqr gaglapvmgq plrdggrqhs gprrhsgpr
tqaqlrvgc vlgtcqvnql 121 shrlwqlmgp agrqdsapvd pssphsyg

(245) In some embodiments of the methods of the disclosure, the wild type human TSPAN5 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_005723.3):

(246) TABLE-US-00053 (SEQ ID NO: 35) 1 aggcggggcgg agcgaggggt
gggagggcgc gcgcgaacgg gcggggcgagc aagcgagcgg 61 cgtctccacc agcatctgcc
gcggccgcct ttccccgaag cccggggagc aaccgacgga 121 ccgaccgcct ggcgacgga
cgcgggcgct cgcttttgtt tcggggctag cgtcggcgag 181 gcttgagctt gcagcgcgcg gcttcctgc
tttctcgcg ccacccggc tccggcgcc 241 tcggcgcgcg aggggctgga ggtgcgggag
ccgctctccg ccggtcggtc cccgcgcggc 301 tgagcccagg ccgccagcgc cgcgccccg
tgcggtgtcc ctgagctct gttccccgc 361 gggctgtctc gagcaacggg gttcggagc tccaaactcg
ggctgccggg gcaagtgtct 421 tcatgaacc agaggatgtc cgggaagcac tacaagggtc ctgaagtcag
ttgtgcatc 481 aaataactca tatttggtt caatgtcata tttggttt tgggaataac atttcttga 541
attggactgt gggcatggaa taaaaagga gttctgtcca acatctctc catcaccgat 601 ctggcggt
ttgaccagt ttgctctc cttgtgtgg gaggagtgt gttcatttg 661 ggatttgag ggtgcattg
agcgctacgg gaaaacactt tcttctcaa gtttttct 721 gtgttctgg gaattattt ctctctggag
ctactgccg gaggcttagc attgtttt 781 aaagactgga tcaaagacca gctgtattt ttataaaca
acaacatcag agcatatcg 841 gatgacattg atttgcaaaa cctcatagac ttcaccagg aatattggca
gtgctgtgg 901 gcttttggag ctgatgatt gaacctaat atttactca attgcacaga ttccaatgca 961
agtcgagagc gatgtggcgt tccattctc tgctgacta aagatccgc agaagatgtc 1021 atcaacactc
agtgtggcta tgatgccagg caaaaaccag aagtgacca gcagattgta 1081 atctacacga aaggctgtgt
gccccagttt gagaagtgtg tgcaggacaa ttaaccatc 1141 gttgttgta tttcatagg cattgcattg
ctgcagatat ttgggatatg cctggcccag 1201 aatttggtta gcgatatcga agctgtcagg gcgagctgg
agacccccctg caaccgtgc 1261 tgcaagacac tggacagacc cagcttccg gaccctccg cgtgccgaac
tgatcttga 1321 gctgcatgga cctaataca gatgcagcct gcagtctgc ctaatggagc tgccattagg 1381
ggagtgtaaa actgggaaat gctgtcact gacagaatta aaaaaaaaaa taaccagtat 1441 gaaagtcgt
gcgccgtgaa tcttactgt agccatgaat ttatggacag ttagatgctt 1501 accaaaaaag aaaaaagg
agggtagggg acccagatgt acttgaatgt gcagaaaata 1561 cattctgtc ctcatctcc gtaattggag
ggctgggaga ggcagctttg ctctcacca 1621 cacctggac ggaccacct ctttctgtc catggcctga
aggagtgcatt ctctcaaag 1681 actcagcccc tcacctggga gggcagtggt ttgtgggcat cctccatgt
acatttagg 1741 aaacactgc aactctatc tgaagaagaa aacaactcat cttgggtc agattttgtg 1801
atggtattca gcaagtcact tgggcgagca cacttggtc atcctggaaa gtctcctat 1861 aagagaagt
gtgtatttca tgtgaccga gcaaggcat tggaagacgt catgaggctg 1921 tattttagca ggactgatc
ttttctaag tagacctgag cttgtttat cagtgaatt 1981 caaggagaaa atgaggtaa tgaagagga
tcagttaaatt atccccctt ttcacctg 2041 caaaaattag cagttggatt ttggaaact ctggaatatt
ctgggtcatt ttgtttgta 2101 tgtttgtgt tttcgtct ccaaagggtga aagctatgat acagttccac ttaaattta
2161 gtgtttctt actcagctca agcattaatt ttgattaag tctaatctg catgacctgt 2221 gaatctgaat
ccatcatctc ctttctgc cagctttct acaaacattg aaatatgta 2281 ttggtcagc acttatttc
taggttaca gcctgggag gttgtggcat gtctcccag 2341 tctggctggg aagagaccag ctgtaccatc
caaatgctc cctggtctg atgatctt 2401 ccagagtcga tctgagtggc ctttctgca cctcccct
ctttctttt gaatggaatt 2461 aaaccaatt tggaaacaac attgaccag tcaaaagctt ctaatgggtt cttttctt
2521 ctccagttt agtttgctt tattaataaa agaaaatagt gcatggccat agctcctca 2581 gttcttat
tgcagactaa ccatcaggat ggtatcaaag cacaataact ttggagggga 2641 atgcgttgaa ctggggcaag
tactctgtaa ccaaagtgg gaaaccact cctggtgctg 2701 ccgtctctgc cccacttta ggtgggagg

acgagttttg ccctctagat ttttaaccag 2761 ctggtgtcca ccgagatgtg ccctcctggg gagcagatat
cagtctgtgg aactctggga 2821 aaaccacagg cacattttc ggtgcggaca gattgccag cacataactg
ggcagccagc 2881 tagaatactt tgtggaaatt aagcgaggtt ttccatttca gccccatggt gcatgggtgt 2941
ggccgatgaa tgtgtcagtc tgctcagaga aaggacaaaa aggaaattat tttcaaaact 3001 gtgttcactg
tttgggtgtg tgtatggctc tgcattgtgt tgttttgtc tctgtatagg 3061 tagaggtatt cacatcttac
tccgactgta aggttgtctt acttcatctc tgccccacc 3121 acagttgcca tttgtaatg tcctccaac
atggagaaga cagcagctct ctccagtgg 3181 catcatttgt ctttttgtt gattgcctca ttctccagtg
aactccatct ggccaattga 3241 tcagaatca ggcaagatcc ctgcccttg gcacatccac tgaaaggcca
aacagcaagt 3301 ccgagtgagt ttaaatatt aattaatcac ctttattt ttacactga gagtgattgt 3361
aataaaggct gtcattaata aacttggtc taccttaaaa aaaaaa

(247) In some embodiments of the methods of the disclosure, the wild type human TSPAN5 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_005714.2):

(248) TABLE-US-00054 (SEQ ID NO: 52) 1 msgkhykpe vsccikyif gfnvifwflg
itflgiglwa wnekglvlni ssitdlggfd 61 pvwlflvvgg vmfilgfagc igalrentfl lkffsvflgi
iffleltagv lafvfkdwik 121 dqlyffinnn irayrddidl qnlidftqey wqccgafgad dwnlniyfnc
tdsnasrerc 181 gvpfscctkd paedvintqc gydarqkpev dqquviytkg cvpqfekwlq dnltivagif 241
igiallqifg iclaqnlvsd ieavrasw

(249) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_032294.2, transcript variant 1):

(250) TABLE-US-00055 (SEQ ID NO: 53) 1 ctgggccccca gcgaggcggg
ggggcggggc ggggcggggc ggggcgcgca gcaggagcga 61 gtggggccgc ccgccggggc
gcggacactg tcgccggcg cccaggttcc caacaaggct 121 acgcagaaga acccccttga ctgaagcaat
ggaggggggt ccagctgtct gctgccagga 181 tcctcgggca gagctggtag aacgggtggc agccatcgat
gtgactcact tggaggaggc 241 agatggtggc ccagagccta ctagaaacgg tgtggacccc ccaccacggg
ccagagctgc 301 ctctgtgatc cctggcagta cttcaagact gctcccagcc cggcctagcc tctcagccag
361 gaagctttcc ctacaggagc ggccagcagg aagctatctg gaggcgcagg ctgggcctta 421
tgccacgggg cctgccagcc acatctcccc ccgggcctgg cggaggccca ccatcgagtc 481 ccaccacgtg
gccatctcag atgcagagga ctgcgtgcag ctgaaccagt acaagctgca 541 gagtgaatt ggcaagggtg
cctacggtgt ggtgaggctg gcctacaacg aaagtgaaga 601 cagacactat gcaatgaaag tcctttcaa
aaagaagtta ctgaagcagt atggctttcc 661 acgtcgccct cccccgagag ggtcccaggc tgcccaggga
ggaccagcca agcagctgct 721 gccctggag cgggtgtacc aggagattgc catcctgaag aagctggacc
acgtgaatgt 781 ggtcaaaactg atcgaggtcc tggatgacct agctgaggac aaccttatt tgggtttga 841
cctcctgaga aaggggcccc tcattggaagt gccctgtgac aagcccttct cggaggagca 901 agctcgccct
tacctgcggg acgtcatcct gggcctcgag tacttgact gccagaagat 961 cgtccacagg gacatcaagc
catccaacct gctcctgggg gatgatgggc acgtgaagat 1021 cgccgacttt ggcgtcagca accagttga
ggggaacgac gctcagctgt ccagcacggc 1081 gggaacccca gcattcatgg cccccaggc catttctgat
tccggccaga gcttcagtgg 1141 gaaggccttg gatgtatggg cactggcgt caggtgtac tgctttgtct
atgggaagtg 1201 ccattcatc gacatttca tctggccct ccacaggaag atcaagaatg agccgtggt 1261
gttctctgag gagccagaaa tcagcgagga gctcaaggac ctgactctga agatgttaga 1321 caagaatccc
gagacgagaa ttgggtgcc agacatcaag ttgcacctt gggtgaccaa 1381 gaacggggag gagcccttc
cttcggagga ggagcactgc agcgtggtgg aggtgacaga 1441 ggaggagggt aagaactcag tcaggctcat
ccccagctgg accacggtga tctggtgaa 1501 gtccatgctg aggaagcgtt ctttgggaa cccgtttgag
cccaagcac ggagggaaga 1561 gcatccatg tctgtccag gaaacctact ggtgaaagaa gggtttggtg
aagggggcaa 1621 gagcccagag ctccccggcg tccaggaaga cgaggctgca tctgagccc ctgcatgcac
1681 ccagggccac ccggcagcac actcatccc cgctccaga ggcccaccc tcattgaaca 1741
gccgccccg caggcagggg gctggggact gcagccccc tcccgcctt ccccatcgt 1801 gctgcatgac
ctccacgcac gcagtcctg ggacagactg gaatgtatgt catttgggt 1861 ctggggggca gggctccac
gaggccatcc tctcttctt ggacctctt ggcctgacct 1921 attctgtggg gaaaccgggt gccatggag

cctcagaat gccaccggc tgggtggcat 1981 ggctggggc aggaggcaga ggcaggagac caagatggca
 ggtggaggcc aggttacca 2041 caacggaaga gacctccgc tggggccggg caggcctggc tcagctgcca
 caggcatatg 2101 gtggagagg ggggtacctg cccacctgg ggtgggtggca ccagagctct tgtctattca 2161
 gacgtggta tgggggctcg gaccctcac tggggacagg gccagtgtg gagaattctg 2221 attcctttt
 tgtgtcttt tactttgtt ttaacctgg ggggtcgggg agaggccctg 2281 cttgggaaca tctcacgagc
 tttctacat ctccgtggt tcccagcaca gcccaagatt 2341 attggcagc caagtggatg gaactaactt
 tctggactg tgttcgcat tcggcggtat 2401 ctggaaagt gactgaacgg aatcaagctc tgagcagagg
 cctgaagcgg aagcaccaca 2461 tcgtccctgc ccatctact ctctccctg atgatgccc tagagctgag
 gctggagaag 2521 acaccagggc tgacttgac cgaggggccat ggacgcgaca ggcctgtggc cctgcgcatg
 2581 ctgaataaac tggaaccag cctctctcc tacaccggc taccatctg ggccaagag 2641
 ctgcactcac actctacaa cgaaggacaa actgtccagg tcggagggat cacgagacac 2701 agaactgga
 ggggtgtgca cgctggcagg tggcctctgc ggcaattgcc tcaccctgag 2761 gacatcagca gtcagcctgc
 tcagagcggg ggtgctggag cgctgcaga cacagctctt 2821 ccggagcagc cttaccttc tctctggat
 cagtgtccgg ctggccgacg tggcattgc 2881 tgaccgaatg ctcatagagg ttgacccca cagggtcacg
 caggactcg acactgccct 2941 ggaaacatgg atggacaagg gctttggcc acaggtgtgg gtgtcctgtt
 ggaggagggc 3001 ttgttgag aaggaggct ggctggggga gaaaccgga tcccgctgca tctccgcgc
 3061 tgtgggtgca tgtcgcgtgc tcatctgtg cacacagctc actcgtatgt cctgcactgg 3121 tacatgcatc
 tgtaatacag ttttacgtc tatttaaggc taggagccga atgtgcccc 3181 ttgtcagtgg gtccacgtt
 ctccccggct cctctgggt aaggcagtgt ggcccgaagc 3241 ttaaaaagtt actcgtact gttttaaga
 acactttat agagttagtg gaaggcaagt 3301 taagagcaa tcactgatcc ccaagtgtt cttgagcatc
 tggctgggg ggaccactt 3361 gatcggacc accctggaa agctcaggg tagggccagg tgggatgctc
 accctgtcac 3421 tgagggttt ggttggcatc gttgttttg aatgtagcac aagcgatgag caaactctat 3481
 aagagtgtt taaaaattaa ctccagga agtgagtaa aaacaataaa agcccttct 3541 tgagttaaa
 agaaaaaaa aaaaaaaaaa aaaaaaaaaa aaa

(251) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_115670.1, transcript variant 1):

(252) TABLE-US-00056 (SEQ ID NO: 54) 1 meggpavccq dpraelerv aaidvthlee
 adggpeptrn gvdppprara asvipgstsr 61 llparpsla rklslqerpa gsyleaqagp yatgpashis
 prawrrptie shhvaisdae 121 dcvqlnqykl qseigkgayg vvrlaynese drhyamkvls kkkllkqygf
 prppprgsq 181 aaqggpakql lplervyqei ailkkldhvn vvklieldd paednlylvf dllrkgpvme 241
 vpcdkpfsee qarlylrdiv lgleylhcqk ivhrdikpsn llgddghvk iadfgvsnf 301 egndaqlsst
 agtpafmape aisdsgqsfs gkaldvwatg vtlycfvygk cpfiddfila 361 lhrkiknepv vfpeepeise
 elkdliikml dknpetrigv pdiklhpwvt kngeelpse 421 eehcsvvevt eeeknsvrl ipswttvilv
 ksmrlkrsgf npfepqarre ersmsapgnl 481 lvkegfgegg kspelpgvqe deaas

(253) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_172206.1, transcript variant 2):

(254) TABLE-US-00057 (SEQ ID NO: 55) 1 agcagaacag agtatgcaat ttgggaagct
 gtggtgtggc tgcagtggag agttcccaac 61 aaggctacgc agaagaacc cttgactga agcaatggag
 gggggtccag ctgtctgctg 121 ccaggatcct cgggcagagc tggtagaacg ggtggcagcc atcgatgtga
 ctacttga 181 ggaggcagat ggtggcccag agcctactag aaacgggtgtg gacccccac cacgggccag
 241 agctgcctct gtgatccctg gcagtactc aagactgtc ccagcccggc ctgcctctc 301
 agccaggaag ctttcctac aggagcggcc agcaggaagc tatctggagg cgcaggctgg 361 gccttatgcc
 acggggcctg ccagccacat ctcccccg gcctggcgga ggcccaccat 421 cgagtcacc
 cacgtggcca tctcatatgc agaggactgc gtgcagctga accagtacaa 481 gctgcagagt gagattggca
 aggggtgccta cgggtgtgtg aggctggcct acaacgaaag 541 tgaagacaga cactatgcaa tgaaagtct
 ttcaaaaaag aagtactga agcagtatgg 601 cttccacgt cgccctccc cgagagggtc ccaggctgcc
 caggaggagc cagccaagca 661 gctgtgccc ctggagcggg tgtaccagga gattgccatc ctgaagaagc
 tggaccacgt 721 gaatgtgtc aaactgatc aggtcctgga tgaccagct gaggacaacc tctattgtt

781 gtttgacctc ctgagaaagg ggccccgtcat ggaagtgcc tgtgacaagc ccttctcgga 841
 ggagcaagct cgcctctacc tgcgggacgt catctgggc ctcgagtact tgcactgcc 901 gaagatcgct
 cacagggaca tcaagccatc caacctgctc ctgggggatg atgggcacgt 961 gaagatcgcc gactttggcg
 tcagcaacca gtttgagggg aacgacgctc agctgtccag 1021 cacggcgga accccagcat tcatggcccc
 cgaggccatt tctgattccg gccagagctt 1081 cagtgggaag gccttggatg tatgggccac tggcgtcacg
 ttgtactgct ttgtctatgg 1141 gaagtgccca ttcactgacg atttcatct ggccctccac aggaagatca
 agaatgagcc 1201 cgtggtgttt cctgaggagc cagaaatcag cgaggagctc aaggacctga tcctgaagat 1261
 gttagacaag aatcccagaga cgagaattgg ggtgccagac atcaagtgc acccttgggt 1321 gaccaagaac
 ggggaggagc cccttcctc ggaggaggag cactgcagcg tggtaggag 1381 gacagaggag gaggttaaga
 actcagtcag gctcatcccc agctggacca cggtgatcct 1441 ggtgaagtc atgctgagga agcgttcct
 tgggaacccg tttgagcccc aagcacggag 1501 ggaagagcga tccatgtctg ctccaggaaa cctactggtg
 aaagaagggt ttggtgaagg 1561 gggcaagagc ccagagctcc ccggcgcca ggaagacgag gctgcacatc
 gagccctgc 1621 atgacccag ggccaccg cagcacactc atcccgcc tccagaggcc caccctcat
 1681 gcaacagccg ccccgagcagg cagggggctg gggactgcag cccactccc gccctccc 1741
 catcgtgctg catgacctc acgcacgcac gtccaggagc agactggaat gtatgtcatt 1801 tggggtcttg
 ggggcagggc tcccagagg ccactcctc cttctggac ctcttggcc 1861 tgaccattc tgtggggaaa
 ccgggtgccc atggagcctc agaaatgcca cccggctggt 1921 tggcatggcc tggggcagga ggcagaggca
 ggagaccaag atggcagggt gaggccaggc 1981 ttaccacaac ggaagagacc tccgctggg
 gccggcagg cctggctcag ctgccacagg 2041 catatggtgg agaggggggt accctgcca cttgggggtg
 gtggcaccag agctctgtc 2101 tattcagacg ctggtatggg ggctcggacc cctcactggg gacagggcc
 gtgttgaga 2161 attctgattc cttttgtt gtctttact tttgttta acctgggggt tcggggagag 2221
 gccctgctg ggaacatctc acgagcttc ctacatctc cgtggttccc agcacagccc 2281 aagattatt
 ggagccaag tggatggaac taacttctt ggactgtgt tcgattcgg 2341 cgttatctgg aaagtggact
 gaacggaatc aagctctgag cagaggcctg aagcggaagc 2401 accacatcgt ccctgcccac ctactctc
 cccttgatga tgccctaga gctgaggctg 2461 gagaagacac cagggtgac ttgaccgag ggccatggac
 gcgacaggcc tgtggccctg 2521 cgcagctga aataactgga accagcctc tctctaca ccggcctacc
 catctgggc 2581 caagagctgc actcacactc ctacaacgaa ggacaaactg tccaggtcgg agggatcacg
 2641 agacacagaa cctggagggg tgtgcacgct ggcaggtggc ctctcggca attgcctcac 2701
 cctgaggaca tcagcagta gcctgctcag agcgggggtg ctggagcgcg tgcagacaca 2761 gctctccg
 agcagcctc acctctctc tgggatcagt gtccggctgg ccgacgtggc 2821 attgctgac cgaatgctc
 tagaggttga cccacagag gtcacgcagg actcggacac 2881 tgccctggaa acatggatg acaagggtt
 ttggccacag gtgtgggtgt cctgttgag 2941 gagggctgt ttggagaagg gaggtggtt gggggagaaa
 cccgatccc gctgcacctc 3001 cgcgcctgtg ggtgcatgt gcgtgctcat ctgtgcaca cagctcactc
 gtatgtcctg 3061 cactggtaca tgcactgta atacagttc tacgtctatt taaggctagg agccaatgt 3121
 gcccattgt cagtgggtcc acgttctcc ccgctctc tgggctaagg cagtgtggcc 3181 cgaagctaa
 aaagtactc ggtactgtt ttaagaacac tttatagag ttagtgaag 3241 gcaagttaag agccaatcac
 tgatcccaa gtgttctg agcatctgt ctggggggac 3301 cactttgat ggaccaccc ttggaaagct
 cagggttagg ccaggtggg atgtcaccc 3361 tgtcactgag ggtttggtt ggcacgtt ttttgatg
 tagcacaagc gatgagcaa 3421 ctctataaga gtgtttaaa aattaactc ccaggaagt agttaaaac
 aataaaagcc 3481 ctttctgag taaaaagaa aaaaaaaaa aaaaaaaaa aaaaaaaaa

(255) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_757343.2, transcript variant 2):

(256) TABLE-US-00058 (SEQ ID NO: 56) 1 mqf gklwgc sgefptrlrr rtplteameg
 gpavccqdr aelverv aai dvthleeadg 61 gpeptrngvd ppraraasv ipgstrllp arpslsarkl
 slgerpagsy leaqagpyat 121 gpashispra wrptieshh vaisdaedcv qlnqyklqse igkgaygvvr
 laynesedrh 181 yamkvlskkk llkqygfpr pprgsqaaq ggpakqllpl ervyqeail kkldhvnvkv 241
 lievlddpae dnlylvfdll rkgpvmevpc dkpfseeqar lyldrivilg eylhcqkivh 301 rdikpsnlll
 gddghvkiad fgvsnfegn daqlsstagt pafmapeais dsqsfsgka 361 ldvwatgvtl ycfvygkcpf
 iddfilalhr kiknepvvfp eepeiseelk dlilkmldkn 421 petrivpdi klhpwvtkng eeplpseeh

csvvveetee vksnvrllps wttvilvksm 481 lrksrfgnfp epqarreers msapgnllvk egfgegksp
elpgvqedeas

(257) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_172207.2, transcript variant 3):

(258) TABLE-US-00059 (SEQ ID NO: 57) 1 ctgggccccca gcgaggcggg
ggggcggggc ggggcggggc ggggcgcgca gcaggagcga 61 gtggggccgc ccgccggggc
gcggacactg tcgccggcg cccagggtcc caacaaggct 121 acgcagaaga acccccttga ctgaagcaat
ggaggggggt ccagctgtct gctgccagga 181 tcctcgggca gagctggtag aacgggtggc agccatcgat
gtgactcact tggaggaggc 241 agatggtggc ccagagccta ctagaaacgg tgtggacccc ccaccacggg
ccagagctgc 301 ctctgtgatc cctggcagta cttcaagact gctcccagcc cggcctagcc tctcagccag
361 gaagctttcc ctacaggagc ggccagcagg aagctatctg gaggcgcagg ctgggcctta 421
tgccacgggg cctgccagcc acatctcccc ccgggcctgg cggaggccca ccatcgagtc 481 ccaccacgtg
gccatctcag atgcagagga ctgcgtgcag ctgaaccagt acaagctgca 541 gagtgagatt ggcaagggtg
cctacggtgt ggtgaggctg gcctacaacg aaagtgaaga 601 cagacactat gcaatgaaag tcctttccaa
aaagaagtta ctgaagcagt atggctttcc 661 acgtcgccct cccccgagag ggtcccaggc tgcccaggga
ggaccagcca agcagctgct 721 gccctggag cgggtgtacc aggagattgc catcctgaag aagctggacc
acgtgaatgt 781 ggtcaaaactg atcgaggctc tggtgaccc agctgaggac aacctctatt tggccctgca
841 gaaccaggcc cagaatatcc agttagattc aacaaatc gccaaagcccc actccctgct 901
tcctctgag cagcaagaca gtggatccac gtgggctgcg cgctcagtgt ttgacctct 961 gagaaagggg
cccgtcatgg aagtgcctg tgacaagccc ttctcggagg agcaagctcg 1021 cctctacctg cgggacgtca
tcctgggcct cgagtactg cactgccaga agatcgcca 1081 caggacatc aagccatcca acctgctct
gggggatgat gggcacgtga agatcgcca 1141 ctttggcgtc agcaaccagt ttgaggggaa cgacgctcag
ctgtccagca cggcgggaac 1201 ccagcattc atggccccg aggccattc tgattccggc cagagctca
gtgggaaggc 1261 cttggatgta tgggccactg gcgtcacgtt gtactgctt gtctatggga agtggccatt 1321
catcgacgat ttcactctgg cctccacag gaagatcaag aatgagccc tggtgttcc 1381 tgaggagcca
gaaatcagcg aggagctcaa ggacctgatc ctgaagatgt tagacaagaa 1441 tcccagacg agaattgggg
tgccagacat caagttgcac cttgggtga ccaagaacgg 1501 ggaggagccc cttcttcgg aggaggagca
ctgcagcgtg gtggaggtga cagaggagga 1561 ggtaagaac tcagtcaggc tcatccccag ctggaccag
gtgatcctgg tgaagtccat 1621 gctgaggaag cgttctttg ggaaccggt tgagcccaa gcacggaggg
aagagcgatc 1681 catgtctgct ccaggaaacc tactggtgta agtactggtg ggccagggac tgccgggac
1741 tcctggagt tgggtgggga ggtctgaggc ccactctcc actctactg tcgttgggcc 1801
aaggccagag cctggggact tggccaggtc tcggtgttg cccatttgc atctctgtcc 1861 ccaaggtag
tcggggctag aagggacctt ttgggccag ctctgttc attctgggg 1921 ccagcatccc tcacacac
acttcaggg atgaggagct cacgcagccc ctccatggga 1981 caggaagacc cttctccat gcagcttgat
gtcactctc cactgggtcc agcccctctg 2041 gggttcaaa tctgtggccc cctcagccct tggcagcctg
gcagaggtt gcagacaggc 2101 tgatgttggc ttctgtagg aggtggcgg gctgtagagg aggggtgctg
gccctctgc 2161 ctggccctgg ggactgttg ctgctctcc aagtggcca ggctgcctgc agccattgct 2221
ggggctctgt gccagtcag cacttttga gtgctgttc agtgagtaag cagggacagg 2281 ctggccggtg
gaccacggga gaggaacccg cattggcca gggctcccta tggtagcca 2341 cgctgtggg ttaccacct
cctaggaggg tccagaaaag cagctccca agcctgtgcg 2401 cctcgtctc agcagatcca cttcttcac
tataataaaa gccagtctgg gatgctaaa 2461 aaaaaaaaa aaaaaaaaa aaaaaaaaa aaaaaaaaa
aaaaaaaaa aaaaaaaaa 2521 aaaaaaaaa aaaaa

(259) In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_757344.2, transcript variant 3):

(260) TABLE-US-00060 (SEQ ID NO: 58) 1 meggpavccq dpraelerv aaidvthlee
adggpeptrn gvdppprara asvipgstsr 61 llparpsla rklsqerpa gsyleaqagp yatgpashis
prawrrptie shhvaisdae 121 dcqlnqyk qseigkgay vvrlaynese drhyamkvl kkkllkqygf
prppprgsq 181 aaqggpakql lplervyqei ailkkldhvn vvklieldd paednlylal qnqaqniql 241

stniakphsl lpseqqdsqs twaarsvfdl lrkpvmevp cdkpfseeqa rlylrdvilg 301 leylhlcqkiv
hrdikpsnll lgddghvkia dfgvsnqfeg ndaqlsstag tpafrmapai 361 sdsgqsfsqk aldvatgvt
lycfvygkcp fiddfilalh rkiknepvfv peepeiseel 421 kdlilkml dk npetrigvpd ikhlpwvtn
geepipseee hcsvvevtee evknsvrli 481 swttvilvks mlrksfngp fepqarreer smsapgnllv
(261) In some embodiments of the methods of the disclosure, the wild type human MMP7 gene of
the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:
NM_002423.4):

(262) TABLE-US-00061 (SEQ ID NO: 59) 1 gaaaacacca aatcaaccat aggtccaaga
acaattgtct ctggacggca gctatgcgac 61 tcaccgtgct gtgtgctgtg tgcctgctgc ctggcagcct
ggccctgccg ctgcctcagg 121 aggcgggagg catgagttag ctacagtggg aacaggctca ggactatctc
aagagatttt 181 atctctatga ctcagaaaca aaaaatgcc aacagttaga agccaaactc aaggagatgc
241 aaaaattctt tggcctacct ataactggaa tgtaaactc ccgcgtcata gaaataatgc 301 agaagcccag
atgtggagtgc ccagatgttg cagaatactc actatttcca aatagcccaa 361 aatggacttc caaagtggctc
acctacagga tcgtatcata tactcgagac ttaccgcata 421 ttacagtggg tcgattagtgc tcaaaggctt
taaactgtg gggcaaagag atccccctgc 481 attcaggaa agttgtatgg ggaactgctg acatcatgat
tggctttgcg cgaggagctc 541 atgggggactc ctaccattt gatggggcag gaaacacgct ggctcatgcc
tttgcgcctg 601 ggacaggtct cggaggagat gctcacttcg atgaggatga acgctggacg gatggttagca
661 gtctagggat taacttctg tatgtgcaa ctcatgaact tggccattct ttgggtatgg 721 gacattcctc
tgatccta at gcagtgtatg atccaacct tggaaatgga gatccccaaa 781 attttaact ttcccaggat
gatattaaag gcattcagaa actatatgga aagagaagta 841 attcaagaaa gaaatagaaa cttcaggcag
aacatccatt cattcattca ttggattgta 901 tatcattgtt gcacaatcag aattgataag cactgttctt
ccactccatt tagcaattat 961 gtcaccctt tttattgcag ttggttttg aatgtctttc actccttta aggataaact
1021 cctttatggt gtgactgtgt cttattcatc tatacttgca gtgggtagat gtcaataaat 1081 gttacataca
caaataaata aatgtttat tccatggtaa atttaaaaaa aaaaaaaaaa 1141 aaaaaaaaaa aaa
(263) In some embodiments of the methods of the disclosure, the wild type human MMP7 gene of
the disclosure consists of or comprises the amino acid sequence (Genbank Accession number:
NP_002414.1):

(264) TABLE-US-00062 (SEQ ID NO: 60) 1 mrltvlcavc llpgslalpl ppeaggmsel
qweqaqdylk rfylydsetk nansleaklk 61 emqkffglpi tgmlnsrvie imqkprcgvp dvaeyslfpn
spkwtskvvt yrivsytrdl 121 phitvdrivs kalnmwgkei plhfrkvvww tadimigfar gahgdsypfd
gpgntlahaf 181 apgtglggda hfdderwtd gsslginfly aathelghsl gmghssdpna vmyptygngd
241 pqnfklsqdd ikgiqklygk rsnsrkk

(265) In some embodiments of the methods of the disclosure, the wild type human TERC gene of
the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number:
NR_001566.1):

(266) TABLE-US-00063 (SEQ ID NO: 61) 1 ggggttgcgga ggggtgggcct
gggaggggtg gtggccattt tttgtctaac cctaactgag 61 aaggcgtag gcgccgtgct tttgctcccc
gcgcgctgtt tttctcgctg actttcagcg 121 ggcggaaaag cctcggcctg ccgccttcca ccgttcattc
tagagcaaac aaaaaatgtc 181 agctgctggc ccgttcgccc ctcccgggga cctgcggcgg gtcgcctgcc
cagcccccca 241 accccgcctg gaggccgcgg tcggcccggg gcttctccgg aggcaccac tgccaccgcg
301 aagagttggg ctctgtcagc cgcggtctc tcgggggcca gggcgaggtt caggccttcc 361
aggccgcagg aagaggaacg gagcgagtcc ccgcgcgcgg cgcgattccc tgagctgtgg 421 gacgtgcacc
caggactcgg ctacacatg c

Definitions

(267) The following definitions are included for the purpose of understanding the present subject
matter and for constructing the appended patent claims. Abbreviations used herein have their
conventional meaning within the chemical and biological arts.

(268) Unless defined otherwise, technical and scientific terms used herein have the same meaning
as commonly understood by a person of ordinary skill in the art. Any methods, devices and
materials similar or equivalent to those described herein can be used in the practice of this

disclosure. The following definitions are provided to facilitate understanding of certain terms used frequently herein and are not meant to limit the scope of the present disclosure.

(269) As used herein, the term “FILD” refers to fibrotic interstitial lung disease.

(270) As used herein, the term “FIP” refers to Familial Interstitial Pneumonia.

(271) As used herein, the term “HRCT” refers to high-resolution CT (HRCT).

(272) As used herein, the term “ILA” refers to asymptomatic interstitial lung abnormalities.

(273) As used herein, the term “IPF” refers to idiopathic pulmonary fibrosis.

(274) As used herein, the term “PBMC” refers to peripheral blood mononuclear cell.

(275) As used herein, the term “alleviate” is meant to describe a process by which the severity of a sign or symptom of a disorder is decreased. Importantly, a sign or symptom can be alleviated without being eliminated. In a preferred embodiment, the administration of pharmaceutical compositions disclosed herein leads to the elimination of a sign or symptom, however, elimination is not required. Effective dosages are expected to decrease the severity of a sign or symptom. A sign is an objective indication of a medical condition that is observable or detectable by a medical professional or lay person (e.g. family member) (for example, with respect to fibrotic pulmonary disease, signs include, but are not limited to, changes in body weight, changes in body temperature and the presence of a fibrotic lesion in one or both lungs detectable by radiography).

(276) A symptom is an indication of disease that may be a sign but may also be exclusively observable or subjectively experienced by the subject (for example, with respect to fibrotic pulmonary disease, symptoms may include but are not limited to, a dry or hacking cough, a sore throat, a tight chest, shortness of breath, and a feeling of exhaustion or malaise).

(277) In one aspect, the terms “co-administered” and “co-administration” as relating to a subject refer to administering to the subject a compound of the invention or salt thereof along with a compound that may also treat the disorders or diseases contemplated within the invention. In one embodiment, the co-administered compounds are administered separately, or in any kind of combination as part of a single therapeutic approach. The co-administered compound may be formulated in any kind of combinations as mixtures of solids and liquids under a variety of solid, gel, and liquid formulations, and as a solution.

(278) As used herein, the term “composition” or “pharmaceutical composition” refers to a mixture of at least one compound useful within the disclosure with a pharmaceutically acceptable carrier. The pharmaceutical composition facilitates administration of the compound to a patient or subject. Multiple techniques of administering a compound exist in the art including, but not limited to, intravenous, oral, aerosol, parenteral, ophthalmic, nasal, pulmonary and topical administration.

(279) A “disease” as used herein is a state of health of an animal or subject wherein the animal or subject cannot maintain homeostasis, and wherein if the disease is not ameliorated then the animal's or subject's health continues to deteriorate.

(280) A “disorder” as used herein in an animal is a state of health in which the animal or subject is able to maintain homeostasis, but in which the animal's or subject's state of health is less favorable than it would be in the absence of the disorder. Left untreated, a disorder does not necessarily cause a further decrease in the animal's or subject's state of health.

(281) As used herein, the terms “effective amount,” “pharmaceutically effective amount” and “therapeutically effective amount” refer to a nontoxic but sufficient amount of an agent to provide the desired biological result. That result may be reduction and/or alleviation of the signs, symptoms, or causes of a disease, or any other desired alteration of a biological system. An appropriate therapeutic amount in any individual case may be determined by one of ordinary skill in the art using routine experimentation.

(282) As used herein, the term “fibrotic lung disease” or “fibroid lung disease” or “pulmonary fibrosis” or “scarring of the lung” refers to a group of diseases characterized by the formation or development of excess fibrous connective tissue (fibrosis) in the lungs. Symptoms of pulmonary fibrosis are mainly: shortness of breath, particularly with exertion; chronic dry, hacking coughing;

fatigue and weakness; chest discomfort; and loss of appetite and rapid weight loss. Pulmonary fibrosis may be a secondary effect of other diseases, most of them being classified as interstitial lung diseases, such as autoimmune disorders, viral infections or other microscopic injuries to the lung. Pulmonary fibrosis can also appear without any known cause (“idiopathic”). Idiopathic pulmonary fibrosis is a diagnosis of exclusion of a characteristic set of histologic/pathologic features known as usual interstitial pneumonia (UIP).

(283) Diseases and conditions that may cause pulmonary fibrosis as a secondary effect include: inhalation of environmental and occupational pollutants (asbestosis, silicosis and gas exposure); hypersensitivity pneumonitis, most often resulting from inhaling dust contaminated with bacterial, fungal, or animal products; cigarette smoking; connective tissue diseases such as rheumatoid arthritis, SLE; scleroderma, sarcoidosis and Wegener's granulomatosis; infections; medications such as amiodarone, bleomycin (pingyangmycin), busulfan, methotrexate, apomorphine and nitrofurantoin; and radiation therapy to the chest.

(284) As used herein, a “subject in need thereof” is a subject suffering from fibrotic lung disease relative to the population at large. For example, the subject is a patient who is or is about to be administered with comprising administering to the subject an effective amount of a therapeutic agent. For example, the subject is asymptomatic and is at risk of developing the fibrotic lung disease. A “subject” includes a mammal. The mammal can be e.g., any mammal, e.g., a human, primate, bird, mouse, rat, fowl, dog, cat, cow, horse, goat, camel, sheep or pig. Preferably, the mammal is a human.

(285) As used herein, the term “pharmaceutically acceptable” refers to a material, such as a carrier or diluent, which does not abrogate the biological activity or properties of the compound, and is relatively non-toxic, i.e., the material may be administered to an individual without causing undesirable biological effects or interacting in a deleterious manner with any of the components of the composition in which it is contained.

(286) As used herein, the term “pharmaceutically acceptable carrier” means a pharmaceutically acceptable material, composition or carrier, such as a liquid or solid filler, stabilizer, dispersing agent, suspending agent, diluent, excipient, thickening agent, solvent or encapsulating material, involved in carrying or transporting a compound useful within the invention within or to the patient such that it may perform its intended function. Typically, such constructs are carried or transported from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation, including the compound useful within the invention, and not injurious to the patient. Some examples of materials that may serve as pharmaceutically acceptable carriers include: sugars, such as lactose, glucose and sucrose; starches, such as corn starch and potato starch; cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients, such as cocoa butter and suppository waxes; oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; glycols, such as propylene glycol; polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; esters, such as ethyl oleate and ethyl laurate; agar; buffering agents, such as magnesium hydroxide and aluminum hydroxide; surface active agents; alginic acid; pyrogen-free water; isotonic saline; Ringer's solution; ethyl alcohol; phosphate buffer solutions; and other non-toxic compatible substances employed in pharmaceutical formulations.

(287) Pharmaceutically acceptable carriers of the disclosure include, but are not limited to, pharmaceutically acceptable materials, compositions or carriers, such as a liquid or solid fillers, stabilizers, dispersing agents, suspending agents, diluents, excipients, thickening agents, solvents or encapsulating materials, involved in carrying or transporting a compound useful within the invention within or to the patient such that it may perform its intended function. Typically, such constructs are carried or transported from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the

other ingredients of the formulation, including the compound useful within the invention, and not injurious to the patient. Some examples of materials that may serve as pharmaceutically acceptable carriers include: sugars, such as lactose, glucose and sucrose; starches, such as corn starch and potato starch; cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients, such as cocoa butter and suppository waxes; oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; glycols, such as propylene glycol; polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; esters, such as ethyl oleate and ethyl laurate; agar; buffering agents, such as magnesium hydroxide and aluminum hydroxide; surface active agents; alginic acid; pyrogen-free water; isotonic saline; Ringer's solution; ethyl alcohol; phosphate buffer solutions; and other non-toxic compatible substances employed in pharmaceutical formulations.

(288) Suitable forms for administration include forms suitable for systemic administration, oral administration, for example by a capsule or tablet. Once formulated, the compositions of the disclosure can be administered directly to the subject.

(289) The term “prevent,” “preventing” or “prevention,” as used herein, means avoiding or delaying the onset of symptoms associated with a disease or condition in a subject that has not developed such symptoms at the time the administering of an agent or compound commences.

(290) Compounds and Compositions

(291) In some embodiments, compounds known to be useful in treating pulmonary fibrosis are useful within the methods of the invention. Non-limiting examples of such compounds are pirfenidone (5-methyl-1-phenylpyridin-2-one, or a salt or solvate thereof) and nintedanib (methyl (3Z)-3-[[[4-(4-methylpiperazin-1-yl)acetyl]amino}phenyl]amino] (phenyl)methylidene}-2-oxo-2,3-dihydro-1H-indole-6-carboxylate, or a salt or solvate thereof).

(292) In some embodiments, the subject identified as having MUC5B promoter polymorphism rs35705950 is administered a compound contemplated within the disclosure. In some embodiments, the subject is a mammal. In other embodiments, the mammal is a human.

(293) Administration/Dosage/Formulations

(294) The regimen of administration may affect what constitutes an effective amount. The therapeutic formulations may be administered to the subject either prior to or after the onset of a disease or disorder contemplated in the invention. Further, several divided dosages, as well as staggered dosages may be administered daily or sequentially, or the dose may be continuously infused, or may be a bolus injection. Further, the dosages of the therapeutic formulations may be proportionally increased or decreased as indicated by the exigencies of the therapeutic or prophylactic situation.

(295) Administration of the compositions of the present disclosure to a patient, preferably a mammal, more preferably a human, may be carried out using known procedures, at dosages and for periods of time effective to treat a disease or disorder contemplated in the invention. An effective amount of the therapeutic compound necessary to achieve a therapeutic effect may vary according to factors such as the state of the disease or disorder in the patient; the age, sex, and weight of the patient; and the ability of the therapeutic compound to treat a disease or disorder contemplated in the invention. Dosage regimens may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation. A non-limiting example of an effective dose range for a therapeutic compound of the invention is from about 1 and 5,000 mg/kg of body weight/per day. One of ordinary skill in the art would be able to study the relevant factors and make the determination regarding the effective amount of the therapeutic compound without undue experimentation. Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be varied so as to obtain an amount of the active ingredient that is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

(296) The precise therapeutically effective amount for a human subject will depend upon the severity of the disease state, the general health of the subject, the age, weight and gender of the subject, diet, time and frequency of administration, drug combination(s), reaction sensitivities and tolerance/response to therapy. This amount can be determined by routine experimentation and is within the judgement of the clinician.

(297) A medical doctor, e.g., physician or veterinarian, having ordinary skill in the art may readily determine and prescribe the effective amount of the pharmaceutical composition required. For example, the physician or veterinarian could start doses of the compounds of the invention employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

(298) A suitable dose of a compound of the disclosure may be in the range of from about 0.01 mg to about 5,000 mg per day, such as from about 0.1 mg to about 1,000 mg, for example, from about 1 mg to about 500 mg, such as about 5 mg to about 250 mg per day. The dose may be administered in a single dosage or in multiple dosages, for example from 1 to 4 or more times per day. When multiple dosages are used, the amount of each dosage may be the same or different. For example, a dose of 1 mg per day may be administered as two 0.5 mg doses, with about a 12-hour interval between doses.

(299) In some embodiments of the methods of the disclosure, the therapeutic agent comprises pirfenidone. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the effective dosage is about 2400 mg/day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject about 800 mg of pirfenidone per day for a first week; (b) administering to the subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments of the escalating dosage regimen, the capsule or tablet comprises 267 mg of pirfenidone.

(300) In some embodiments of the methods of the disclosure, the therapeutic agent comprises nintedanib. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 300 mg/day. In some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is administered according to a modified or interrupted dosage regimen. In some embodiments, the modified or interrupted dosage regimen comprises (a) administering to the subject about 300 mg of nintedanib per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject about 200 mg of nintedanib per day until the subject presents the control level of liver enzymes; and (c) administering to the subject about 300 mg of nintedanib per day for the remainder of the treatment;

wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the modified or interrupted regimen comprises (a) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject two capsules or tablets comprising about 100 mg twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; and (c) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment.

(301) In some embodiments, the compositions of the invention are formulated using one or more pharmaceutically acceptable excipients or carriers. In one embodiment, the pharmaceutical compositions of the invention comprise a therapeutically effective amount of a compound of the invention and a pharmaceutically acceptable carrier.

(302) The carrier may be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. The proper fluidity may be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms may be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal and the like. In many cases, it is preferable to include isotonic agents, for example, sugars, sodium chloride, or polyalcohols such as mannitol and sorbitol, in the composition.

(303) It is also noted that the term “comprising” is intended to be open and permits but does not require the inclusion of additional elements or steps. When the term “comprising” is used herein, the terms “consisting essentially of” and “consisting” of are thus also encompassed and disclosed. Throughout the description, where compositions or combinations are described as having, including, or comprising specific components or steps, it is contemplated that compositions or combinations also consist essentially of, or consist of, the recited components. Similarly, where methods or processes are described as having, including, or comprising specific process steps, the processes also consist essentially of, or consist of, the recited processing steps.

(304) All publications and patent documents cited herein are incorporated herein by reference as if each such publication or document was specifically and individually indicated to be incorporated herein by reference.

EXAMPLES

(305) In order that the invention disclosed herein may be more efficiently understood, examples are provided below. It should be understood that these examples are for illustrative purposes only and are not to be construed as limiting the invention in any manner.

Example 1: Genetic Background of Asymptomatic Siblings of FIP Subjects

(306) Asymptomatic siblings (>50 years old) of patients with established FIP underwent HRCT scan of the chest. HRCT scans were assessed for FILD by blinded thoracic radiologists; when possible, specific radiographic patterns were identified. PBMCs RNA and DNA were isolated. Genotyping for rs35705950 and microarray analysis were performed (SurePrint G3 Human Gene Expression Microarray). Data were analyzed using Partek Genomics Suite and RStudio. Four-hundred eighty-eight FIP siblings from 271 families were evaluated, 25 HRCT scans were excluded due to technical inadequacy, leaving 463 to be interpreted. Of these, 19% (n=88) met criteria for FILD. A subset of the positive FILD scans (n=58) were evaluated for specific interstitial patterns: the predominant radiographic finding was Usual Interstitial Pneumonia (UIP), documented as possible (n=37), probable (n=6), or definite (n=5) in 82.8% of these cases. DNA was available for 443 subjects (358 without and 85 with FILD). The minor allele (T) frequency

(MAF) of rs35705950 was higher among those with evidence of FILD (MAF=0.29) than among those with normal appearing HRCT scans (MAF=0.21, $p=0.005$). The rs35705950 variant was associated with the presence of FILD (OR=1.90, 95% CI 1.10-3.30, $p=0.02$), and FILD was associated with age (OR=1.09, 95% CI 1.06-1.12, $p=7.24 \times 10^{-9}$), male sex (OR=1.81, 95% CI 1.04-3.16, $p=0.04$), and history of smoking (OR=1.94, 95% CI 1.11-3.40, $p=0.02$). Microarray analysis on PBMC RNA from 40 subjects with FILD and 105 unaffected siblings revealed 1,272 differentially expressed genes (FDR<0.05, fold-change>2); hierarchical clustering performed on the top 194 differentially expressed probes illustrates segregation of FILD subjects from unaffected siblings (FIG. 1).

Example 2: Role of MUC5B in Pathogenesis of IPF

(307) Common genetic variants play major and similar roles in the development of both familial and sporadic IPF (Table 3), indicating a similar etiology for familial and sporadic IPF. A common gain-of-function MUC5B promoter variant rs35705950 is a strong risk factor (genetic and otherwise), accounting for at least 30% of the total risk of developing IPF (10) confirmed in independent studies, including a GWAS (OR for T (minor) allele=4.51; 95% CI=3.91-5.21; $P=7.21 \times 10^{-95}$); 3) rs35705950 may be used to identify individuals with PrePF and is predictive of radiographic progression of PrePF. MUC5B promoter variant rs35705950 is present in over 50% of non-Hispanic white (NHW) patients with IPF and is associated with unique biological and clinical IPF phenotypes. PrePF can be predicted using a combination of clinical risk factors, the MUC5B promoter variant rs35705950, and a panel of biomarkers.

(308) TABLE-US-00064 TABLE 1 Common IPF risk variants identified by targeted sequencing of risk loci in 3,642 IPF cases and 4,442 unaffected controls

Chrm	Variant	Gene Annotation	sup.a	sup.b	sup.c
3q26	rs2293607	TERC 3' UTR C	0.2999	1.30 (1.18-1.43)	1.79 (1.49-2.15)
4q22	rs2609260	FAM13A Intronic C	0.2289	1.35 (1.22-1.50)	1.96 (1.56-2.47)
5p15	rs4449583	TERT Intronic T	0.2641	0.68 (0.62-0.75)	0.46 (0.39-0.55)
6p24	rs2076295	DSP Intronic G	0.5428	1.27 (1.14-1.42)	2.08 (1.83-2.37)
7q22	rs6963345	ZKSCAN1 Intronic A	0.4444	1.35 (1.22-1.50)	1.73 (1.51-1.99)
10q24	rs2488000	OBFC1 Intronic T	0.08	0.70 (0.62-0.79)	7.13 $\times 10^{-9}$
11p15	rs35705950	MUC5B Promoter T	0.3533	5.45 (4.91-6.06)	18.68 (13.34-6.17)
13q34	rs1278769	AK025511 3' UTR A	0.1996	0.77 (0.70-0.85)	0.69 (0.56-0.86)
15q15	rs35700143	IVD — C	0.4118	0.76 (0.68-0.84)	0.63 (0.55-0.71)
19p13	rs12610495	DPP9 Intronic G	0.3398	1.22 (1.11-1.35)	1.59 (1.36-1.87)

OR, odds ratio. The minor allele is defined as the minor allele in the combined case and control group. .sup.aBased on SNPDOC; .sup.bP value adjusted for sex; .sup.cOR resulting from dominant test.

(309) MUC5B is predicted is involved in the pathogenesis of IPF. FIG. 5 shows that MUC5B promoter variant is associated with enhanced MUC5B expression in both unaffected subjects and in patients with IPF and in IPF, MUC5B message and protein are expressed in bronchoalveolar epithelia (FIG. 6) and honeycomb cysts. In mice, the concentration of Muc5b is directly related to the fibroproliferative response to bleomycin (FIG. 7), Muc5b protein is expressed in the injured lung following bleomycin challenge, and enhanced production of Muc5b in mice appears to initiate endoplasmic reticulum (ER) stress in peripheral airways (FIGS. 8 and 9). Preliminary studies, also show that mucociliary clearance is decreased in mice that overexpress Muc5b (SFTPC-Muc5b.sup.Tg) and in humans with IPF (FIG. 10).

(310) Interstitial lung abnormalities on HRCT scans show asymptomatic relatives of patients with familial IPF and in the elderly. Similar to patients with IPF, interstitial lung abnormalities in asymptomatic subjects are associated with advanced age, cigarette smoking, reduced lung volume and decreased exercise tolerance. Moreover, the MUC5B promoter variant rs35705950 is associated with a higher prevalence of interstitial lung abnormalities on HRCT scan and is

predictive of radiographic progression. Suggesting that interstitial lung abnormalities on HRCT scan are a precursor of IPF. However, interstitial lung abnormalities are not specific and include non-fibrotic and fibrotic HRCT defects, and consequently, the prevalence of interstitial lung abnormalities (>5% in the general population ≥ 50 years of age is orders of magnitude higher than IPF.

(311) To address the non-specificity of interstitial lung abnormalities, a novel entity—Preclinical Pulmonary Fibrosis (PrePF) was used. PrePF is reported more frequently among smokers and in families with two or more cases of pulmonary fibrosis. In the Framingham population, data shows that PrePF is present in 1.8% of the general population ≥ 50 years of age (in contrast, interstitial lung abnormalities were seen in 6.7%) and that the MUC5B promoter variant rs35705950 is predictive of those with PrePF (OR=6.3 per allele [95% CI 3.1-12.7]). As shown herein, among asymptomatic first-degree family members of familial interstitial pneumonia (FIP) 14% have fibrotic interstitial changes on CT scan and 35% have interstitial abnormalities on transbronchial biopsy. Moreover, in the Framingham population, it is shown that rs35705950 is predictive of radiographic progression of PrePF (OR=2.8 per allele [95% CI 1.8-4.4]) which is associated with a greater FVC decline ($P=0.0001$) and an increased risk of death (HR=3.7 [95% CI 1.3, 10.7]; $P=0.02$), indicating that in addition to having radiographic features of IPF, PrePF has similar risk factors (age, gender, smoking, and MUC5B variant) and a progressive clinical course. While the MUC5B promoter variant is predictive of PrePF, rs35705950 is present in $\approx 19\%$ (minor allele frequency (MAF)=0.09) of the NHW population, however IPF occurs infrequently ($<0.1\%$). Thus, additional biomarkers may be used in combination with rs35705950 identify PrePF within at-risk populations.

(312) The data provided herein suggest that 1) IPF is under-diagnosed; 2) PrePF is prevalent in at-risk populations; 3) approximately 75% of the cases of PrePF are progressive; 4) radiographic progression of PrePF is associated with increased morbidity and mortality; and 5) MUC5B variant rs35705950, peripheral blood biomarkers, clinical/biological, and radiographic screening should be useful in identifying those with PrePF (FIG. 11). While IPF takes years to develop, most patients with IPF are diagnosed in the advanced stage when little can be done to influence survival. Once the lung has undergone remodeling, the non-compliant, stiff lung matrix causes additional remodeling through activation of myofibroblasts, resulting in a feed-forward loop of lung remodeling. Earlier diagnosis of IPF detects subjects with a lower burden of fibrotic lung disease.

(313) This disclosure provides a strategic approach to screening for early forms of IPF needs to be established (FIG. 11). While the MUC5B promoter variant is predictive of PrePF (defined as chest HRCT consistent with probable or definite fibrosis (e.g., bilateral subpleural reticular changes, honeycombing, or traction bronchiectasis) occurring in asymptomatic subjects ≥ 40 years of age that emerge from at-risk populations), the MUC5B promoter variant is present in $\approx 19\%$ of the NHW population and IPF occurs infrequently ($<0.1\%$). To study at-risk populations (asymptomatic siblings ≥ 40 years of age of patients with family or sporadic IPF), identification of genetic variants and biomarkers that increase the yield of patients with PrePF are used to establish screening tools and approaches that identify early stages of IPF. This approach changes the way IPF is diagnosed and treated, and is critical to developing interventions to prevent PrePF progression to established IPF. The methods provided in this disclosure fundamentally alter the clinical approach to patients with IPF from palliative to preventive (FIG. 10).

Example 3: Predictive Biomarker Profile for Established IPF

(314) To address the development of a peripheral blood biomarker profile for IPF, an assay of the expression levels of >3700 plasma proteins was performed on plasma from 70 patients with established IPF and 70 controls. After controlling for multiple comparisons and appropriate co-variables, 57 proteins were up-regulated >1.5-fold (including surfactant proteins, MMP7, and C3) in the plasma of patients with IPF and 12 were significantly down-regulated (FIG. 2).

Example 4: Predictive Biomarker Profile for Early IPF

(315) To evaluate a predictive biomarker profile in cases of preclinical pulmonary fibrosis (PrePF) derived from families with familial IPF (≥ 2 cases of IPF in a family), HRCT scans were performed on 496 asymptomatic family members ≥ 40 years of age previously phenotyped as unaffected from 263 families with familial IPF. PrePF, consistent with the operational definition (defined as abnormalities on chest HRCT consistent with probable or definite fibrosis (e.g., bilateral subpleural reticular changes, honeycombing, or traction bronchiectasis) occurring in asymptomatic subjects ≥ 40 years that emerge from at-risk populations), was present in 77 (15.5%) of 496 asymptomatic individuals from families with familial IPF. The minor allele frequency (MAF) of the MUC5B promoter variant was 0.29 in those with PrePF versus 0.21 in those without fibrosis ($P=0.025$). Preliminary analysis of PBMC gene expression profiles evaluated by microarrays from 38 cases of PrePF and 187 subjects without fibrosis identified 16 genes significantly differentially expressed between the two groups (p -value <0.05 and >1.5 fold change). Among genes differentially expressed in PrePF are those involved in innate immunity and inflammatory responses (SIGLEC14), antibacterial effects (ADM2), growth and motility (TSPAN5), and protein phosphorylation (CAMKK1). Moreover, PBMC gene expression appears to contribute to the ability to predict PrePF in an at-risk population (FIG. 3).

(316) Additionally, RNA-sequencing analysis was performed on 40 PrePF subjects and 80 subjects with a normal HRCT scan. Sequencing of the polyA-enriched libraries was prepared using Illumina TrueSEQ reagents and multiplexing 10 samples on each lane of HiSEQ4000 to obtain on average 35-40 million reads per sample. This high coverage allows for the consideration of a broad dynamic range of mRNA transcripts for biomarker selection. Platform selection of serum and plasma samples from the same subjects are used for proteomic analysis.

Example 5: Biomarker Identification

(317) To examine for association between each biomarkers and PrePF, a multivariable logistic regression model for PrePF with biomarkers and covariates is used for inclusion and a step-wise forward selection procedure is constructed. Variables stay in the model if associated at $P \leq 0.01$ after adjustment for the variables already in the model. Protein biomarkers that are significantly associated with established IPF and the top 20 differentially expressed genes in PrePF are considered for inclusion in a multivariable model. The number of potential biomarkers allowed in the joint model is restricted to approximately 20 given the number cases of PrePF expected. Secondly, interactions between MUC5B genotype and the other biomarkers are tested for, which allow for the possibility that different biomarker profiles are diagnostic in IPF patients with/without the MUC5B risk allele.

Example 6: Predictive Ability of Biomarkers

(318) To test the predictive value of the combination of biomarkers associated with PrePF, the observed expression and other biomarker values from those associated with PrePF in the siblings of FIP patients is used to obtain the probability, for each sibling, having PrePF.

(319) Following, a construct receiver operating characteristic (ROC) curves (see M. S. Pepe et al., Phases of biomarker development for early detection of cancer. Journal of the National Cancer Institute 93, 1054-1061 (2001)), is used to choose the probability threshold that maximizes the area under the ROC curve. This probability threshold is used to classify each individual as predicted to have PrePF or not, allowing calculation of the sensitivity, specificity, positive predictive value, and negative predictive value of the predictive model. The properties of the predictive model(s) in the independent set of siblings of patients with IPF are evaluated. Different aliquots are run for 10 samples for each assay at each time the assays is run in order to use those 10 samples to evaluate the need for standardization of the absolute values for each assay over time. Either the raw or standardized values, for a given model, is used to observe biomarker values in the PrePF siblings and non-PrePF siblings to obtain the probability of being in the disease group based on the model parameters developed using the FIP siblings. The thresholds identified among the FIP siblings are used to classify each individual as predicted to have PrePF or not. This categorization allows for

the calculation of the sensitivity, specificity, positive predictive value, and negative predictive value of the predictive model among the siblings of independent cases of IPF to that observed in the siblings of FIP cases.

(320) Power is calculated to detect differences between those with and without PrePF assuming 500 siblings and 10% (N=50) with PrePF. Assuming $\alpha=0.00005$ (conservatively correcting for up to 1000 independent tests), we have 80% (90%) power to detect differences in protein or expression level of 0.74 (0.80) standard deviation between PrePF and unaffected siblings. These differences are larger than previously-observed protein and gene-expression levels in IPF patients and controls (see I. V. Yang et al., The peripheral blood transcriptome identifies the presence and extent of disease in idiopathic pulmonary fibrosis. PLoS One 7, e37708 (2012). With 50 PrePF and 450 unaffected, there is 90% power to bound the sensitivity of the biomarker-based classification of PrePF with a margin of error of 11% if the sensitivity is 65%, and 6.5% if the sensitivity is 95%; the margins of error for 65% and 95% sensitivity are 4.5% and 2.5%, respectively.

Example 7: MUC5B Promoter Variant rs35705950 is a Risk Factor for Rheumatoid Arthritis—Interstitial Lung Disease

(321) Methods

(322) Study Cohorts

(323) This study included a discovery cohort and multi-ethnic replication cohorts. The discovery cohort included patients with RA, with and without ILD (RA-noILD) as assessed by chest HRCT, and controls, from the French RA-ILD network. The multi-ethnic replication cohorts were obtained from six countries (China, Greece, Japan, Mexico, the Netherlands and United States). This included patients with RA-ILD and RA-noILD patients, and controls. All cases fulfilled the 2010 European League Against Rheumatism-American College of Rheumatology (EULAR-ACR) and/or 1987 ACR revised criteria for RA. The ILD status of patients with RA was established by chest HRCT images that were centrally reviewed by experienced readers for each participating cohort. There was one cohort, the RA-noILD cases from the USA1 cohort, which was determined by self-report. The chest HRCT ILD pattern was classified as UIP, possible UIP or inconsistent with UIP according to international criteria and all readers were blinded to the clinical and genetic data. The institutional review boards at each institution approved all protocols, and all patients provided written informed consent.

(324) Genotyping

(325) Genotyping of the MUC5B rs35705950 single nucleotide polymorphism (SNP) involved use of Taqman Genotyping Assays (Applied Biosystems, Foster City, CA, USA) as previously reported, by direct Sanger Sequencing or imputation from genome-wide association study data.

(326) The additional common IPF risk variants on 3q26, 4q22, 5p15, 6p21.3, 6p24, 7q22, 10q24, 11p15.5, 13q34, 15q14-15, and 19p13 were genotyped by Taqman qPCR (Thermo Fisher Scientific, California) per the manufacturer's instructions.

(327) Lung Tissue Analysis

(328) In order to determine if MUC5B was expressed in RA-ILD lung tissue, we analyzed lung tissue was analyzed from nine patients with RA-ILD undergoing lung transplantation (University of California, San Francisco) compared to six unaffected controls with ILD (NHLBI Lung Tissue Research Consortium; <https://ltrcpublic.com>) or concordant expression of other relevant markers of pulmonary fibrosis. The tissue was formalin fixed, paraffin embedded and cut in 4 um sections. Tissue sections were deparaffinized in xylene, followed by dehydration in series of ethanol. Following citrate buffer antigen retrieval, slides were incubated overnight with primary antibodies against MUC5B (1:4000, Santa Cruz, Dallas, TX). Secondary antibody diluted 1:1000 tagged with HRP (Life Technologies) was visualized using an Aperio CS2 slide scanner (Leica, Buffalo Grove, IL).

(329) Results

(330) Study Cohorts

(331) This case-control genetic study included 620 RA-ILD cases, 614 RA-noILD cases and 5448 unaffected controls. The discovery cohort included 118 RA-ILD cases, 105 RA-noILD cases and 1229 unaffected controls. The multi-ethnic replication sample included 502 RA-ILD, 509 RA-noILD cases and 4219 unaffected controls.

(332) Characteristics of the Discovery Cohort

(333) As compared with RA-noILD, patients with RA-ILD were more frequently male, older and more frequently smoked cigarettes (54.7% versus 36.1%) (FIG. 13). However, after adjusting for sex, the relationship between RA-ILD and cigarette smoking was no longer statistically significant (FIG. 13). After adjustment, RA-ILD and RA-noILD patients did not differ in rheumatoid factor (RF) and/or anti-citrullinated protein antibody (ACPA) positivity, erosive status of RA, exposure to methotrexate or the mean RA duration from diagnosis at inclusion in the cohort. Overall, 41% of patients with RA-ILD had a UIP or possible UIP HRCT pattern.

(334) MUC5B Promoter Variant and Risk of Rheumatoid Arthritis-Associated Interstitial Lung Disease

(335) Comparison of RA-noILD and controls revealed that none of the cohorts (discovery cohort and multi-ethnic cohorts) demonstrated a significant difference in the frequency of the MUC5B promoter variant (FIG. 14; FIG. 16A), suggesting a lack of association between the MUC5B promoter variant and RA. In the discovery cohort, the minor allele frequency (MAF) of the MUC5B promoter variant was 10.9% in unaffected controls and 32.6% in cases of RA-ILD; this variant was in Hardy-Weinberg equilibrium (HWE) in both study groups. In the discovery population, after controlling for sex we detected a significant association between the MUC5B promoter variant and RA-ILD when compared to non-RA controls ($OR_{adj}=3.8$; 95% CI, 2.8 to 5.2; $P=9.7\times 10^{-17}$) (FIG. 14). Similar to the discovery population, the MUC5B promoter variant was significantly over-represented among the cases of RA-ILD compared to unaffected non-RA controls in all of the multi-ethnic study case series, except in the two Asian case series (FIG. 14). Given that the MUC5B promoter variant is under-represented in Asian populations compared to non-Hispanic whites (FIG. 14; www.ncbi.nlm.nih.gov/projects/SNP/snp_ref.cgi?rs=35705950), a likely explanation, especially given the consistent point estimates, for the absence of a significant relationship between the MUC5B promoter variant and RA-ILD is that the analysis of the two Asian case series is likely underpowered. The relationship between the MUC5B promoter variant and RA-ILD in combined multi-ethnic study case series ($OR_{adj}=4.7$; 95% CI, 3.9 to 5.8; $P=1.3\times 10^{-49}$) (FIG. 14) (FIG. 16B) validated the observed association between the MUC5B promoter variant and RA-ILD in the discovery study population. In addition, the cases of RA-ILD in the study populations from Greece and USA-1 were not in HWE, suggesting (as has been observed in cases of IPF 14), that the MUC5B promoter variant and/or common variants in high or complete linkage disequilibrium with the MUC5B promoter variant should be considered as causative in these cases of RA-ILD. For the comparison with non-RA controls, the best-fitting genetic model for the three study populations (discovery population, combined multi-ethnic case series, and combined analysis) for the association of the MUC5B

(336) MUC5B RS35705950 and Risk of Interstitial Lung Disease Among Patients with Rheumatoid Arthritis

(337) To further investigate whether the MUC5B promoter variant rs35705950 contributes to the risk of ILD among patients with RA, we compared RA-ILD and RA-noILD patients, adjusting for sex, age at inclusion and cigarette smoking. In the discovery cohort, the MUC5B variant was associated with RA-ILD ($OR_{adj}=3.1$; 95% CI, 1.6 to 6.3; $P=9.4\times 10^{-4}$), and this finding was replicated in the aggregate multi-ethnic cohort ($OR_{adj}=2.9$; 95% CI, 1.1 to 8.4; $P=0.04$) and the combined analysis ($OR_{adj}=3.1$; 95% CI, 1.8 to 5.4; $P=7.4\times 10^{-5}$) (FIG. 14; FIG. 16C). For the comparison of RA-ILD with RA-noILD, the best-fitting genetic model for the three study cohorts (discovery population, combined multi-ethnic case series, and combined analysis) was dominant. After adjusting for covariates, no association between tobacco smoking and the risk of ILD among

patients with RA was found and no interaction of tobacco smoke exposure with the MUC5B promoter variant was observed (OR_{adj}=0.7; 95% CI, 0.3 to 1.9; P=0.51).

(338) MUC5B RS35705950 and UIP on HRCT Scan

(339) Limiting the RA-ILD cases to those with radiographic evidence of definite or possible UIP on HRCT scan, the association observed in the discovery cohort (OR_{adj}=5.0; 95% CI, 2.1 to 12.3; P=3.0×10⁻⁴), was replicated in the combined multi-ethnic cohort (OR_{adj}=9.2; 95% CI, 2.3 to 38.7; P=1.8×10⁻³) (FIG. 16C), and was observed in the combined cohort analysis (OR_{adj}=6.1; 95% CI, 2.9 to 13.1; P=2.5×10⁻⁶) (FIG. 16C). In the combined analysis, the comparison of odds ratios for UIP RA-ILD vs RA-noILD (OR_{adj}=6.1; 95% CI, 2.9 to 13.1; P=2.5×10⁻⁶) to non-UIP RA-ILD vs RA-noILD (OR_{adj}=1.3; 95% CI, 0.6 to 2.8; P=0.46) was statistically significant (P=0.02), suggesting that the effect of the MUC5B promoter variant was restricted to the UIP RA-ILD sub-phenotype (FIG. 16C). Finally, consistent with our previous findings, the MUC5B promoter variant was found to increase the risk of developing a UIP pattern among patients with RA-ILD through a dominant model in the discovery, replication and combined analysis; the odds of having a UIP and possible UIP pattern for patients with RA-ILD carrying at least one MUC5B rs35705950 T risk allele were 2.9 times greater than individuals having the GG genotype (OR_{adj}=2.9; 95% CI, 1.7 to 4.8; P=5.1×10⁻⁵) (FIG. 15; FIG. 16C). After adjusting for covariates, tobacco smoking exposure did not contribute to a specific HRCT pattern for RA-ILD and no interaction with the MUC5B rs35705950 variant was detected.

(340) Sites of MUC5B Expression in Ra-ILD

(341) We performed immunohistochemical staining for MUC5B in nine RA-ILD lung tissue explants (5 GG and 4 GT) and 6 unaffected controls (3 GG and 3 GT). Similar to what has been reported in IPF, RA-ILD lung tissue demonstrated MUC5B in the cytoplasm of the bronchioles and in areas of microscopic honeycombing, including staining of the metaplastic epithelia lining the honeycomb cysts and the mucous within the cyst (FIG. 17). The controls demonstrated MUC5B expression in the bronchioles only. There were no obvious differences in MUC5B expression by genotype.

(342) Exploratory Genetic Association Study of 12 Common IPF Risk Variants in Ra-ILD

(343) Having provided evidence for the contribution of the dominant IPF genetic risk variant, i.e. the MUC5B promoter variant, to RA-ILD, we decided to test the association of 12 additional common IPF risk variants with RA-ILD (FIG. 29). This exploratory study included 272 RA-ILD and 242 RA-noILD patients from the France, USA-1 and Mexico case series. Taking into account the relatively small sample size and related low power of detection corresponding P-values, Odds Ratio and 95% CI for the 12 candidate variants were considered as descriptive and Bonferoni correction was therefore not applied (Table 4). Comparison between RA-ILD and RA-noILD revealed that 2 common IPF risk variants, TOLLIP rs5743890 and IVD rs2034650, were significantly associated with RA-ILD. The TOLLIP rs5743890 minor allele was associated with increased risk of RA-ILD and the IVD rs2034650 minor allele was associated with decreased risk of RA-ILD (OR_{adj}=2.13; 95% CI, 1.13 to 4.10; P=0.02 and OR_{adj}=0.59; 95% CI, 0.38 to 0.89; P=0.01, respectively) and the directionality of these relationships is consistent with what has been observed for IPF.^{16,17} No association with RA-ILD was detected for the 10 other IPF risk variants (FIG. 29).

Example 8: MUC5B Promoter Variant is Associated with Visually and Quantitatively Detected Preclinical Pulmonary Fibrosis

(344) Better understanding and recognition of early pulmonary fibrosis is critical because medical therapies have been shown to slow progression, not to reverse or even stabilize established fibrosis—therefore, intervention before irreversible fibrosis has become extensive has the potential to improve quality of life and decrease morbidity. While IPF affects approximately 5 million people worldwide, between 1.8 and 14% of the general population ≥50 years of age have radiologic findings of undiagnosed pulmonary fibrosis. Large cohort studies indicate that interstitial lung

abnormalities, postulated to represent early pulmonary fibrosis, are associated with increased mortality, and that most of these abnormalities progress over time. Members of families with 2 or more cases of pulmonary fibrosis (FIP, Familial Interstitial Pneumonia) have been identified as an “at-risk” population. In a previous study of FIP relatives, 14% had interstitial lung abnormalities on high resolution computed tomography (HRCT), and 35% had an abnormal transbronchial biopsy indicating interstitial lung disease.

(345) HRCT provides visualization of the lung parenchyma and plays a key role in the diagnosis of the Idiopathic Interstitial Pneumonias (ITIPs), including IPF. Currently, visual diagnosis by thoracic radiologists, in conjunction with multidisciplinary clinical conference, is the gold standard for diagnosing ITIPs. However, visual assessment is imprecise and hampered by inter-observer variation. Quantitative HRCT (qHRCT) evaluation provides measures of fibrosis extent that, in subjects diagnosed with IPF, correlate with degree of physiologic impairment at baseline, and may be more sensitive to subtle changes in disease status than routinely used physiological metrics. The design and utility of quantitative methods in the context of early forms of fibrotic ILD requires further study. Deep learning methods have been increasingly used in imaging to identify and classify CT patterns, and may be particularly valuable in detection of early lung fibrosis.

(346) This study aims to: (1) examine risk factors, including two common fibrosis-associated genetic variants in MUC5B and TERT, for undiagnosed pulmonary fibrosis (PrePF) in FIP first-degree relatives; and (2) determine the utility of a deep learning, texture-based qHRCT method in the detection of early fibrosis in this cohort.

Materials and Methods

(347) FIP Relatives Screening:

(348) As part of a study of FIP conducted at the University of Colorado, National Jewish Health, and Vanderbilt University (COMIRB #15-1147; NJH IRB 1441a; Vanderbilt IRB #020343), non-Hispanic white (NHW) relatives of FIP patients, defined as those in families with two or more cases of pulmonary fibrosis, were contacted for enrollment. First-degree relatives without a known prior diagnosis of pulmonary fibrosis and greater than 40 years of age were offered HRCT scans of the chest and asked to undergo peripheral blood draw. Study subjects younger than 40 years of age or older than 40 years of age who reported on pre-study questionnaires to be personally affected by pulmonary fibrosis were excluded (FIG. 18).

(349) Visual CT Review:

(350) HRCT scans were interpreted by study radiologists and examined for the presence of fibrotic ILD. “PrePF” was defined as the presence of “probable” or “definite” fibrotic ILD on HRCT in FIP relatives who had no known diagnosis of pulmonary fibrosis at the time of study enrollment (FIGS. 18, 19).

(351) Quantitative CT:

(352) Inspiratory HRCT series with slice thickness ≤ 1.25 mm and spacing ≤ 2.0 mm were selected for quantitative analysis. This included 212 volumetric series with thin, contiguous sections (slice thickness and spacing both ≤ 1.25 mm) and 191 non-volumetric scans (56 with slice spacing > 1.25 mm and < 10 mm, 65 with slice spacing of 10 mm and 70 with slice spacing = 20 mm). Scans identified as technically inadequate were omitted. In addition, 100 inspiratory volumetric HRCT of never-smoking control subjects from the COPDGene cohort were analyzed (FIG. 20). The lungs were segmented in a semi-automatic fashion using open source software followed by manual editing, if necessary, performed by trained analysts. Examples of the categorization of different parts of CT scans are shown in FIG. 21. Some studies were acquired with contiguous thin axial sections while others used 1 or 2 cm intervals. Also, reconstruction kernel, a parameter that affects image sharpness and noise, was not standardized.

(353) Fibrosis quantification on CT scans was performed using a deep learning technique, with a convolutional neural network (CNN) algorithm trained with image regions of normal and abnormal lung identified by expert radiologists. Training data and an earlier algorithm version were described

previously. Here, a more complex CNN architecture was employed that classifies image regions using pixel and texture features extracted by multiple convolutional layers at different scales. Classification categories included normal lung, airways, reticular abnormality, honeycombing and ground glass. An additional category, “not normal”, was also included for lung regions not classified into any of the named categories. Further, pixels in the “not normal” category were split into two subcategories: “not normal” low density and “not normal” high density using the threshold value of -650 Hounsfield Units (HU). Subject level scores were computed as the percentage of total lung volume classified in each category. HRCT fibrosis score was defined as the sum of CNN classification scores for reticular abnormality, honeycombing, ground glass, and “not normal high density” (FIG. 21).

(354) A simpler previously described densitometric analysis of HRCTs was also performed for comparison. Percent high attenuation area (% HAA), the percentage of total lung volume with HRCT pixel intensity greater than -600 HU and less than -250 HU, has been used as a measure of interstitial lung disease on CT.

(355) Statistical Analysis:

(356) Analysis of the effect of specific alleles on PrePF risk was performed using minor allele frequency (MAF) for comparison of variant prevalence in the study groups; statistical significance was determined utilizing either a z-score test for proportions or a mixed effects logistic regression model when controlling for other clinical factors (age, sex, and history of smoking) and family [random effect]) in both dominant and log-additive models.

(357) Distribution of qHRCT fibrosis scores was left skewed as was % HAA, and therefore these values were log transformed prior to analyses. Log of qHRCT fibrosis score (hereafter, “fibrosis score”) and log (% HAA) were compared with visual scores using ANOVA and Tukey's honest significant difference (HSD) test. To determine the ability of qHRCT scores to predict visual diagnosis of PrePF, receiver-operating characteristic (ROC) analysis was performed. Optimal threshold for discriminating visual diagnosis of fibrotic ILD was determined with Youden's method. Five-fold cross-validation was performed to test detection accuracy, sensitivity and specificity, and consistency of optimal threshold. Linear regression was performed to test association between the MUC5B genotype and qHRCT fibrosis score and log (% HAA).

(358) A p-value of <0.05 was considered statistically significant for differences between groups as well as for associations between individual variables and outcomes in linear and logistic regression modeling. Statistical analyses were performed using RStudio (Version 0.99.473).

(359) Results

(360) Study Cohort Characteristics

(361) A total of 1,090 FIP relatives were contacted, and 523 eligible subjects were recruited and underwent HRCT screening (FIG. 18). Of the 523 subjects, 26 were excluded due to technical inadequacy of images and one for an equivocal consensus read by study radiologists. The remaining 496 subjects from 263 families were included in the final analyses. The mean age of study subjects was 57 years (95% CI: 56.5-58), 189 (38%) were male, and 148 (29%) were either current or former smokers. The minor allele (T) frequency of the MUC5B promoter polymorphism rs35705950 was 0.22 in this cohort; 45% of the subjects in this cohort had one or two copies of the minor allele (FIG. 22). The minor allele (C) frequency of the TERT variant rs2736100 was 0.47 in the entire cohort; 69% of the subjects in the cohort having one or two copies of the minor allele (FIG. 22).

(362) Prevalence of Preclinical Pulmonary Fibrosis (PrePF) in FIP Relatives

(363) Of the 496 HRCT scans, 401 showed no CT evidence of interstitial lung disease (ILD), and 95 showed evidence of ILD, either fibrotic (27 probable and 50 definite) or non-fibrotic ($n=18$). Therefore, among these 496 subjects who reported being personally unaffected by pulmonary fibrosis, the PrePF prevalence was 15.5% ($n=77$) (FIG. 18).

(364) The CT patterns noted in PrePF subjects (FIG. 23) show that possible, probable, or definite

UIP pattern was the most commonly considered ($n=59$, 77% of all PrePF cases). NSIP was considered in 45 subjects (58% of all PrePF cases). The fibrotic changes were most commonly lower-lobe predominant and subpleural in nature, consistent with a UIP pattern (FIG. 23). Non-fibrotic ILD scans, on the other hand, generally had more diffuse, upper-lobe predominant abnormalities.

(365) There were 402 study subjects with HRCT scans that were technically adequate for quantitative assessment. 212 of the scans had both slice thickness and spacing ≤ 1.25 mm (thin, contiguous); of the remaining 191 scans, 56 had slice spacing >1.25 mm and <10 mm, 65 had slice spacing = 10 mm, and 70 had slice spacing = 20 mm. Volumetric HRCT scans on an additional 100 COPD Gene subjects were included as normal controls. Fibrosis score means were significantly different ($p < 0.0001$) across groups defined by visual diagnosis (FIG. 24). Comparison of means showed fibrosis score were significantly different comparing each group (all between-group comparisons $p < 0.01$). Means of log (% HAA) scores were also significantly different across visual scoring groups ($p < 0.0001$), and individual between-group comparisons showed log (% HAA) was significantly different in most comparisons ($p < 0.0001$), except between the “probable” and “definite” visual scores ($p = 0.35$).

(366) ROC analysis showed that fibrosis score discriminates subjects with visual diagnosis of PrePF (FIG. 25B). Average area under the curve (AUC) in five-fold cross validation was 0.85 (range 0.83-0.87) and average accuracy, sensitivity, and specificity in the test partitions were 0.83 (range 0.74-0.86), 0.74 (range 0.56-0.92), and 0.84 (range 0.76-0.89), respectively. Optimal threshold for fibrosis score ranged from 1.40-1.42, corresponding to 4.1% fibrotic area in examined lung. Utilizing a cutoff of 1.40 for fibrosis score on the entire dataset, the sensitivity was 74%, specificity was 82%, and accuracy was 81%; the negative predictive value of this test was 95%, exceeding its positive predictive value (42%) (FIG. 25C).

(367) Compared to the classification achieved with the CNN as described above, ROC analysis of log % HAA had lower mean AUC 0.80 (range 0.79-0.81) and average accuracy, sensitivity, and specificity of 0.67 (range 0.63-0.70), 0.82 (range 0.75-0.91) and 0.64 (range 0.62-0.70) respectively (FIG. 25A). Optimal threshold for log % HAA ranged from 1.49-1.57. Utilizing a cutoff of 1.49 for log % HAA, the sensitivity was 88%, specificity was 55%, and accuracy was 60%; the negative predictive value of this test was 96%, exceeding its positive predictive value (26%).

(368) Risk Factors for PrePF

(369) Subjects with PrePF were older (mean age 65.8 years, 95% CI 63.5-68.1) than those without fibrosis (mean age 55.8, 95% CI 54.9-56.6, $p = 6.36 \times 10^{-13}$); they were also more likely to have ever smoked (43% versus 27%, $p = 0.007$), and to be male (48% versus 36%, $p = 0.05$). However, there was no difference in breathlessness between the PrePF and subjects without fibrosis (mean score 0.5 versus 0.6, $p = 0.24$, FIG. 26). When fibrosis was defined by quantitative fibrosis score cutoff (1.4), there was a significant difference between groups in terms of mean breathlessness score (0.39 versus 0.78, $p = 0.003$). Quantitative fibrosis score was positively associated with breathlessness score ($p = 0.001$), even after controlling for age ($p = 1.9 \times 10^{-9}$), male sex ($p = 0.7$), and smoking history ($p = 0.8$).

(370) Screening for autoantibodies in this cohort revealed that there were no differences between PrePF and No Fibrosis subjects in terms of overall seropositivity or individual antibodies' testing in this cohort. For quantitatively defined fibrosis, there was no significant difference between groups in terms of auto-antibody testing, with similar overall seropositivity rates (11% versus 16%, $p = 0.30$).

(371) The MUC5B promoter polymorphism rs35705950 was associated with the visual diagnosis of PrePF (present in 40% of those without fibrosis versus 53% with PrePF; MAF 0.29 versus 0.21, respectively, $p = 0.03$, FIG. 22). After age 60, there was a statistically significant difference in the proportion of subjects with visually diagnosed PrePF when the cohort was stratified by MUC5B genotype (23.8% versus 39.8% prevalence, $p = 0.02$) (FIG. 27).

(372) MUC5B variant carriers, regardless of their visual CT diagnosis, had significantly higher qHRCT fibrosis scores (1.3 [95% CI 1.2-1.5] versus 1.1 [95% CI 1.0-1.2], $p=0.02$). The association between MUC5B genotype and fibrosis score was significant even when controlling for age and male sex in linear regression ($p=0.03$, FIG. 28). Age was significantly associated with fibrosis score ($p=2.17 \times 10^{-9}$), but male sex ($p=0.63$) and smoking ($p=0.94$) were not. To determine whether individual textural components were driving the association of the composite fibrosis score with genotype, each score component was tested individually for association with the MUC5B variant, controlling for age and sex. Quantitative scores for reticulation, honeycombing, and ground glass were significantly associated with the MUC5B variant ($p=0.02$, $p=0.02$, $p=0.04$, respectively), while “not normal high density” was not ($p=0.18$). The simpler quantitative scoring method, log % HAA, was not significantly different in MUC5B variant carriers ($p=0.4$).

(373) In contrast to the MUC5B variant, the common IPF-associated TERT polymorphism (rs2736100) was not significantly associated with PrePF assessed either qualitatively (MAF 0.47 in PrePF versus 0.46 in unaffected, $p=0.77$) or quantitatively (MAF 0.50 fibrotic versus 0.47 not fibrotic, $p=0.40$).

(374) When these factors were examined individually for their contributions to risk of PrePF in our study cohort, we used a mixed effects logistic regression model to test the independent effects of age sex, smoking, and MUC5B or TERT genotypes while controlling for family. Age and the MUC5B genotype remained statistically significantly associated with PrePF (OR 1.15, 95% CI 1.09-1.22, $p=7.34 \times 10^{-7}$ and OR 2.18, 95% CI 1.00-4.73, $p=0.05$, respectively) (FIG. 22). The common TERT polymorphism (rs2736100) associated with fibrotic idiopathic interstitial pneumonia (29) was not significantly associated with PrePF (MAF was 0.45 in PrePF versus 0.45 in unaffected, $p=0.88$) or in a log-additive model controlling for age, sex, and smoking history ($p=0.57$).

(375) Given the presence of non-fibrotic ILD ($n=18$, FIG. 18) in the “No Fibrosis” cohort, secondary analyses were performed that (1) excluded non-fibrotic ILDs and (2) compared all ILD (inclusive of non-fibrotic ILD) to those without any ILD. When non-fibrotic ILDs were excluded from analyses, PrePF subjects were older ($p=4.7 \times 10^{-13}$), more commonly male ($p=0.04$), more often had a smoking history ($p=0.003$), and had a higher prevalence of the MUC5B promoter variant (MAF 0.29 versus 0.20, $p=0.02$). However, when controlling for family relatedness and the other risk factors in a mixed effects logistic regression, only age and the MUC5B promoter variant were significantly associated with PrePF with odds ratios 1.15 (95% CI 1.09-1.22, $p=9.5 \times 10^{-7}$) and 2.16 (95% CI 1.00-4.75, $p=0.05$), respectively. Another secondary analysis of the data was performed in which all subjects with CT findings of ILD (fibrotic or non-fibrotic) were compared to those without any evidence of ILD. Those with CT evidence of ILD were older (mean age 64.3 years, 95% CI 62.2-66.3) compared to those without any evidence of ILD (mean age 55.7 years, 95% CI 54.8-56.6, $p=4.1 \times 10^{-12}$), more likely to be male ($p=0.01$), more likely to have smoked ($p=0.0003$), and more likely to carry the MUC5B promoter variant (MAF 0.21 versus 0.30, $p=0.006$). When controlling for family relatedness in a mixed effects logistic regression model, age (OR 1.10, 95% CI 1.07-1.14, $p=1.21 \times 10^{-9}$), smoking history (OR 1.72, 95% CI 1.00-2.99, $p=0.04$), and the MUC5B promoter variant (OR 1.73, 95% CI 1.08-2.76, $p=0.02$) were significantly associated with risk of ILD.

OTHER EMBODIMENTS

(376) It is to be understood that while the disclosure has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the disclosure, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. A method of treating a fibrotic lung disease in an asymptomatic human subject within an at-risk population, wherein the fibrotic lung disease is familial interstitial pneumonia (FIP), pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), interstitial lung disease (ILD), an interstitial lung abnormality (ILA), an asymptomatic ILA, fibrotic interstitial lung disease (FILD), or rheumatoid arthritis-associated interstitial lung disease (RA-ILD), and wherein the subject has a blood relative with a fibrotic lung disease selected from the group consisting of familial interstitial pneumonia (FIP), pre-clinical pulmonary disease, pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), an asymptomatic ILA, interstitial lung disease (ILD), fibrotic interstitial lung disease (FILD) and rheumatoid arthritis-associated interstitial lung disease (RA-ILD), the method comprising: a) identifying pre-clinical pulmonary fibrosis (PrePF) in the subject by use of quantitative high resolution computed tomography (qHRCT) comprising the use of a convolutional neural network to quantify the extent of fibrosis in the subject and wherein the convolutional neural network architecture classifies image regions using pixel and texture features extracted by multiple convolutional layers at different scales; b) determining that the subject has the T allele of the MUC5B rs35705950 polymorphism; c) administering a therapeutic agent in an amount effective for the treatment of fibrotic lung disease to the subject, wherein the therapeutic agent: (i) prevents the onset or development of a sign or symptom of the fibrotic lung disease; (ii) delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the therapeutic agent; or (iii) agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.
 2. The method of claim 1, wherein the subject presents radiographic Usual Interstitial Pneumonia (UIP).
 3. The method of claim 1, wherein the subject is greater than 40 years in age.
 4. The method of claim 1, wherein the blood relative is a sibling.
 5. The method of claim 1, wherein the MUC5B rs35705950 polymorphism is encoded by a sequence comprising SEQ ID NO: 7.
 6. The method of claim 1, wherein the therapeutic agent comprises a N-acetylcysteine, pirfenidone, or nintedanib.
 7. The method of claim 1, wherein PrePF is identified in the subject when a fibrosis score in the subject is above a predetermined cutoff value.
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